Falchook_NoWriting

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Executive Summary

Methods

For our methodology,

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.1.6
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.1     v tibble    3.3.0
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.2.0
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom

Attaching package: 'zoo'


The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

Question1	Question2	P_value	Significant	test	Adj_P	Sig_Adj
Q13d	Q12	0.007	TRUE	fisher	0.2835	FALSE
Q13h	Q2	0.0055	TRUE	fisher	0.2835	FALSE
Q18	Q11	0.0145	TRUE	fisher	0.3915	FALSE
Q16d	Q3	0.0235	TRUE	fisher	0.4759	FALSE
Q13a	Q1	0.043	TRUE	fisher	0.4891	FALSE
Q13d	Q10	0.0785	FALSE	fisher	0.4891	FALSE
Q13d	Q5	0.0545	FALSE	fisher	0.4891	FALSE
Q13f	Q12	0.069	FALSE	fisher	0.4891	FALSE
Q14a	Q5	0.0445	TRUE	fisher	0.4891	FALSE
Q14c	Q3	0.0555	FALSE	fisher	0.4891	FALSE
Q15a	Q3	0.074	FALSE	fisher	0.4891	FALSE
Q17	Q10	0.0365	TRUE	fisher	0.4891	FALSE
Q18	Q10	0.0715	FALSE	fisher	0.4891	FALSE
Q13b	Q2	0.088	FALSE	fisher	0.5091	FALSE
Q16c	Q3	0.1019	FALSE	fisher	0.5503	FALSE
Q13e	Q10	0.1209	FALSE	fisher	0.6121	FALSE
Q13c	Q1	0.2134	FALSE	fisher	0.6410	FALSE
Q13d	Q11	0.2354	FALSE	fisher	0.6410	FALSE
Q13e	Q12	0.2549	FALSE	fisher	0.6410	FALSE
Q13f	Q10	0.1674	FALSE	fisher	0.6410	FALSE
Q13g	Q1	0.2844	FALSE	fisher	0.6410	FALSE
Q13g	Q5	0.2469	FALSE	fisher	0.6410	FALSE
Q13h	Q12	0.2589	FALSE	fisher	0.6410	FALSE
Q13h	Q1	0.2159	FALSE	fisher	0.6410	FALSE
Q13h	Q5	0.1939	FALSE	fisher	0.6410	FALSE
Q13i	Q10	0.2774	FALSE	fisher	0.6410	FALSE
Q13i	Q11	0.2849	FALSE	fisher	0.6410	FALSE
Q13i	Q5	0.1539	FALSE	fisher	0.6410	FALSE
Q14b	Q5	0.2764	FALSE	fisher	0.6410	FALSE
Q14b	Q3	0.2094	FALSE	fisher	0.6410	FALSE
Q14c	Q5	0.1474	FALSE	fisher	0.6410	FALSE
Q14d	Q5	0.2404	FALSE	fisher	0.6410	FALSE
Q16a	Q3	0.2394	FALSE	fisher	0.6410	FALSE
Q16a	Q1	0.1959	FALSE	fisher	0.6410	FALSE
Q16c	Q1	0.2719	FALSE	fisher	0.6410	FALSE
Q17	Q11	0.2079	FALSE	fisher	0.6410	FALSE
Q13i	Q1	0.2999	FALSE	fisher	0.6565	FALSE
Q13c	Q5	0.3228	FALSE	fisher	0.6567	FALSE
Q14d	Q3	0.3113	FALSE	fisher	0.6567	FALSE
Q16b	Q3	0.3243	FALSE	fisher	0.6567	FALSE
Q13b	Q11	0.3933	FALSE	fisher	0.6974	FALSE
Q13c	Q10	0.3748	FALSE	fisher	0.6974	FALSE
Q13c	Q11	0.3793	FALSE	fisher	0.6974	FALSE
Q13e	Q11	0.3993	FALSE	fisher	0.6974	FALSE
Q13e	Q5	0.3558	FALSE	fisher	0.6974	FALSE
Q13f	Q5	0.4133	FALSE	fisher	0.6974	FALSE
Q13i	Q12	0.4123	FALSE	fisher	0.6974	FALSE
Q18	Q1	0.3758	FALSE	fisher	0.6974	FALSE
Q17	Q1	0.4468	FALSE	fisher	0.7386	FALSE
Q13g	Q12	0.4908	FALSE	fisher	0.7951	FALSE

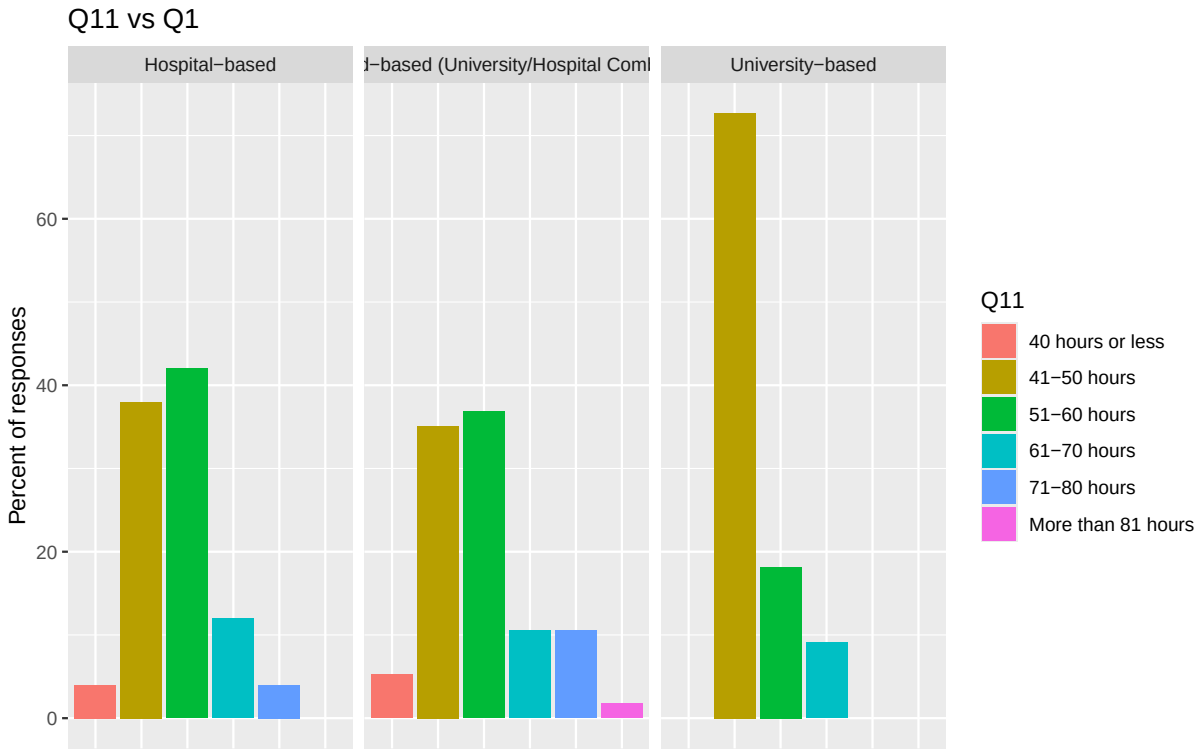
Question1	Question2	P_value	Significant	test	Adj_P	Sig_Adj
Q10	Q2	0.2104	FALSE	fisher	0.9615	FALSE
Q12a	Q2	0.8151	FALSE	fisher	0.9615	FALSE
Q12b	Q2	0.8686	FALSE	fisher	0.9615	FALSE
Q12c	Q2	0.9615	FALSE	fisher	0.9615	FALSE
Q12d	Q2	0.7986	FALSE	fisher	0.9615	FALSE
Q12e	Q2	0.954	FALSE	fisher	0.9615	FALSE
Q12f	Q2	0.8501	FALSE	fisher	0.9615	FALSE
Q12g	Q2	0.4478	FALSE	fisher	0.9615	FALSE
Q12h	Q2	0.1439	FALSE	fisher	0.9615	FALSE
Q12i	Q2	0.8326	FALSE	fisher	0.9615	FALSE
Q21	Q2	0.1189	FALSE	fisher	0.9615	FALSE
Q22a	Q2	0.6617	FALSE	fisher	0.9615	FALSE
Q22b	Q2	0.6502	FALSE	fisher	0.9615	FALSE
Q22c	Q2	0.3793	FALSE	fisher	0.9615	FALSE

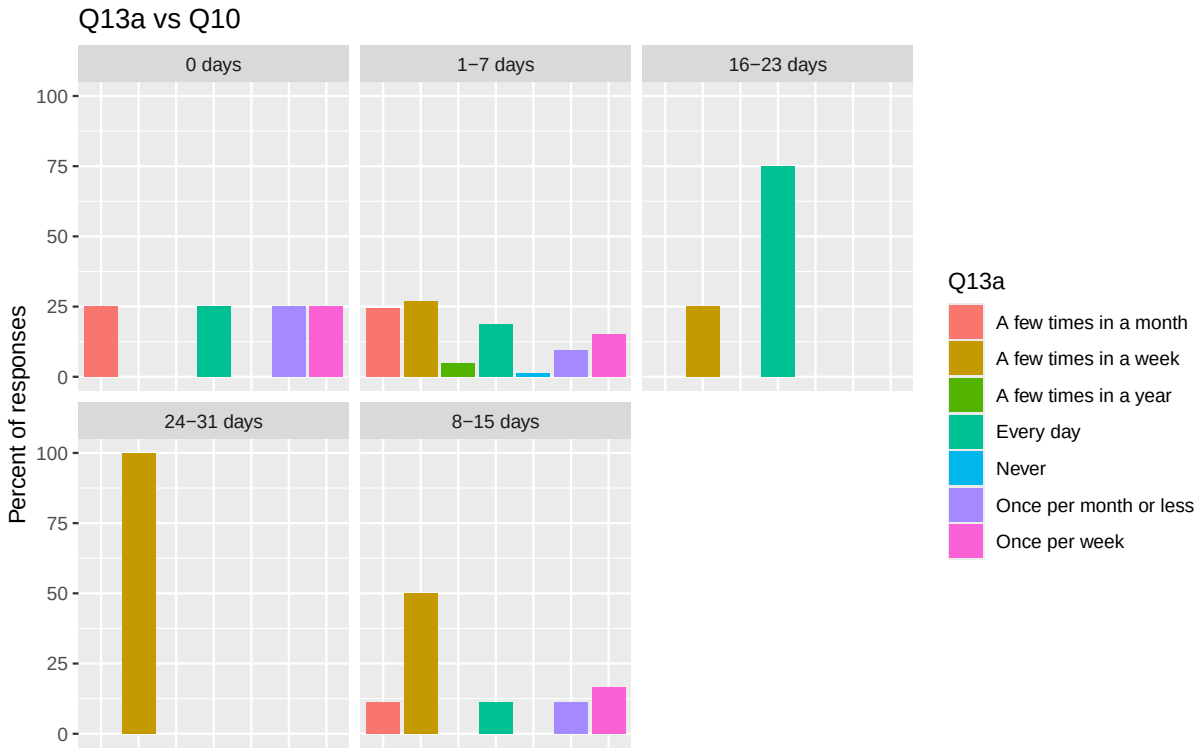
Question1	Question2	P_value	Significant	test	Adj_P	Sig_Adj
Q26	role	1e-04	TRUE	fisher	0.0051	TRUE
Q12	role	5e-04	TRUE	fisher	0.0064	TRUE
Q14b	role	5e-04	TRUE	fisher	0.0064	TRUE
Q21	role	5e-04	TRUE	fisher	0.0064	TRUE
Q3	role	0.0015	TRUE	fisher	0.0096	TRUE
Q13a	role	0.001	TRUE	fisher	0.0096	TRUE
Q13b	role	0.0015	TRUE	fisher	0.0096	TRUE
Q22b	role	0.0015	TRUE	fisher	0.0096	TRUE
Q14a	role	0.002	TRUE	fisher	0.0102	TRUE
Q20b	role	0.002	TRUE	fisher	0.0102	TRUE
Q13c	role	0.003	TRUE	fisher	0.0128	TRUE
Q20a	role	0.003	TRUE	fisher	0.0128	TRUE
Q13f	role	0.0035	TRUE	fisher	0.0137	TRUE
Q20d	role	0.004	TRUE	fisher	0.0146	TRUE
Q22a	role	0.0045	TRUE	fisher	0.0153	TRUE
Q19	role	0.006	TRUE	fisher	0.0191	TRUE
Q22c	role	0.014	TRUE	fisher	0.0420	TRUE
Q15a	role	0.0155	TRUE	fisher	0.0439	TRUE
Q14c	role	0.0305	TRUE	fisher	0.0819	FALSE
Q16b	role	0.0345	TRUE	fisher	0.0880	FALSE
Q9	role	0.0585	FALSE	fisher	0.1244	FALSE
Q13d	role	0.0585	FALSE	fisher	0.1244	FALSE
Q13h	role	0.054	FALSE	fisher	0.1244	FALSE
Q14d	role	0.061	FALSE	fisher	0.1244	FALSE
Q20c	role	0.0605	FALSE	fisher	0.1244	FALSE
Q13e	role	0.0675	FALSE	fisher	0.1275	FALSE
Q24d	role	0.0675	FALSE	fisher	0.1275	FALSE
Q23a	role	0.0785	FALSE	fisher	0.1430	FALSE
Q17	role	0.09	FALSE	fisher	0.1530	FALSE
Q24f	role	0.089	FALSE	fisher	0.1530	FALSE
Q24b	role	0.0955	FALSE	fisher	0.1571	FALSE
Q27	role	0.1039	FALSE	fisher	0.1656	FALSE
Q13g	role	0.1104	FALSE	fisher	0.1706	FALSE
Q15b	role	0.1339	FALSE	fisher	0.1951	FALSE
Q23d	role	0.1314	FALSE	fisher	0.1951	FALSE
Q23c	role	0.1734	FALSE	fisher	0.2456	FALSE
Q13i	role	0.2054	FALSE	fisher	0.2831	FALSE
Q8	role	0.2234	FALSE	fisher	0.2998	FALSE
Q23b	role	0.2419	FALSE	fisher	0.3163	FALSE
Q18	role	0.2529	FALSE	fisher	0.3224	FALSE
Q1	role	0.3188	FALSE	fisher	0.3914	FALSE
Q15c	role	0.3223	FALSE	fisher	0.3914	FALSE
Q11	role	0.3863	FALSE	fisher	0.4582	FALSE
Q24c	role	0.4518	FALSE	fisher	0.5237	FALSE
Q16a	role	0.4778	FALSE	fisher	0.5415	FALSE
Q10	role	0.4913	FALSE	fisher	0.5447	FALSE
Q24e	role	0.7436	FALSE	fisher	0.8069	FALSE
Q25	role	0.931	FALSE	fisher	0.9892	FALSE
Q16c	role	1	FALSE	fisher	1.0000	FALSE
Q16d	role	1	FALSE	fisher	1.0000	FALSE

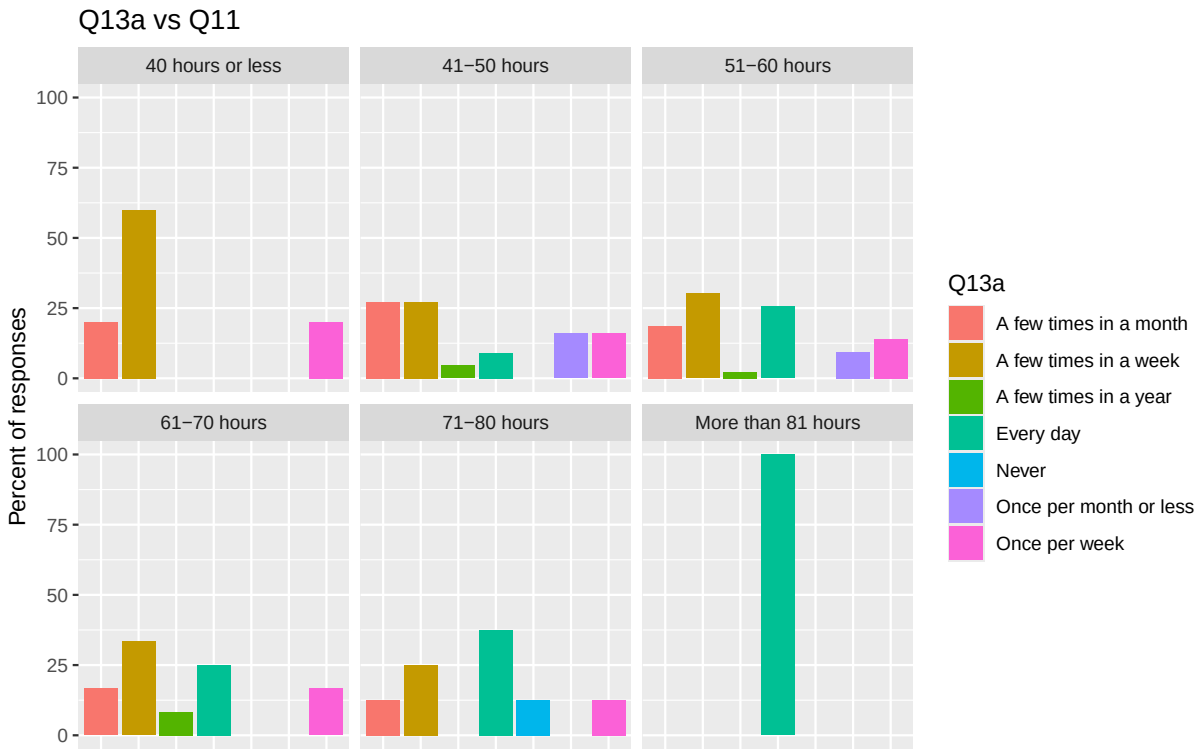
Residents Results Table

Directors Results Table

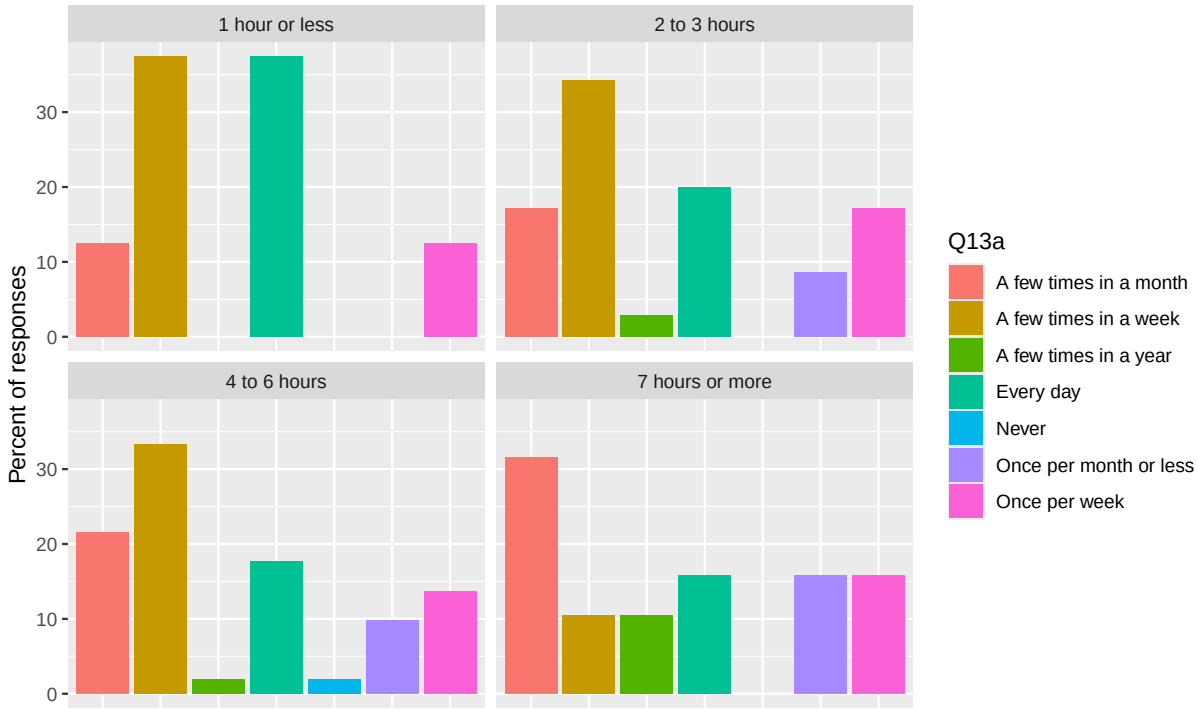
Appendix A: Plots



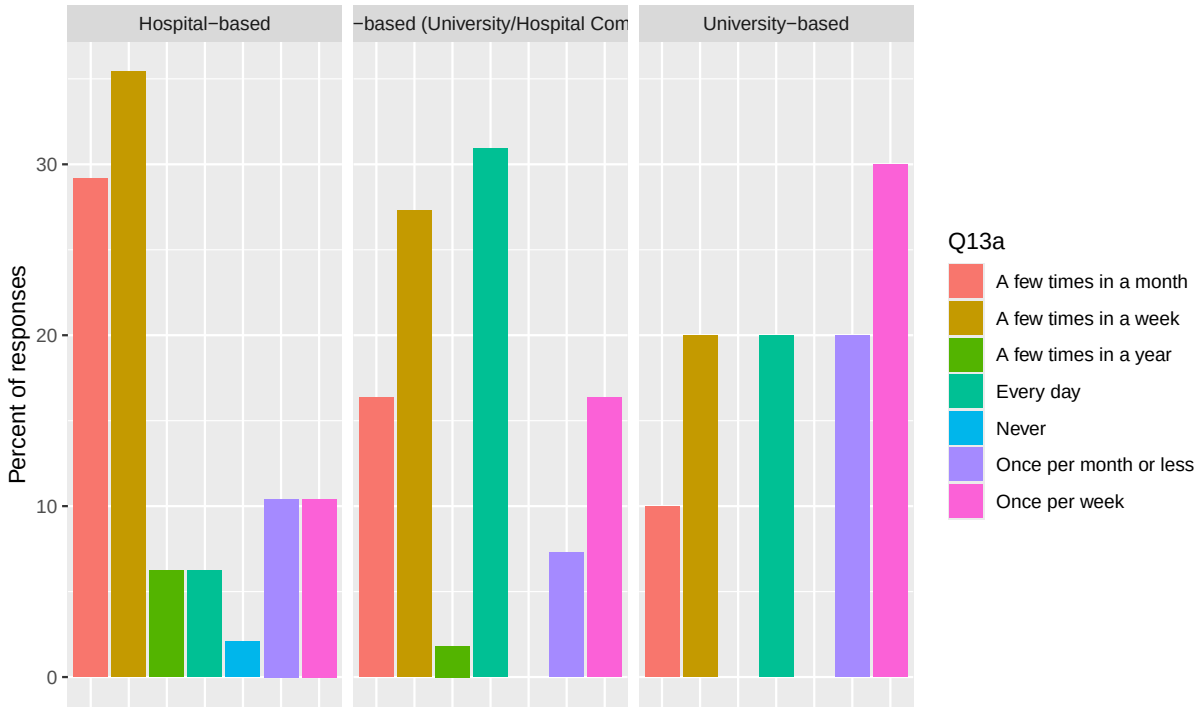

Q13a vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13a
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week


Q13a vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13a
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

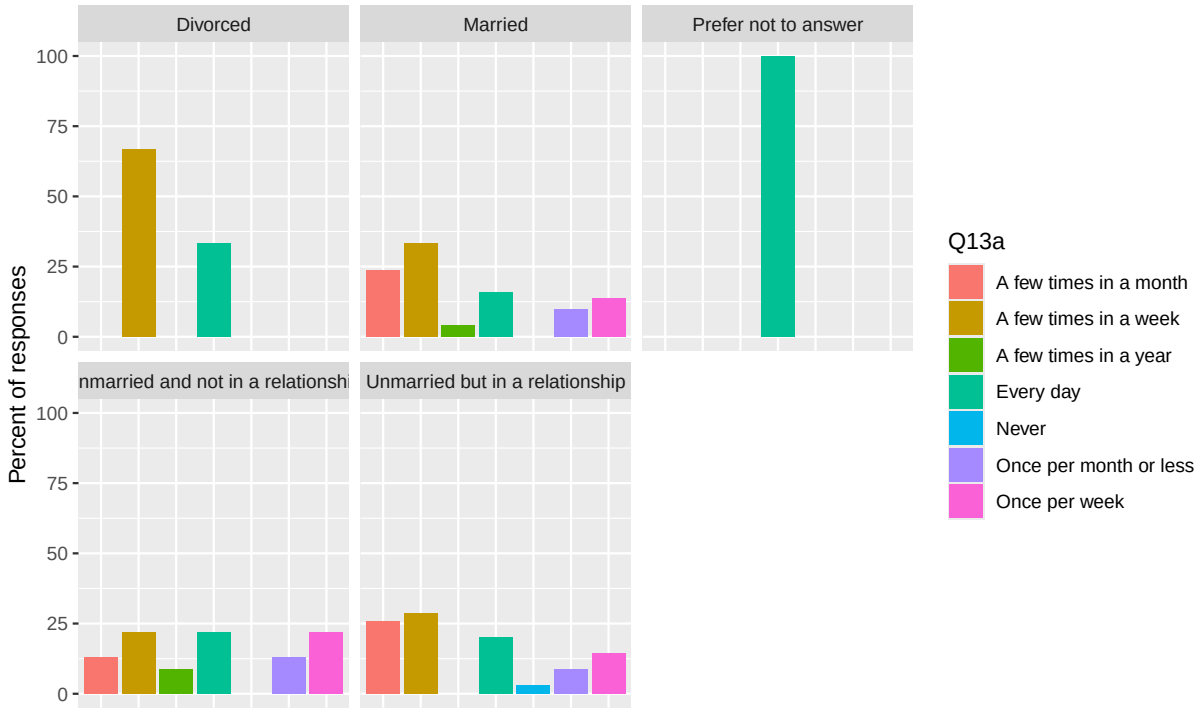
Q13a vs Q12



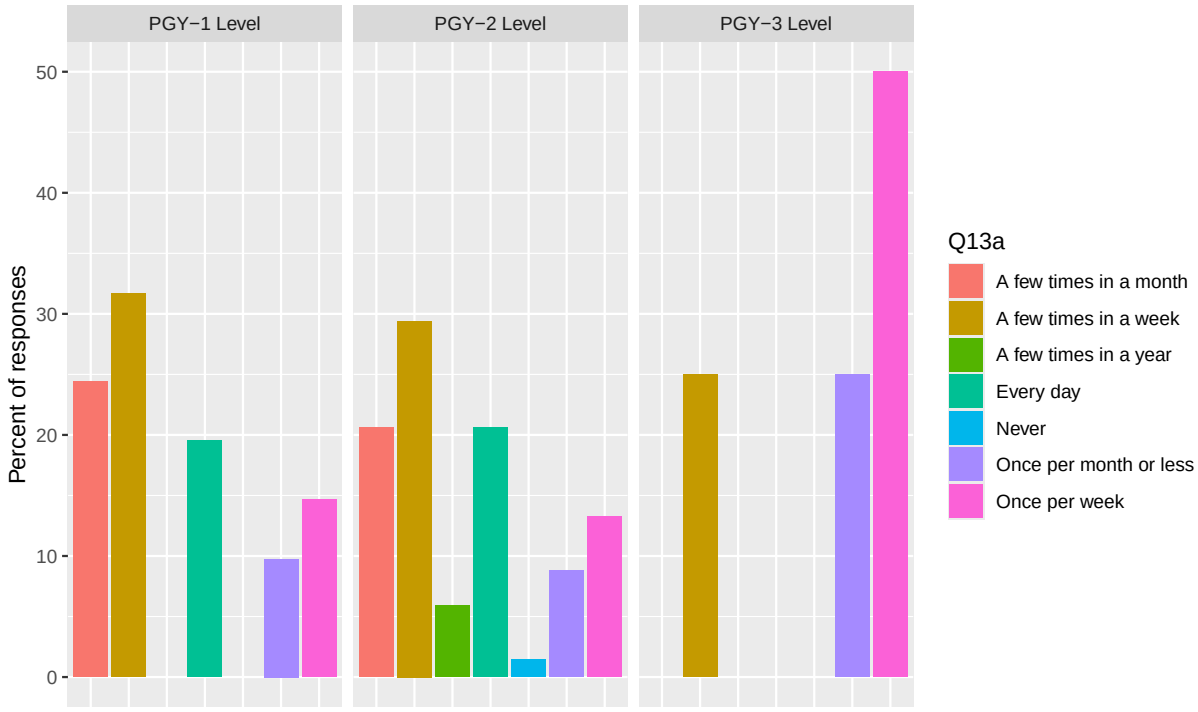
Q13a vs Q1

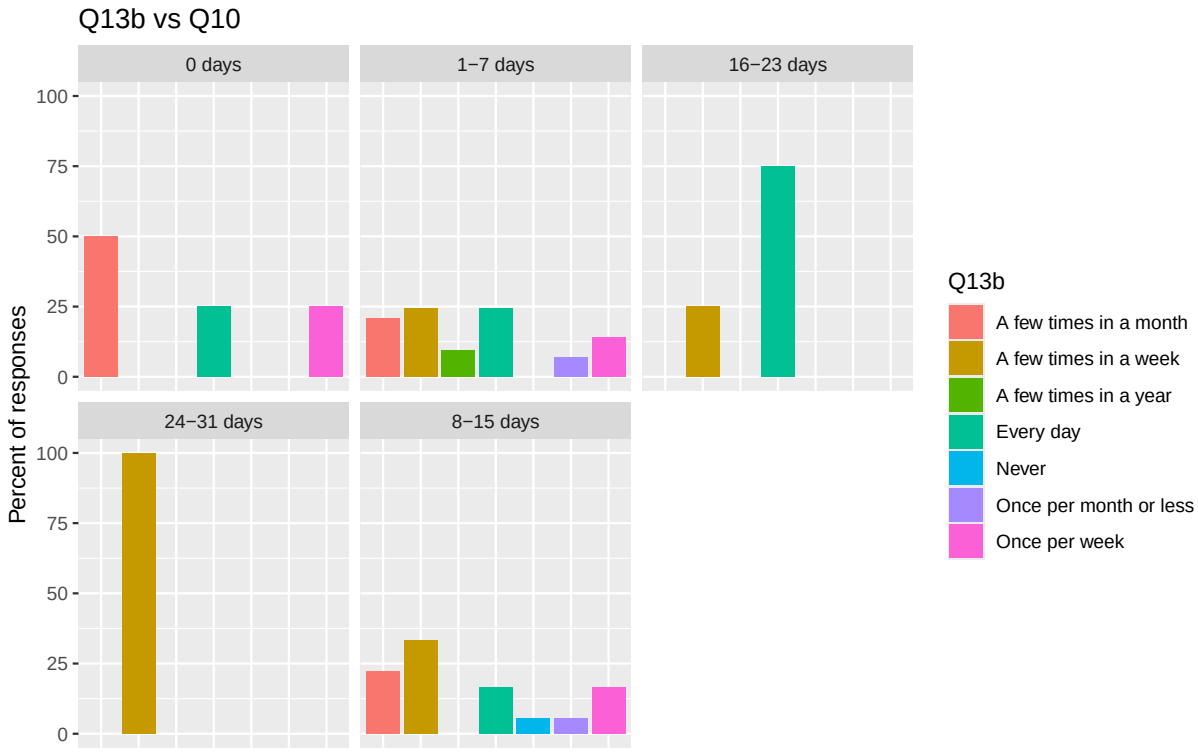


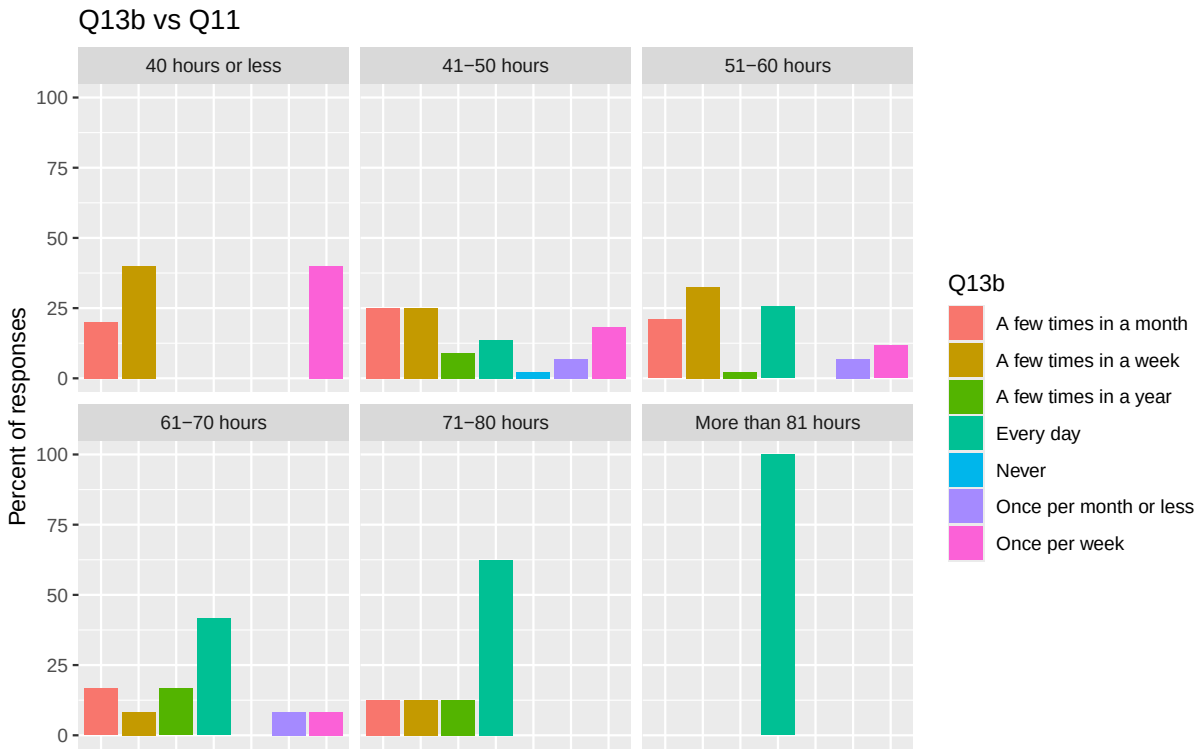
Q13a vs Q5

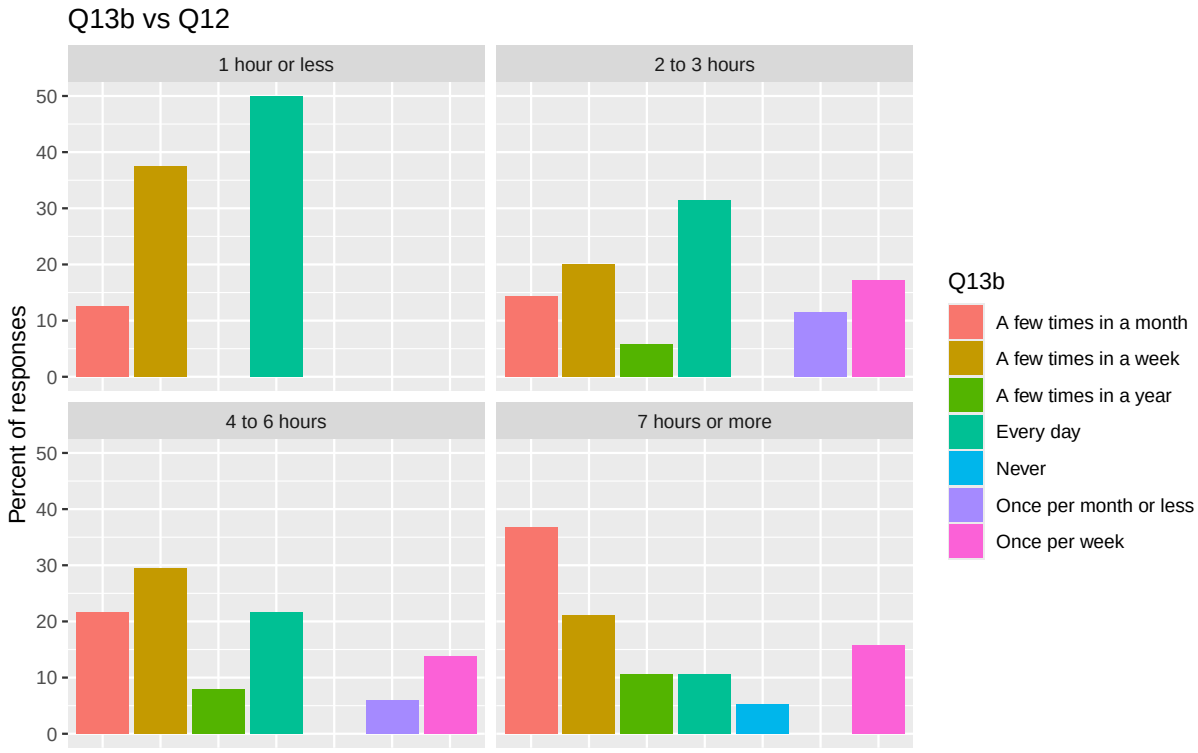


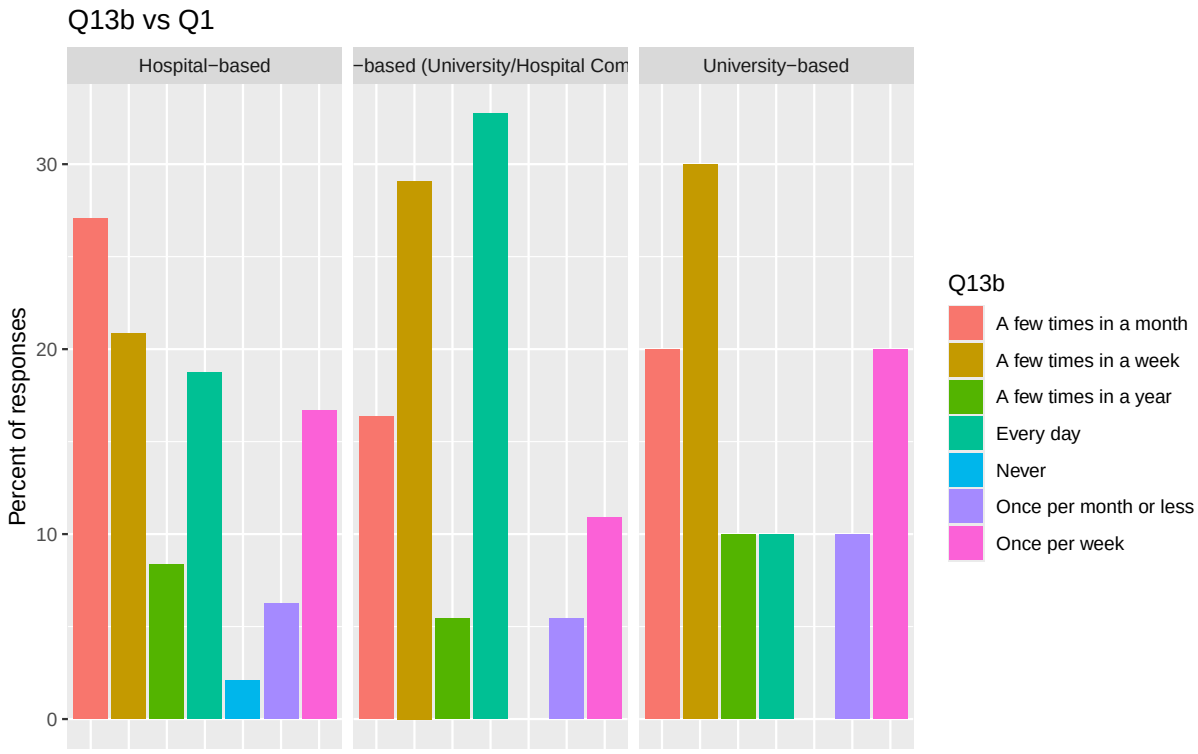
Q13a vs Q2



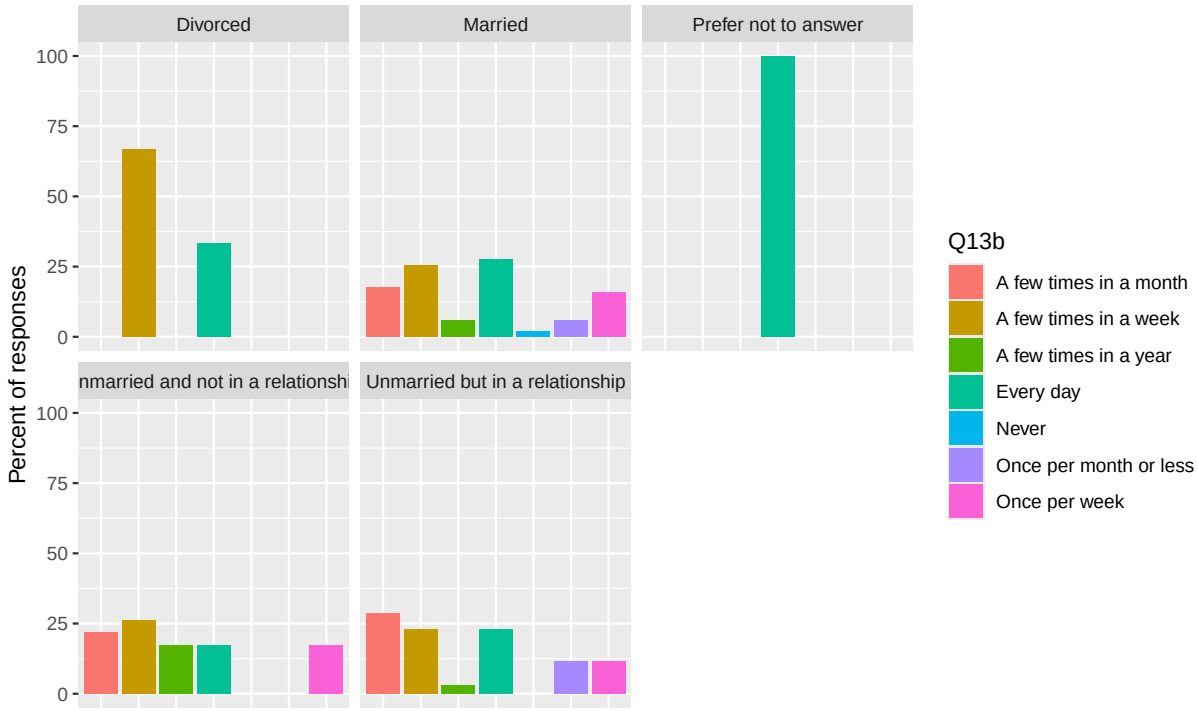

Q13b vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week


Q13b vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

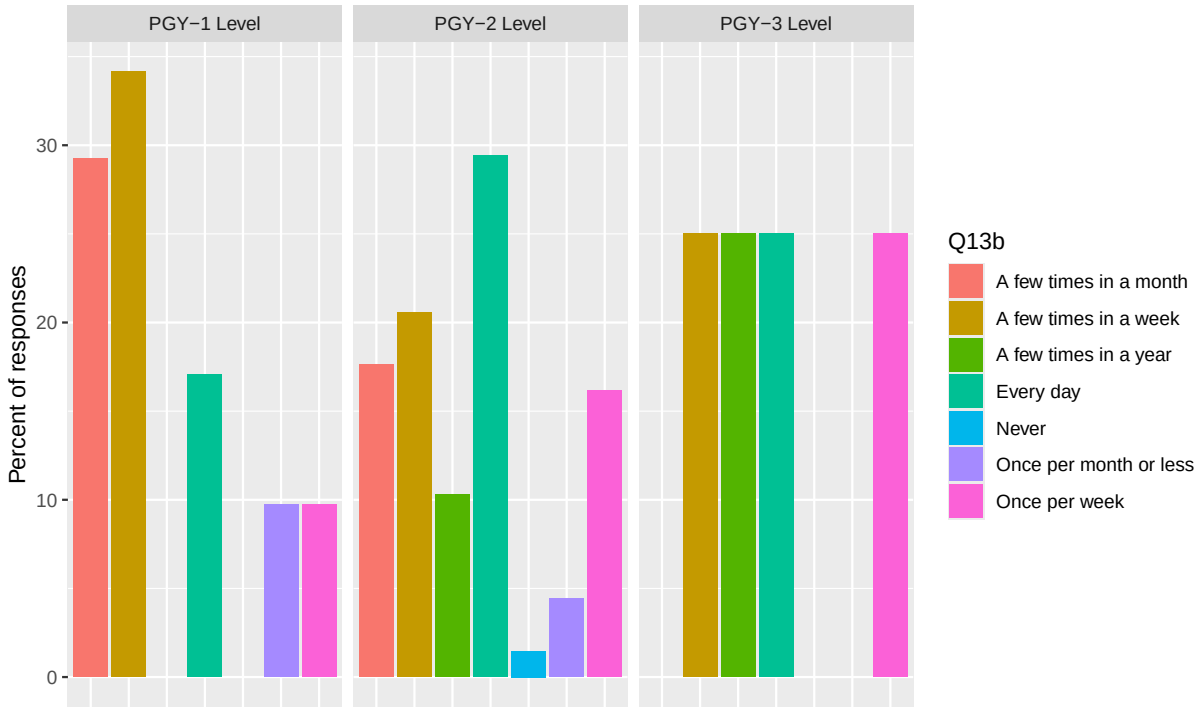

Q13b vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
50
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

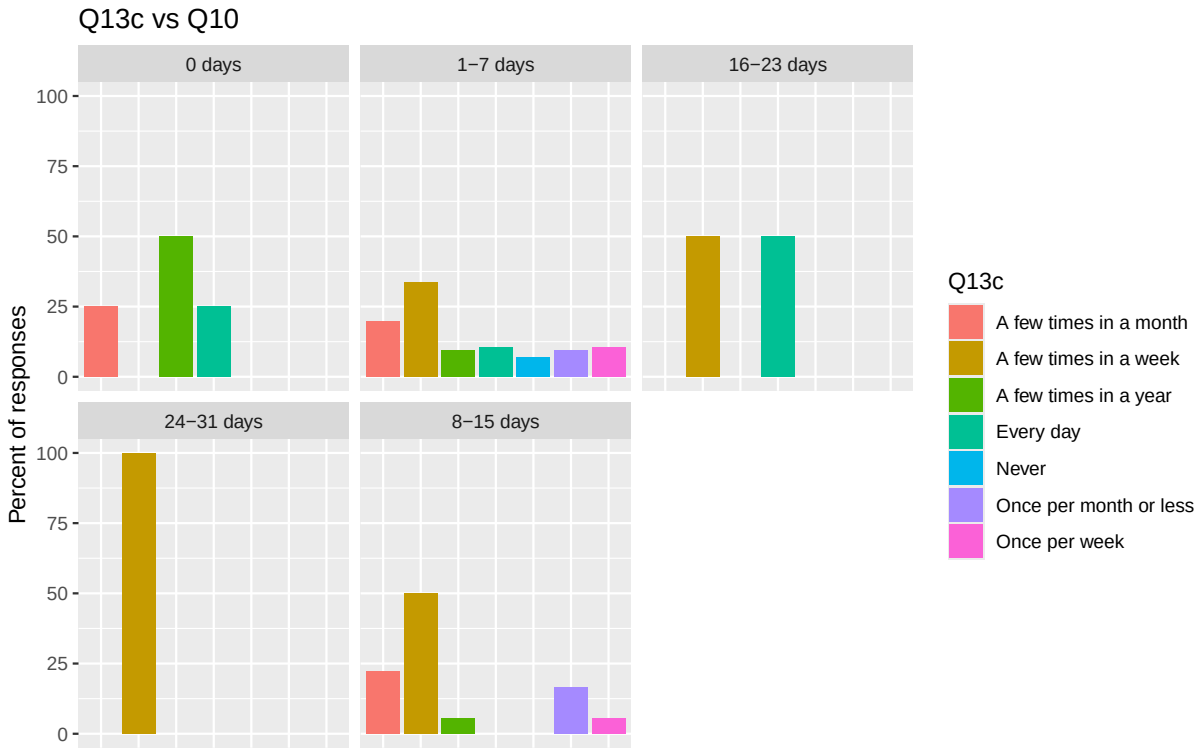

Q13b vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

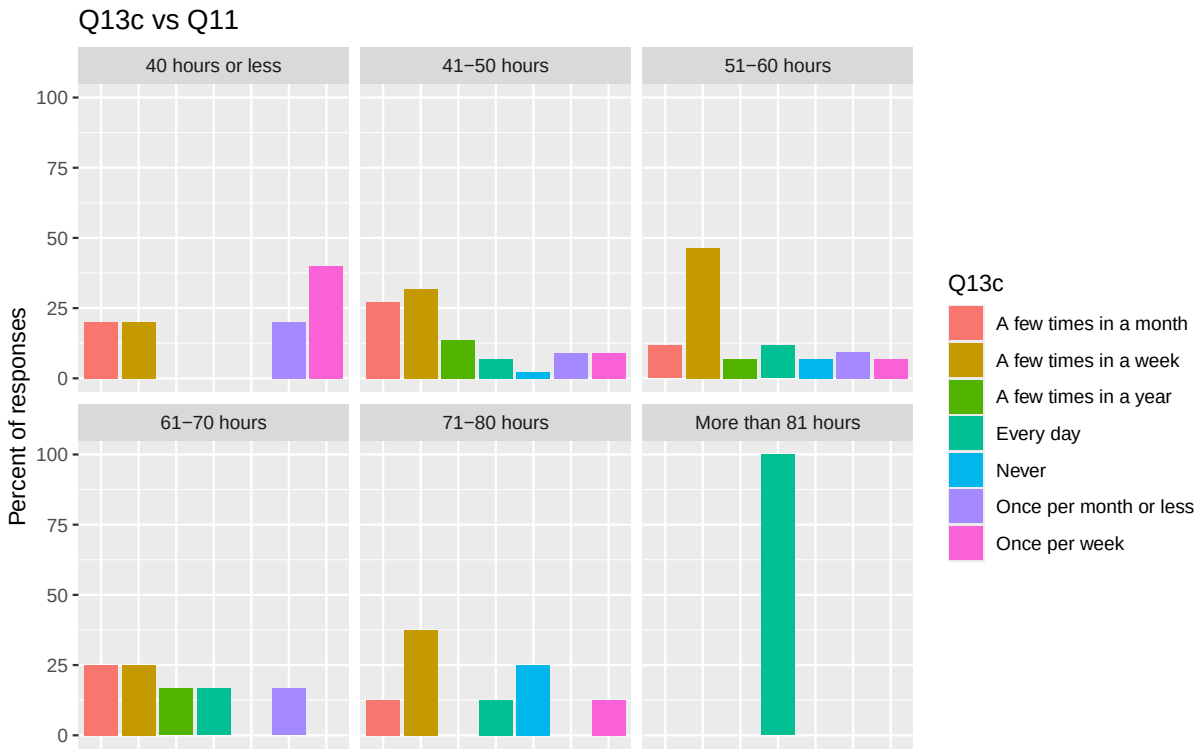
Q13b vs Q5

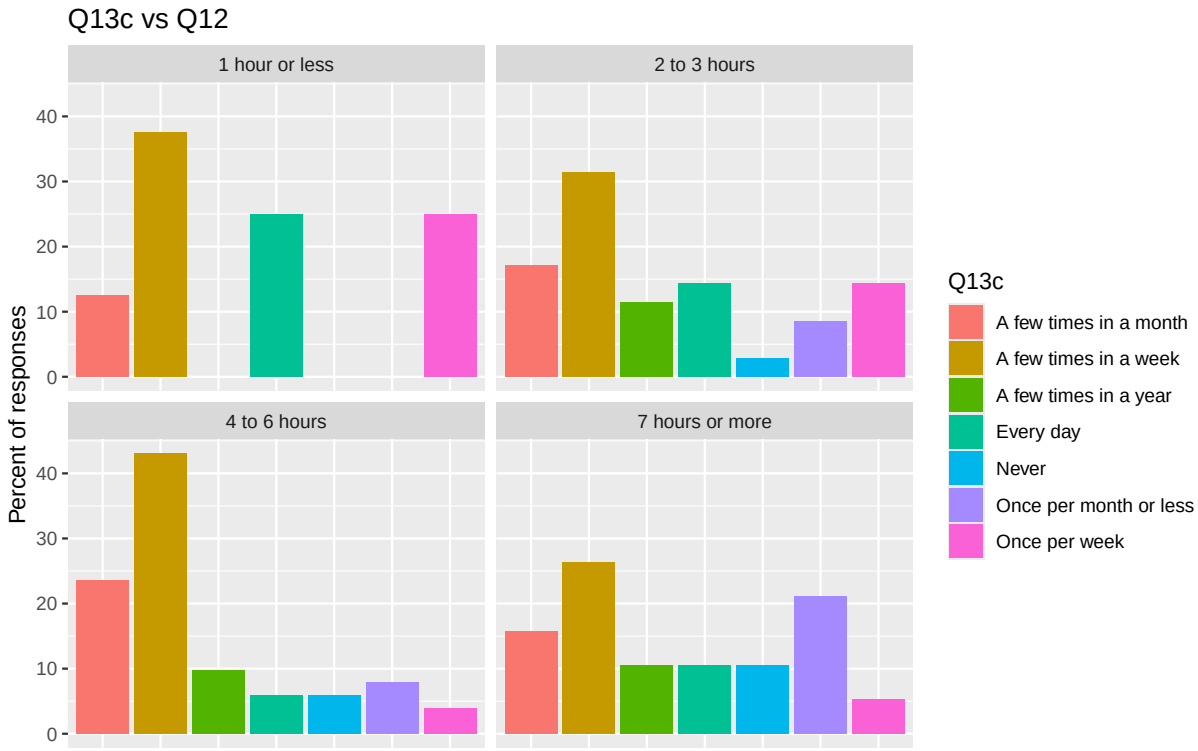


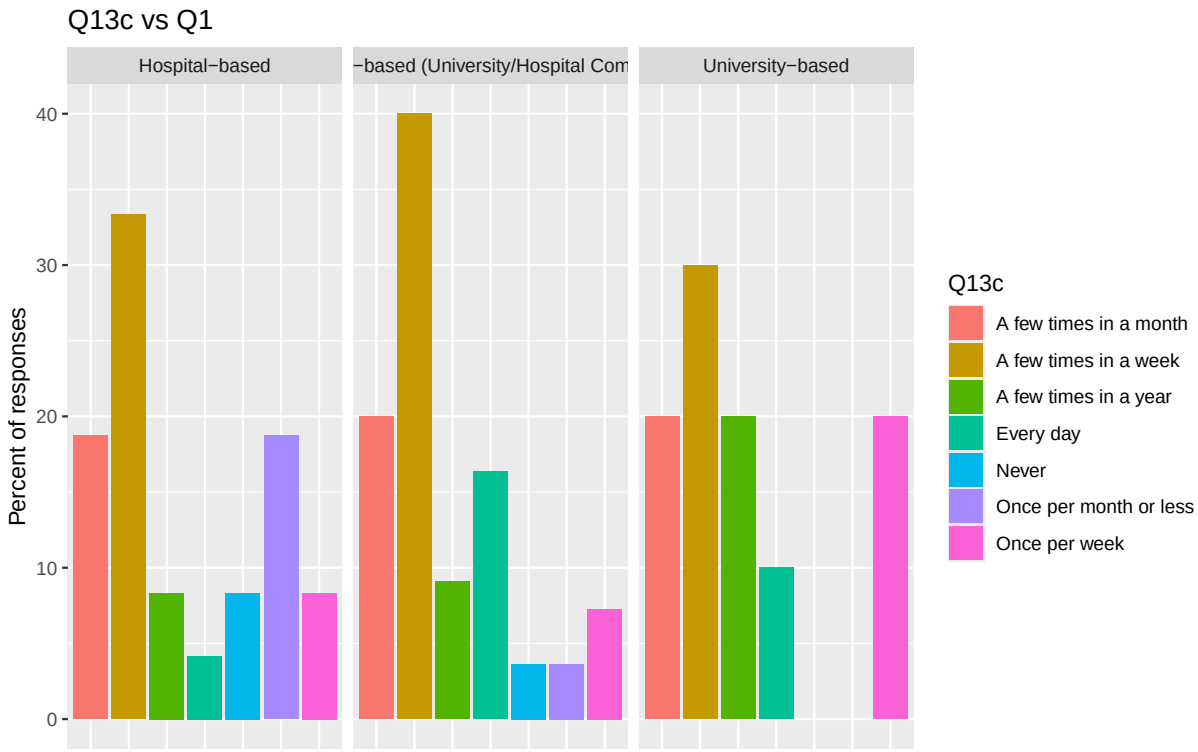
Q13b vs Q2



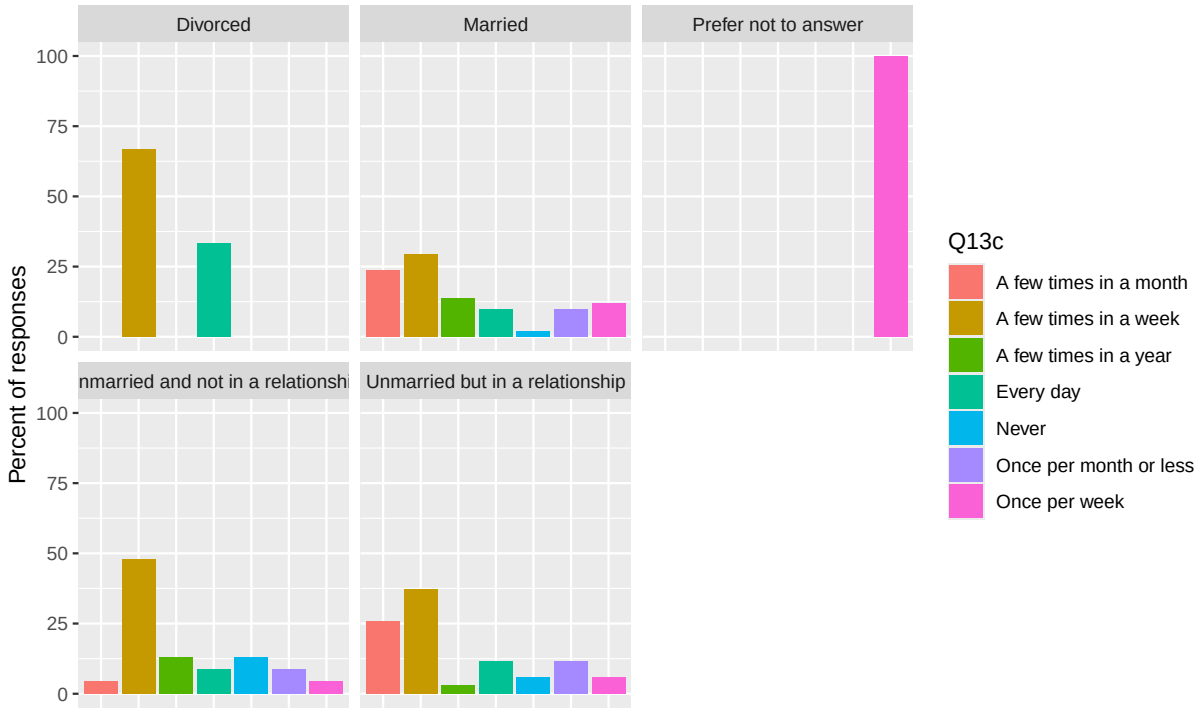

Q13c vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week


Q13c vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

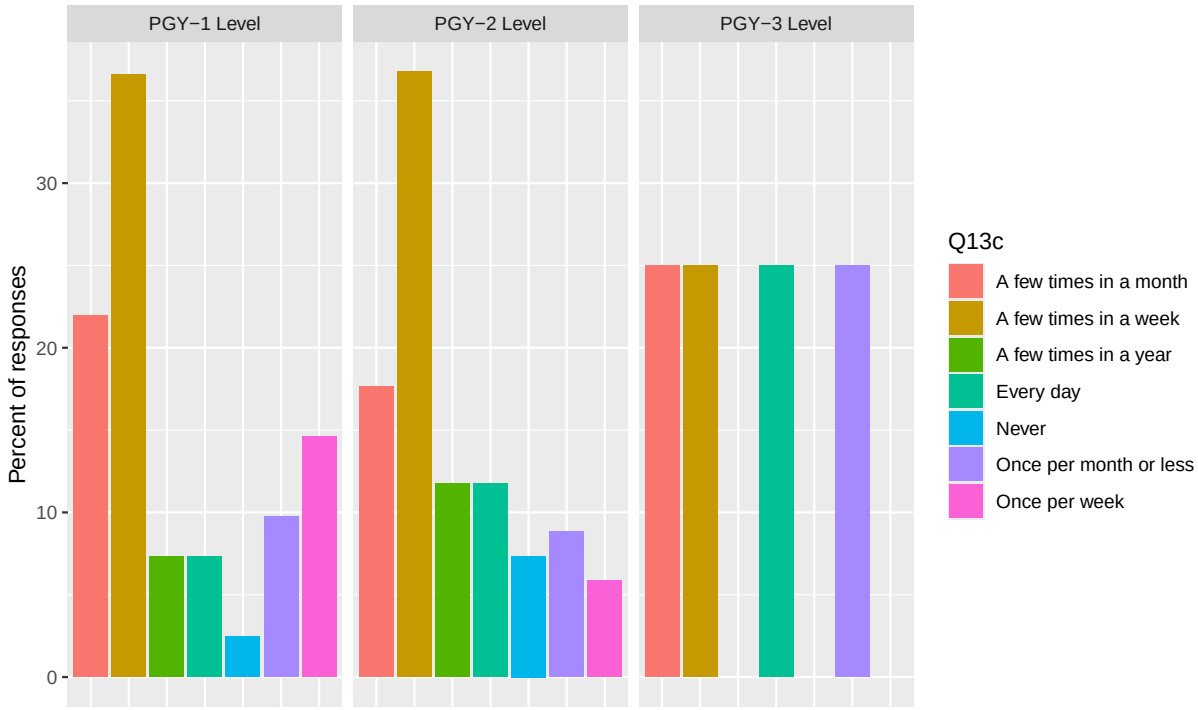

Q13c vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

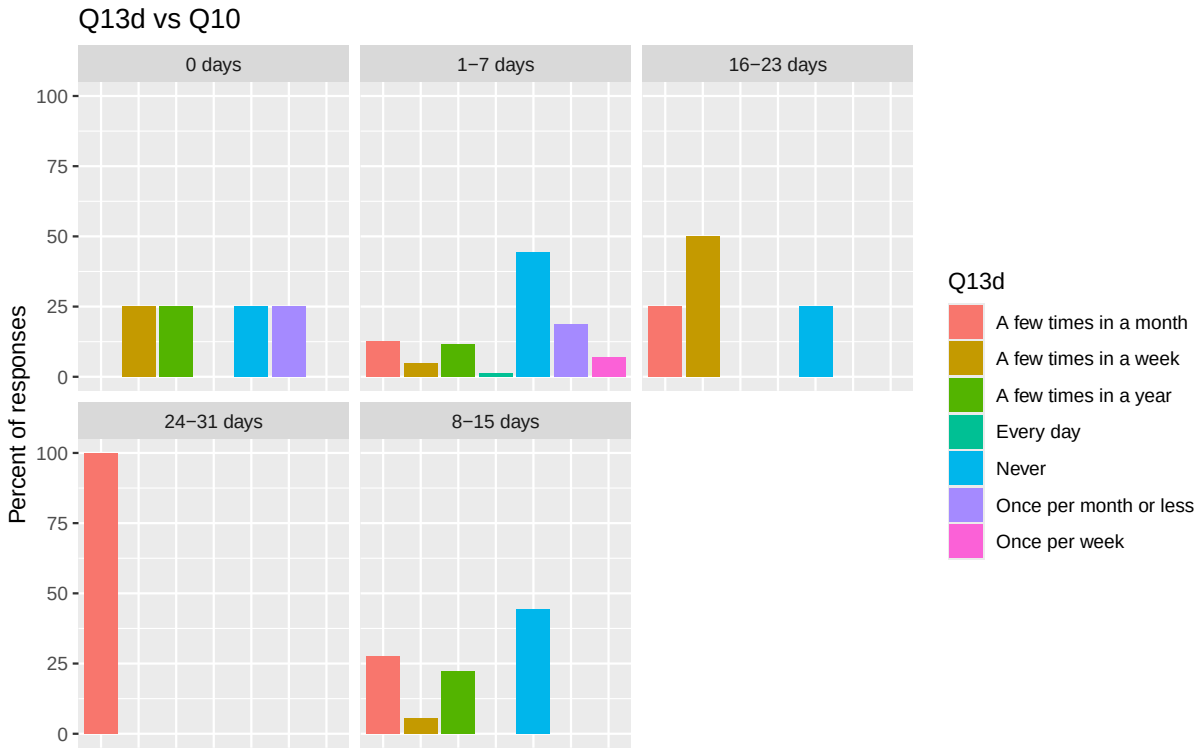

Q13c vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
40
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

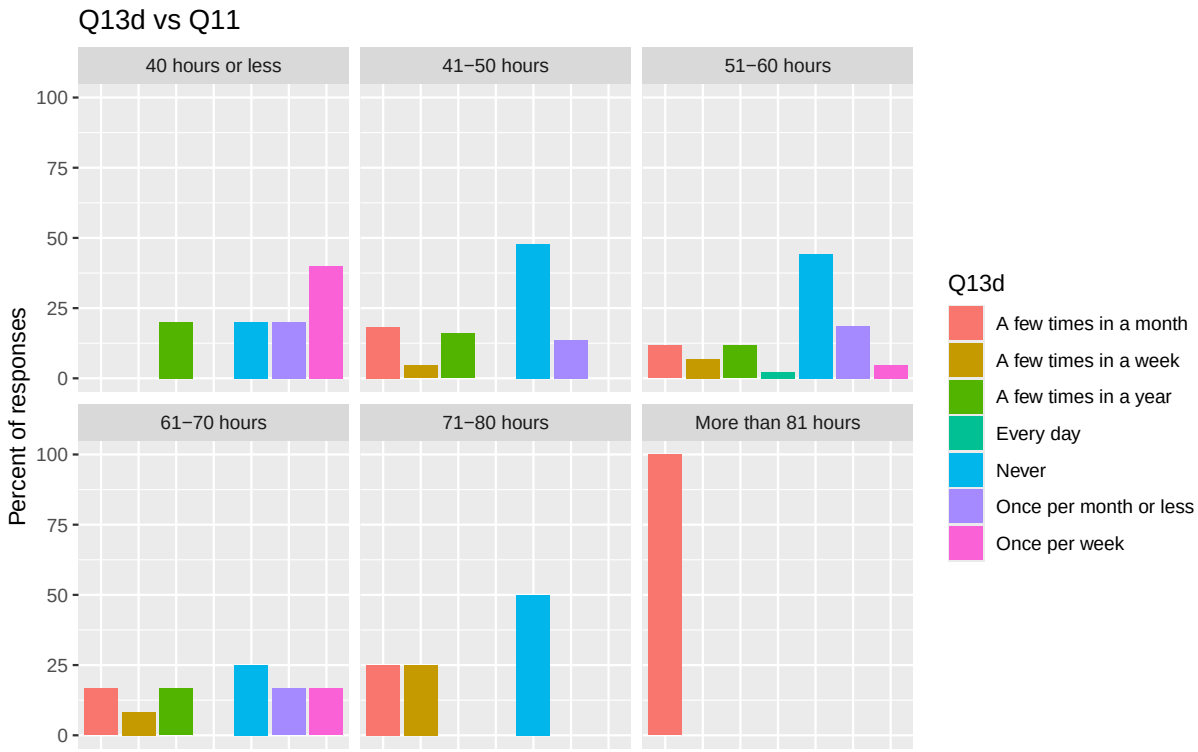
Q13c vs Q5



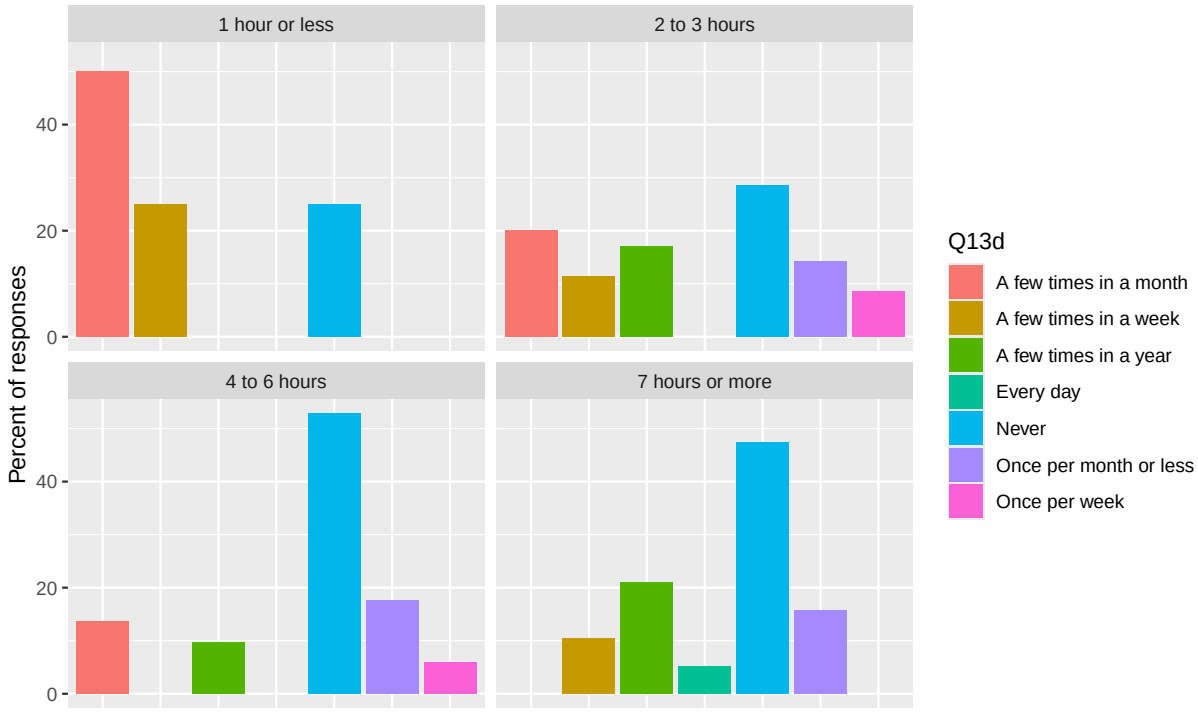
Q13c vs Q2



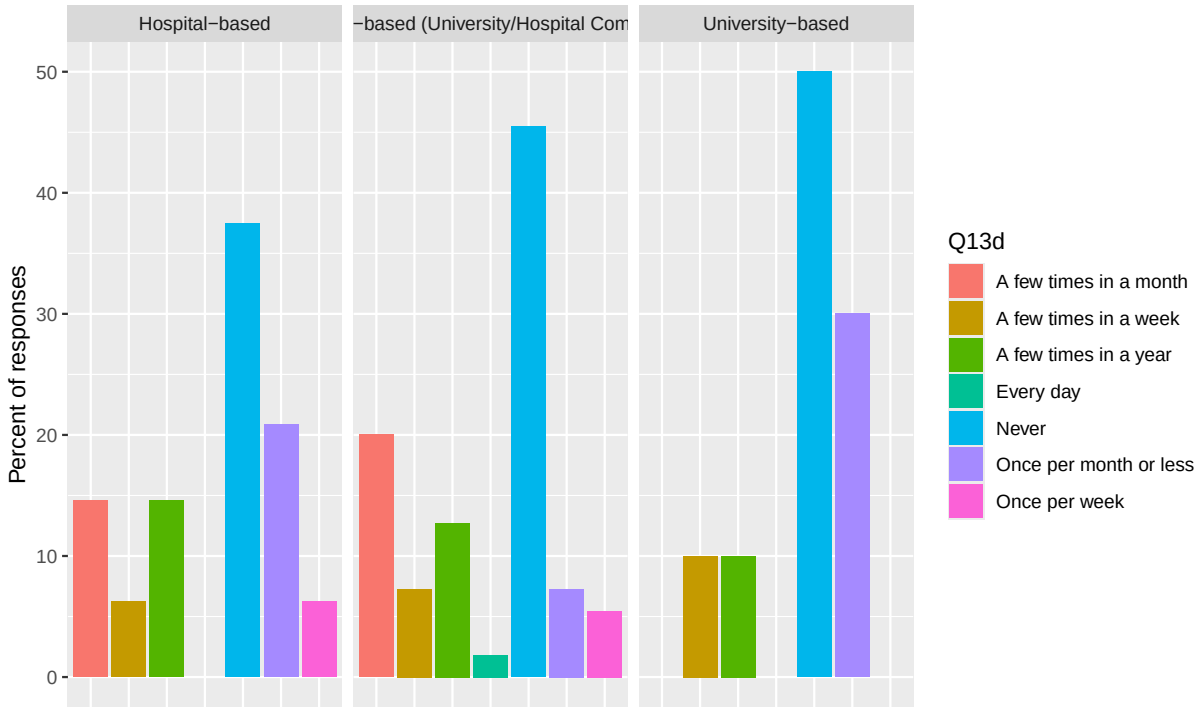

Q13d vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13d
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week


Q13d vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13d
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

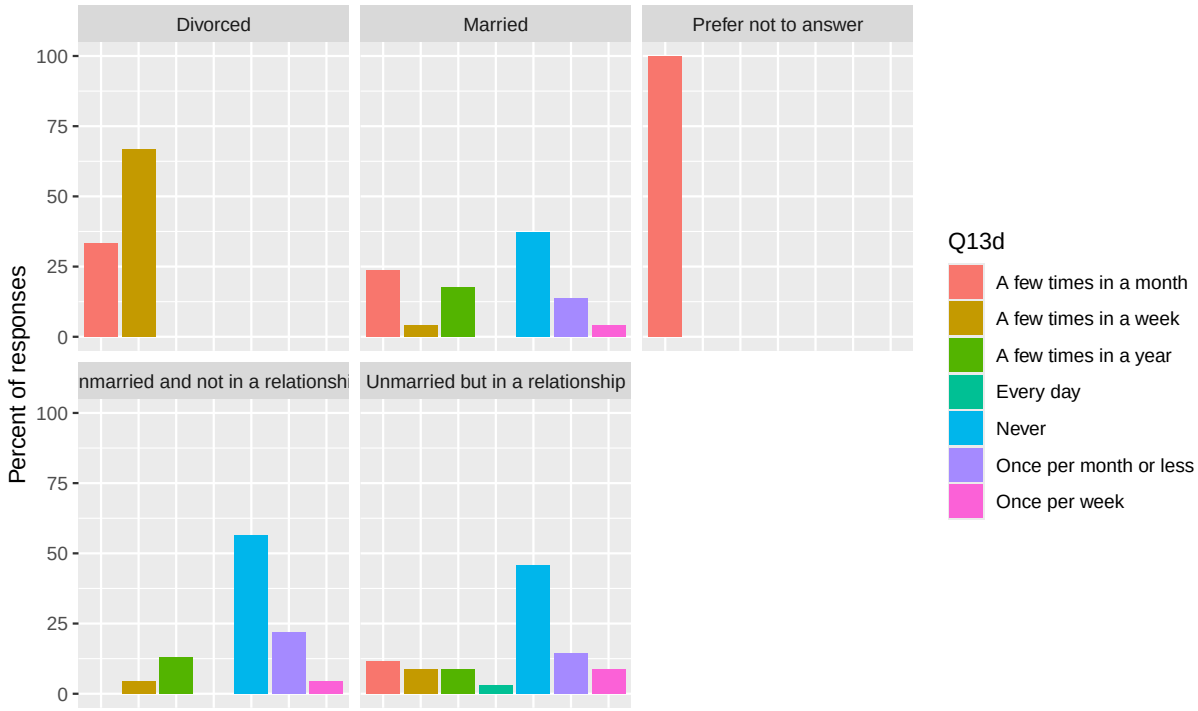
Q13d vs Q12



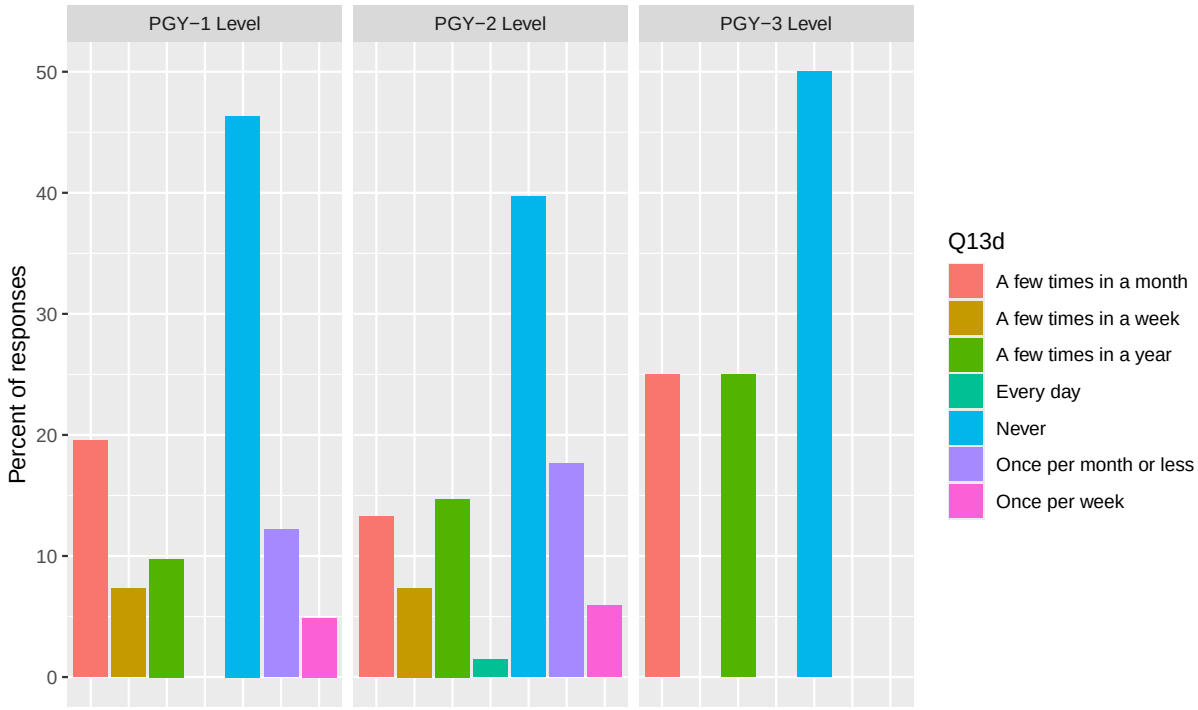
Q13d vs Q1

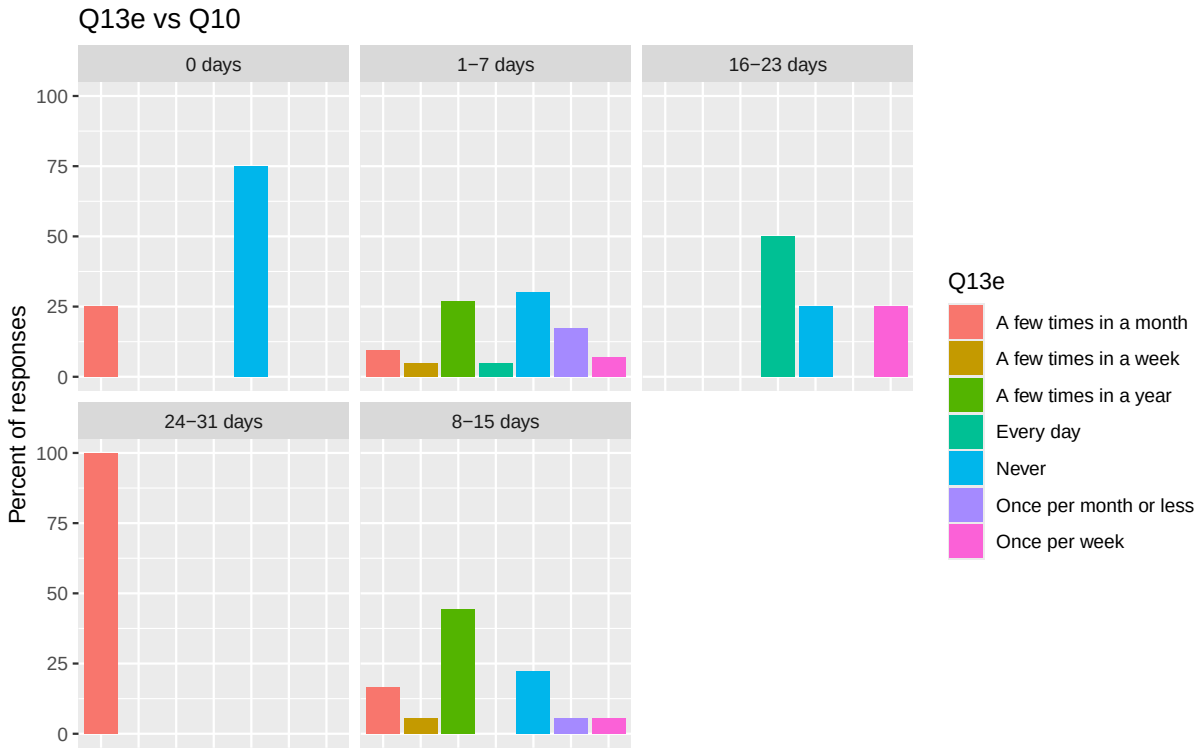


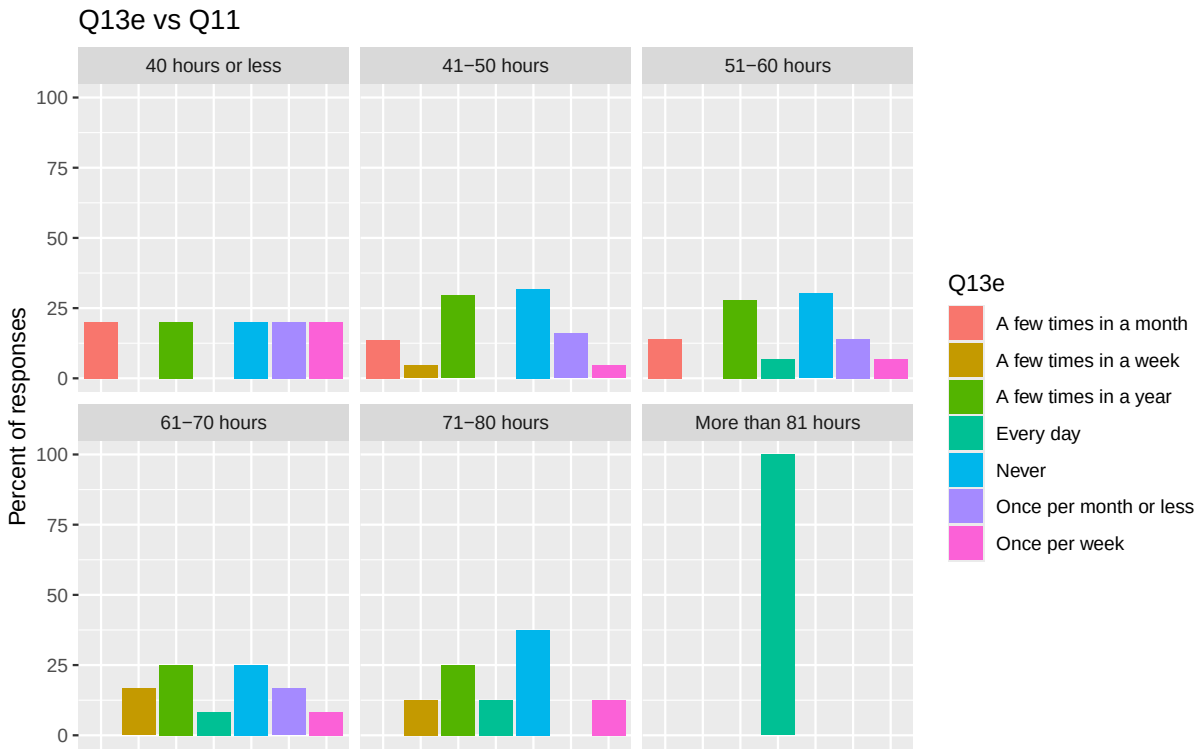
Q13d vs Q5

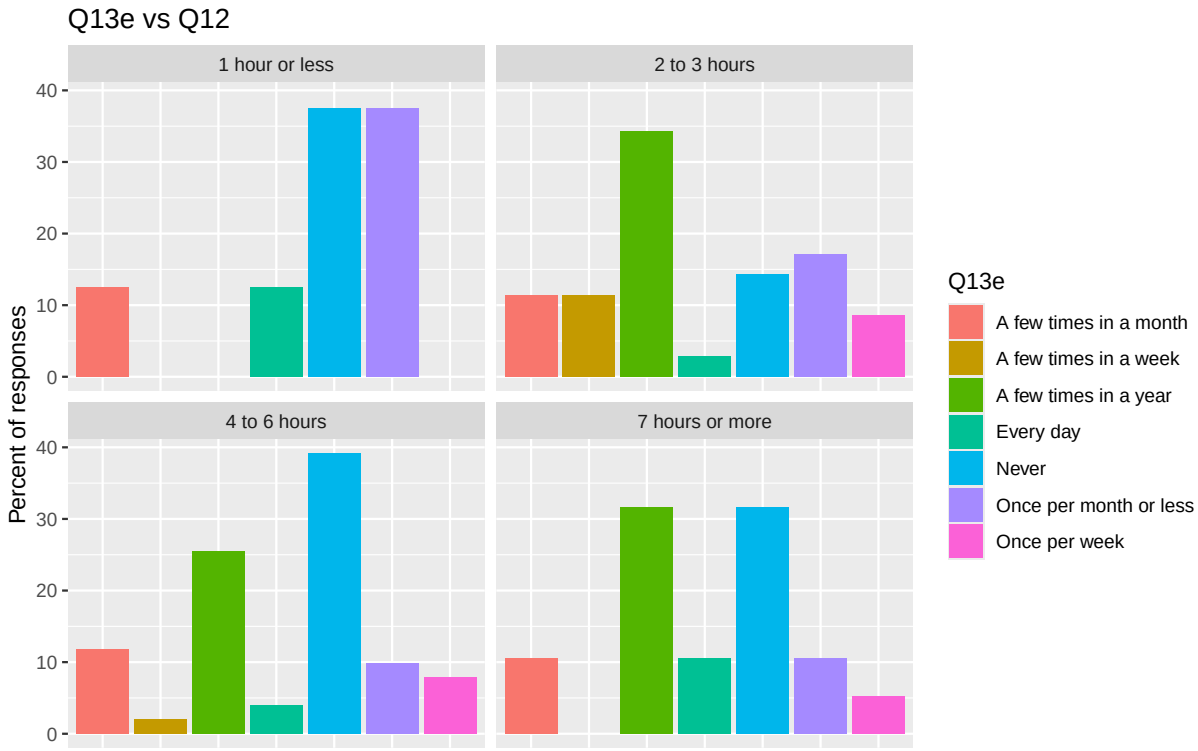


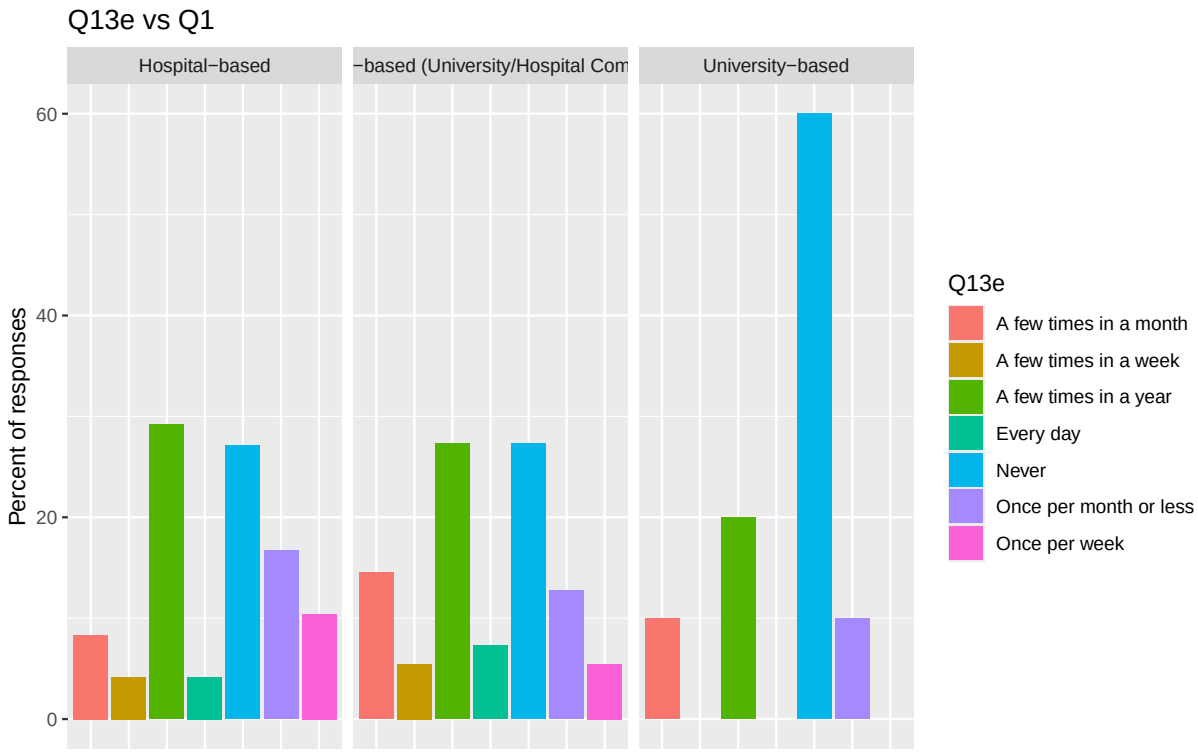
Q13d vs Q2



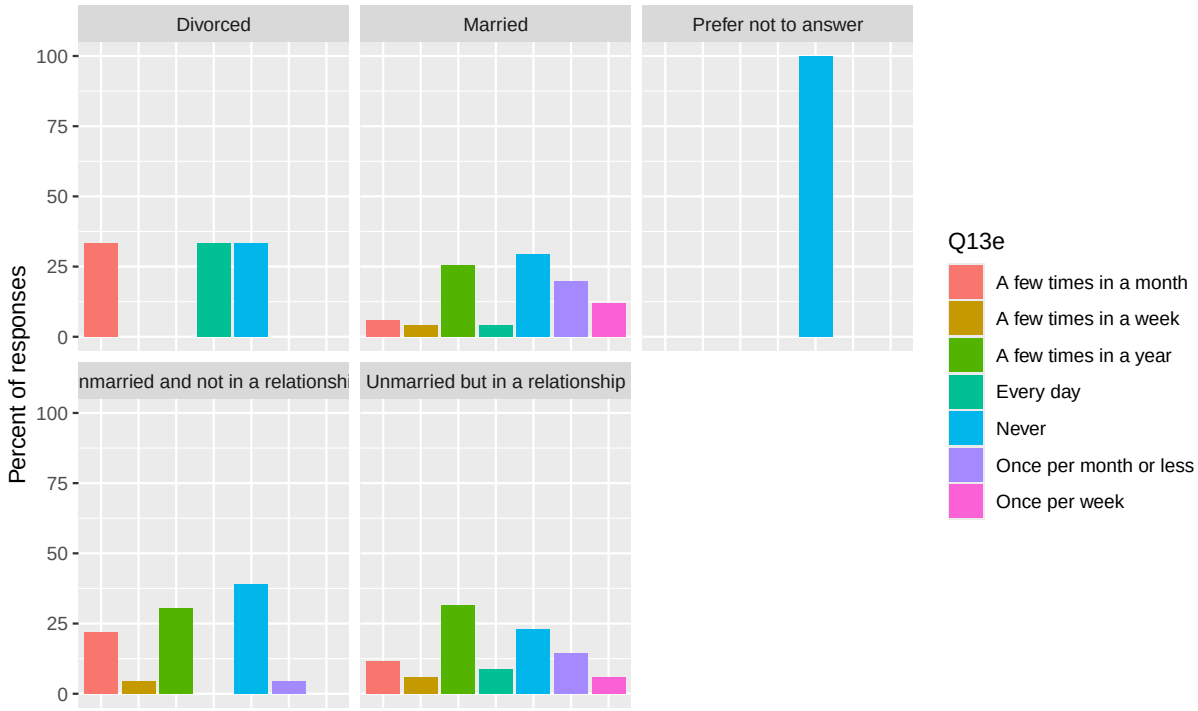

Q13e vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week


Q13e vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

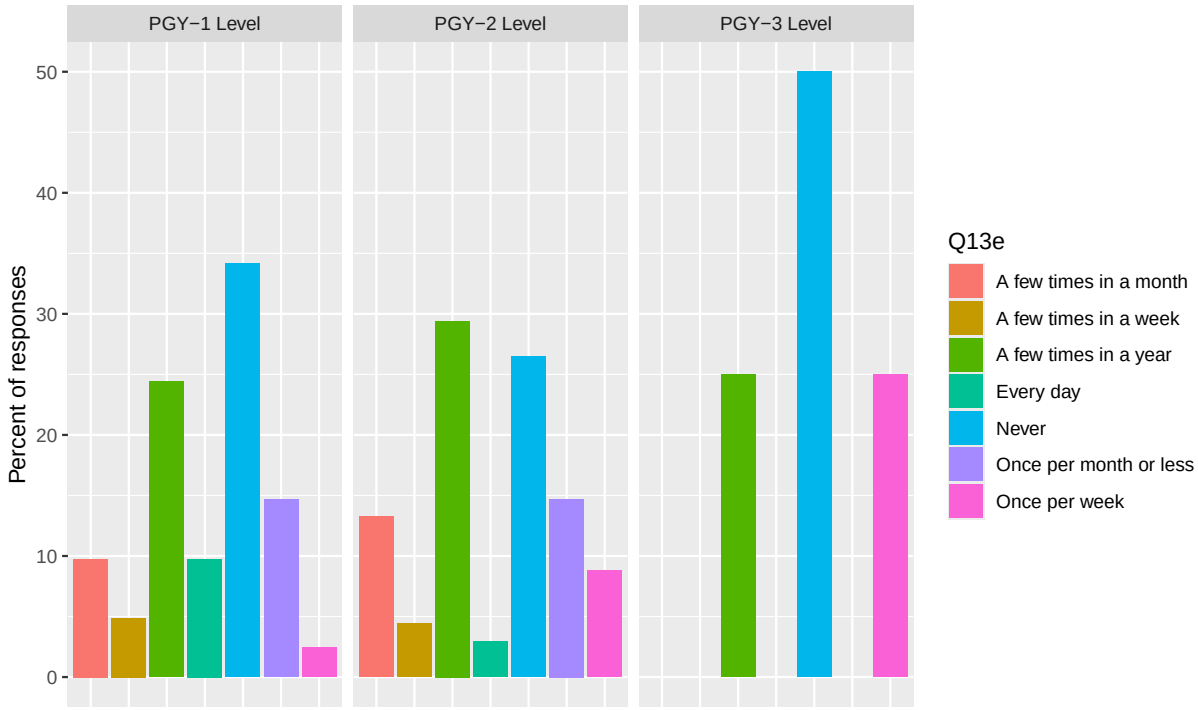

Q13e vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

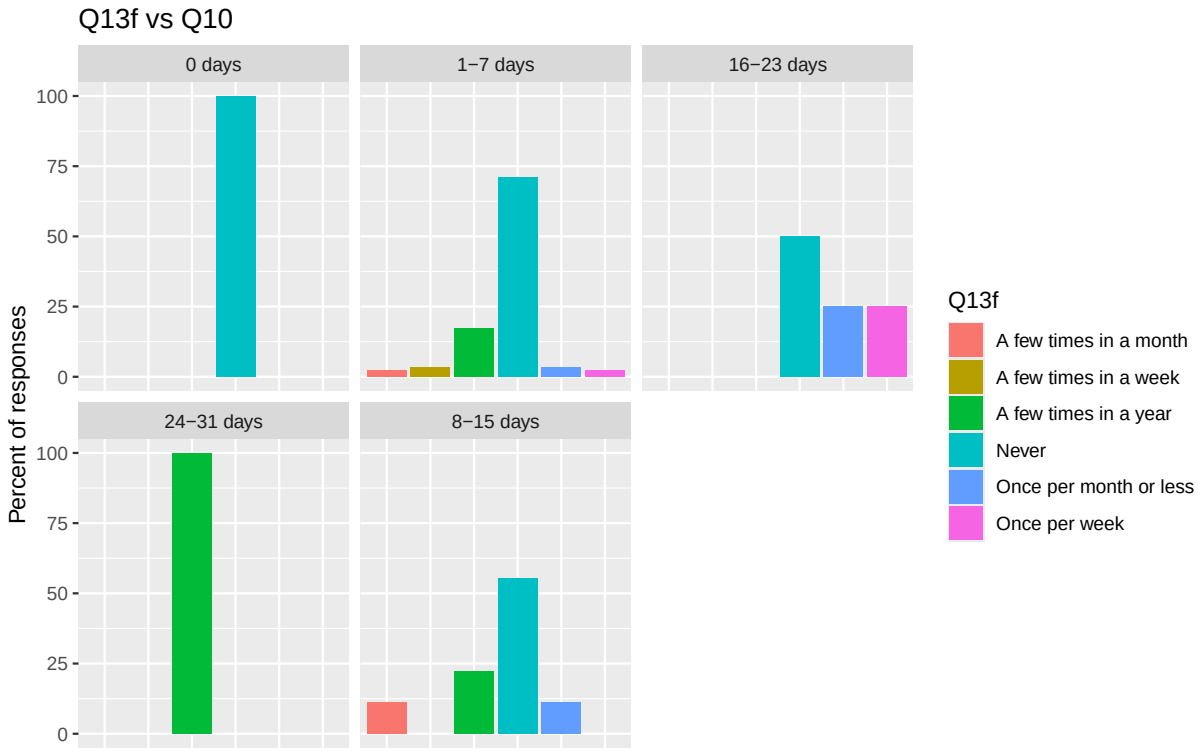

Q13e vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
20
40
60
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

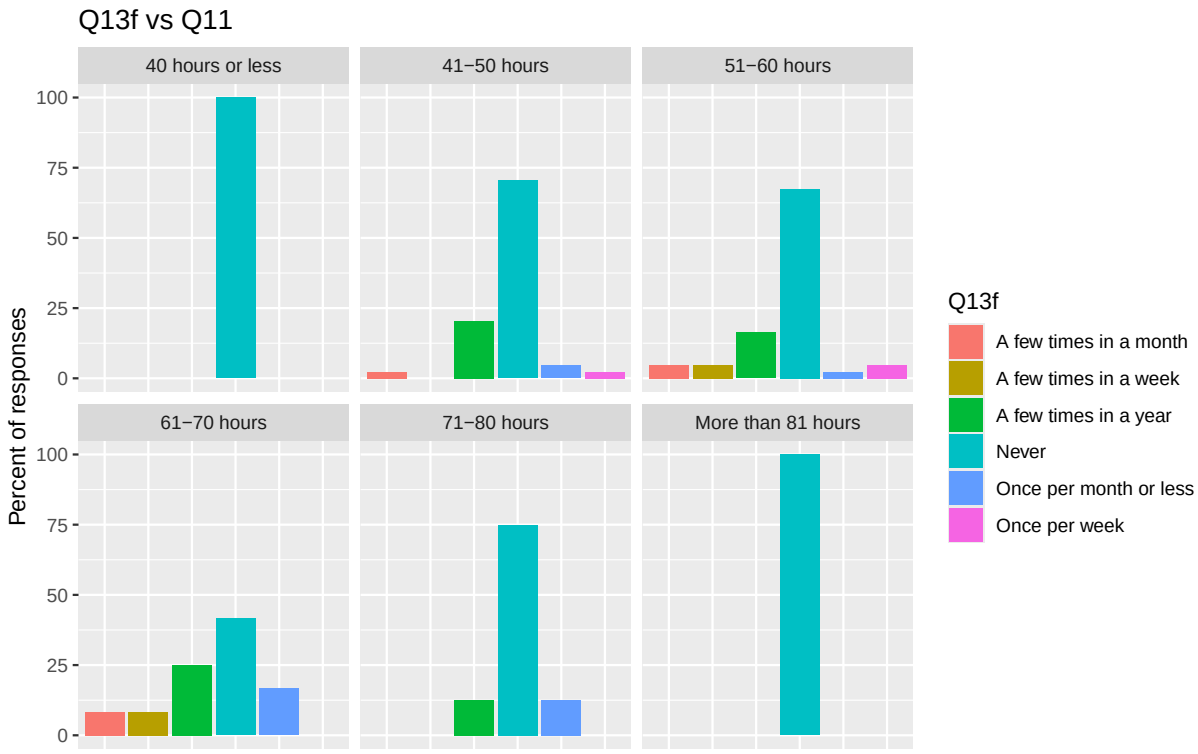
Q13e vs Q5

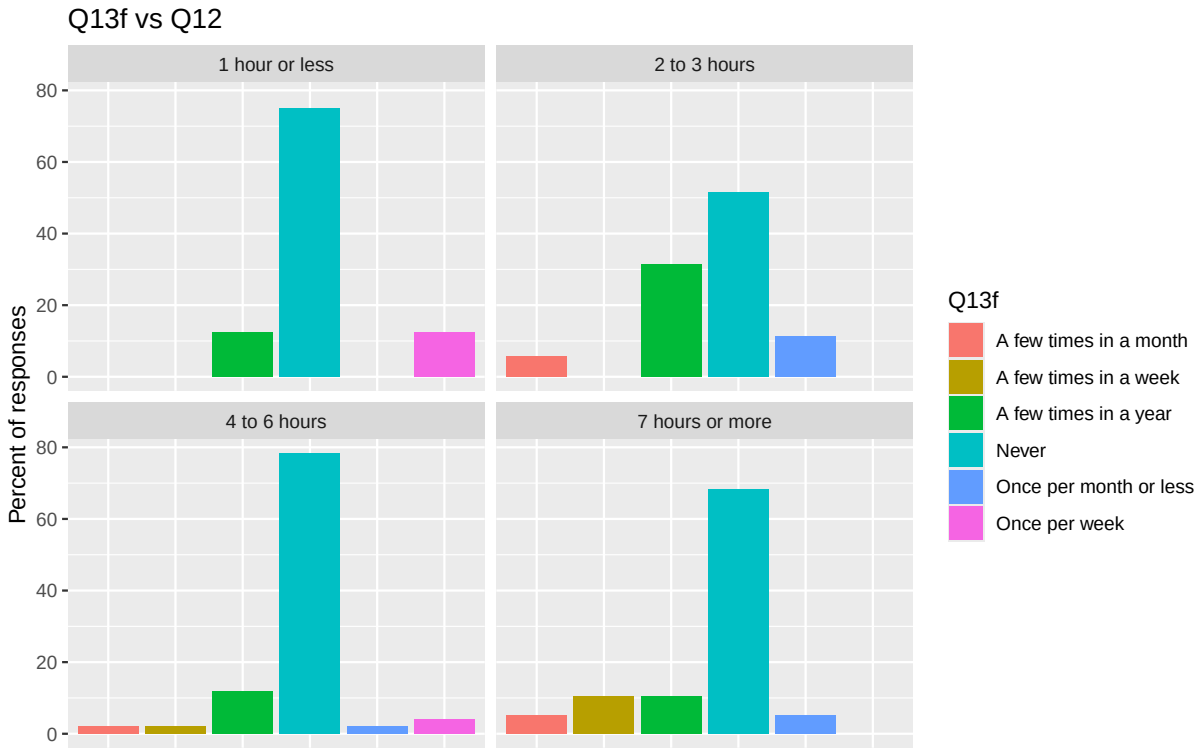


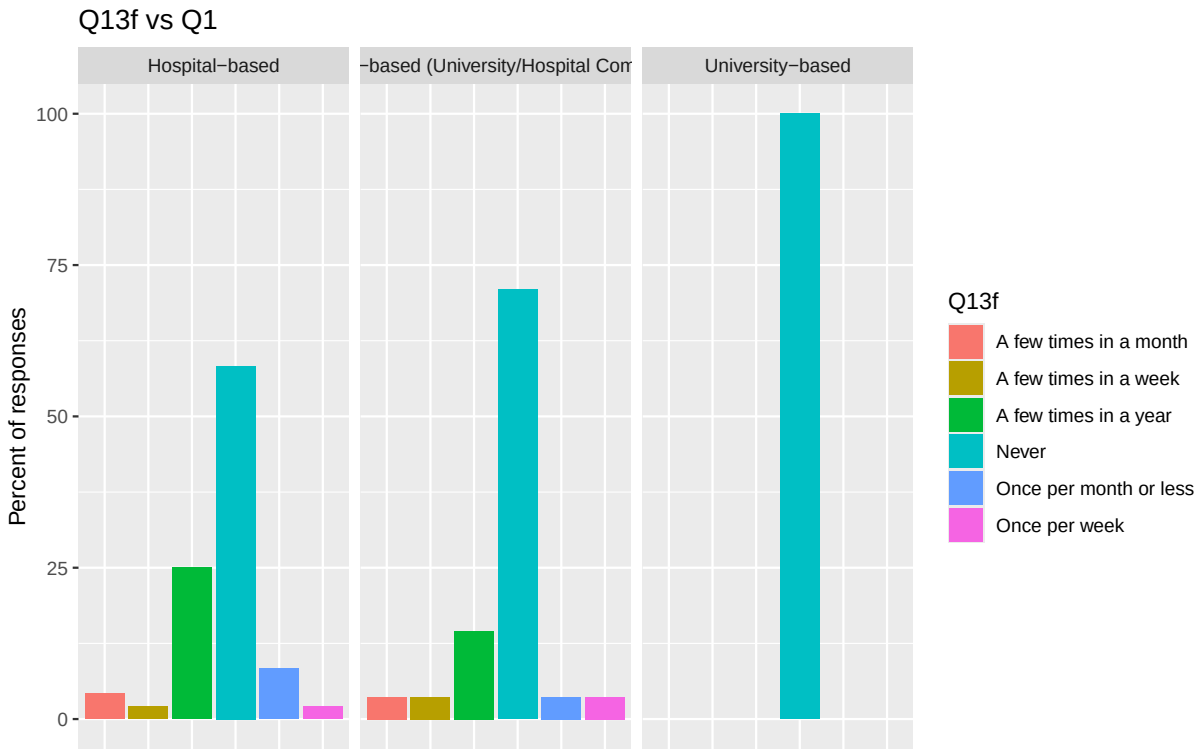
Q13e vs Q2



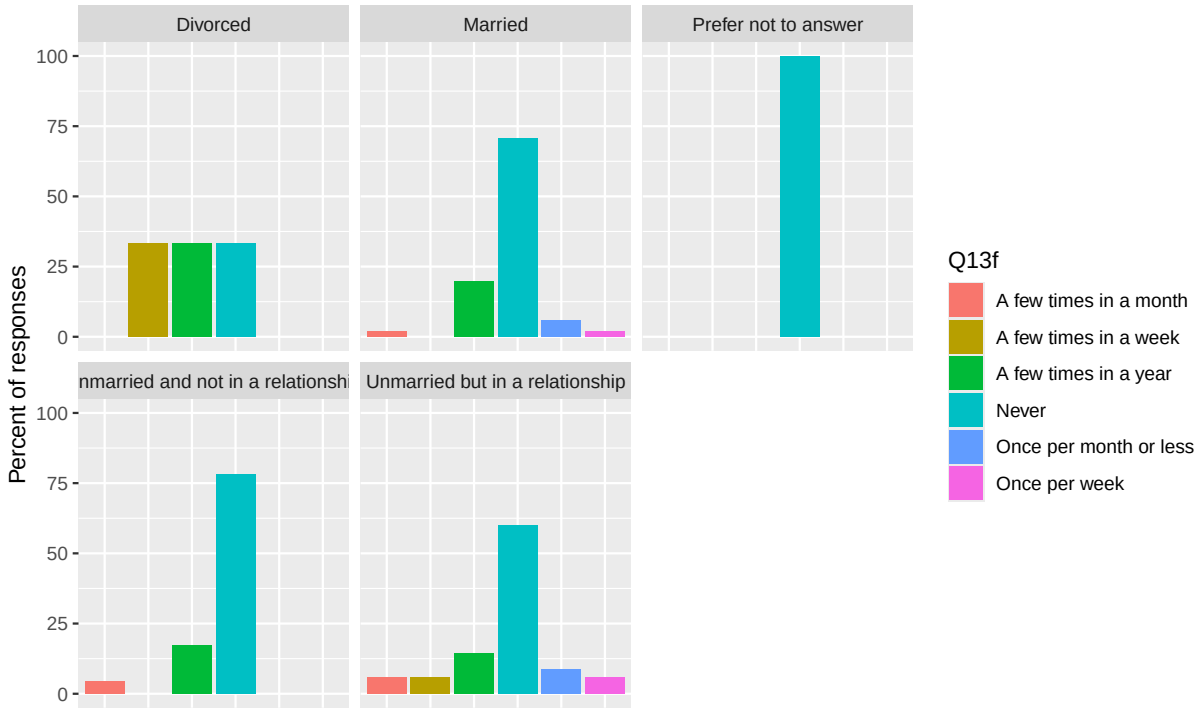

Q13f vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week


Q13f vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week

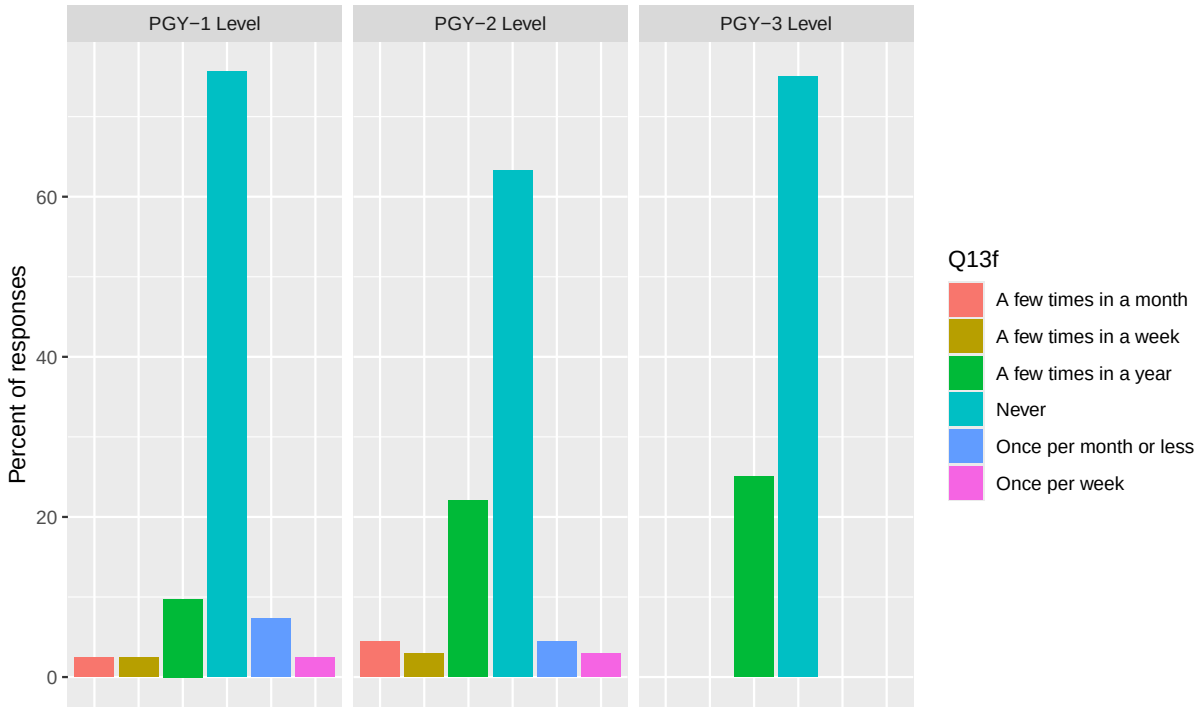

Q13f vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
20
40
60
80
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week

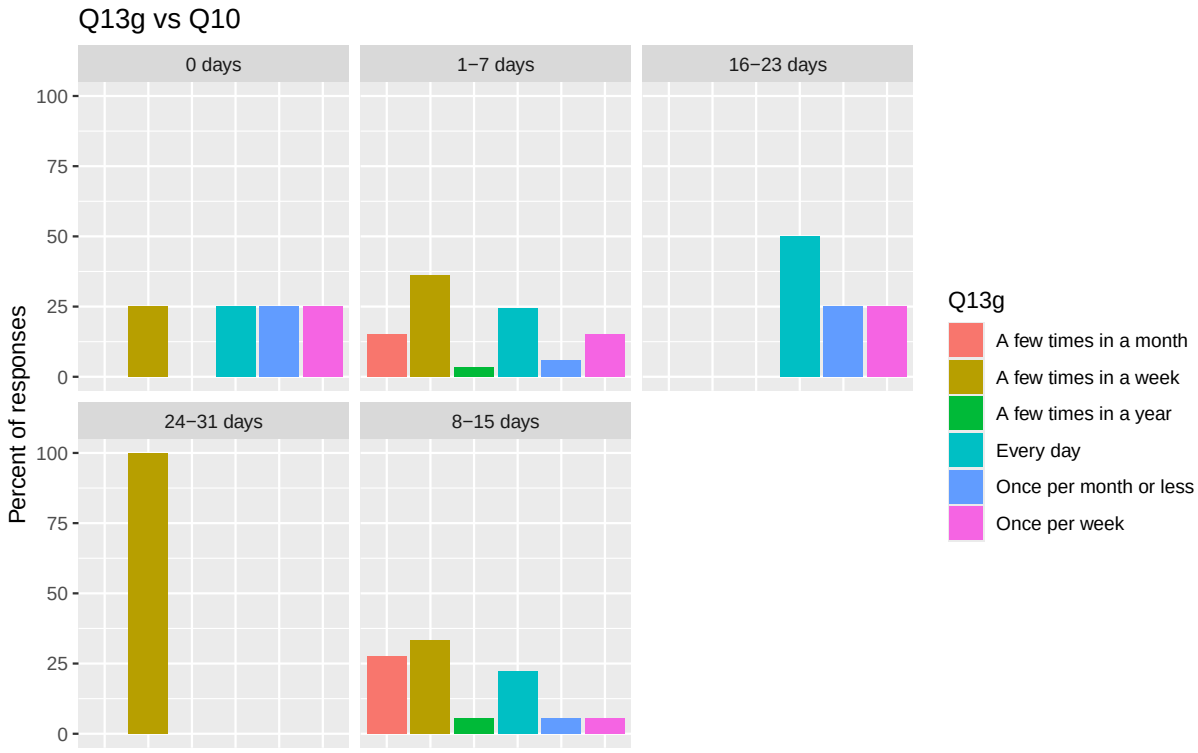

Q13f vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
25
50
75
100
Q13f
A few times in a month
A few times in a week
A few times in a year
Never
Once per month or less
Once per week

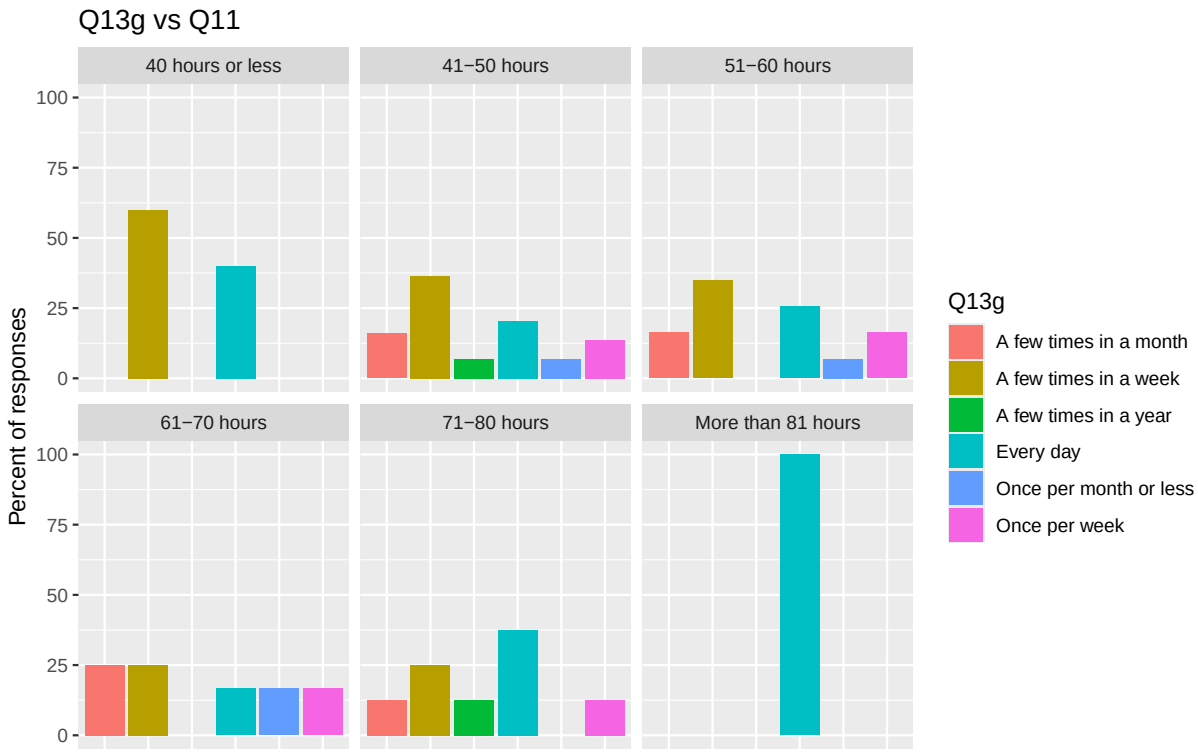
Q13f vs Q5

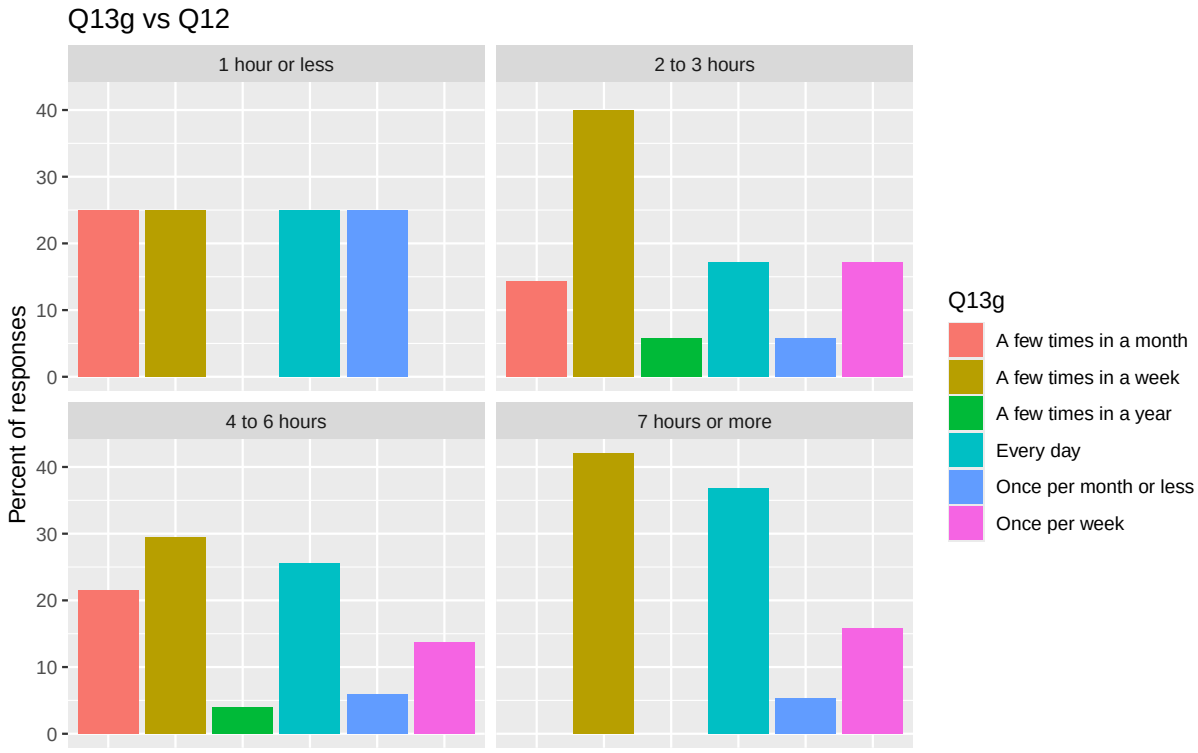


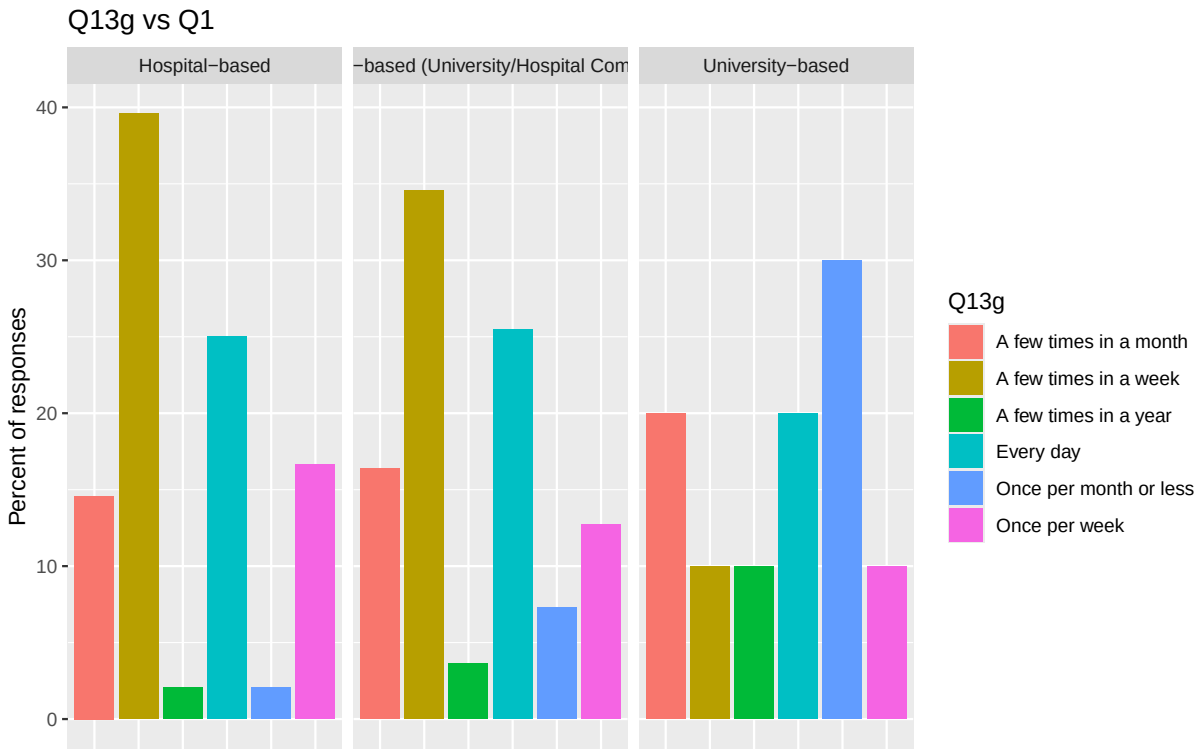
Q13f vs Q2

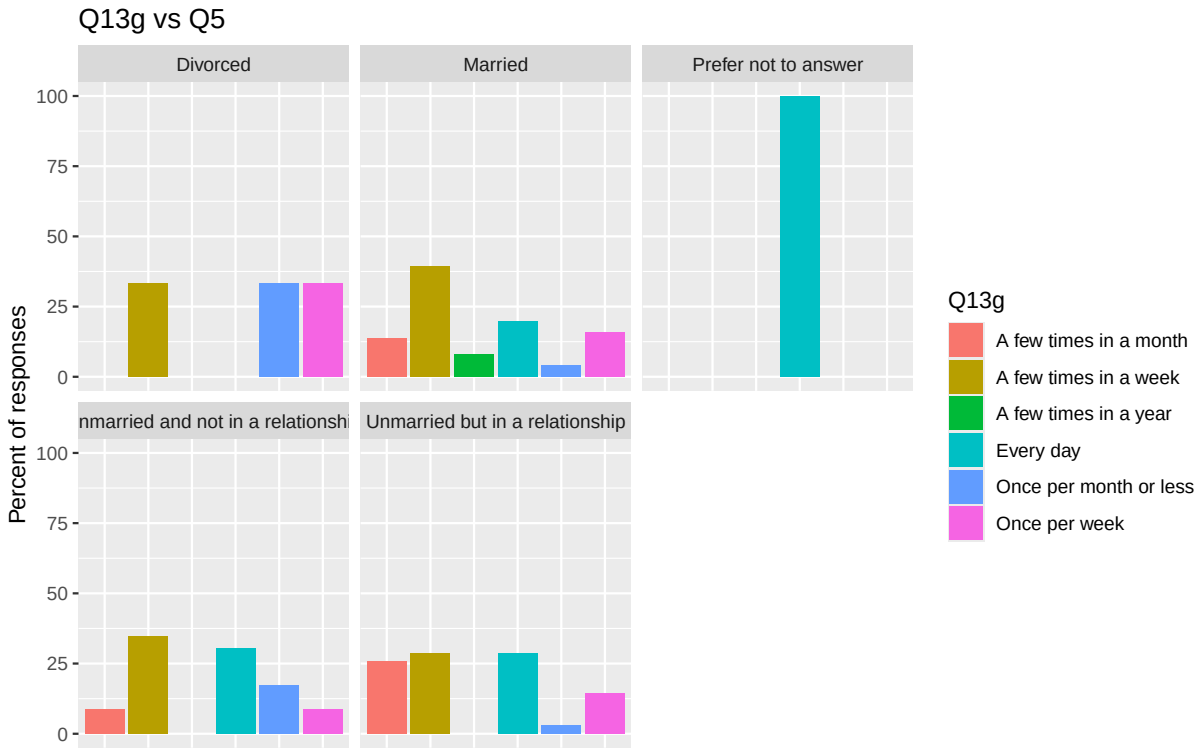


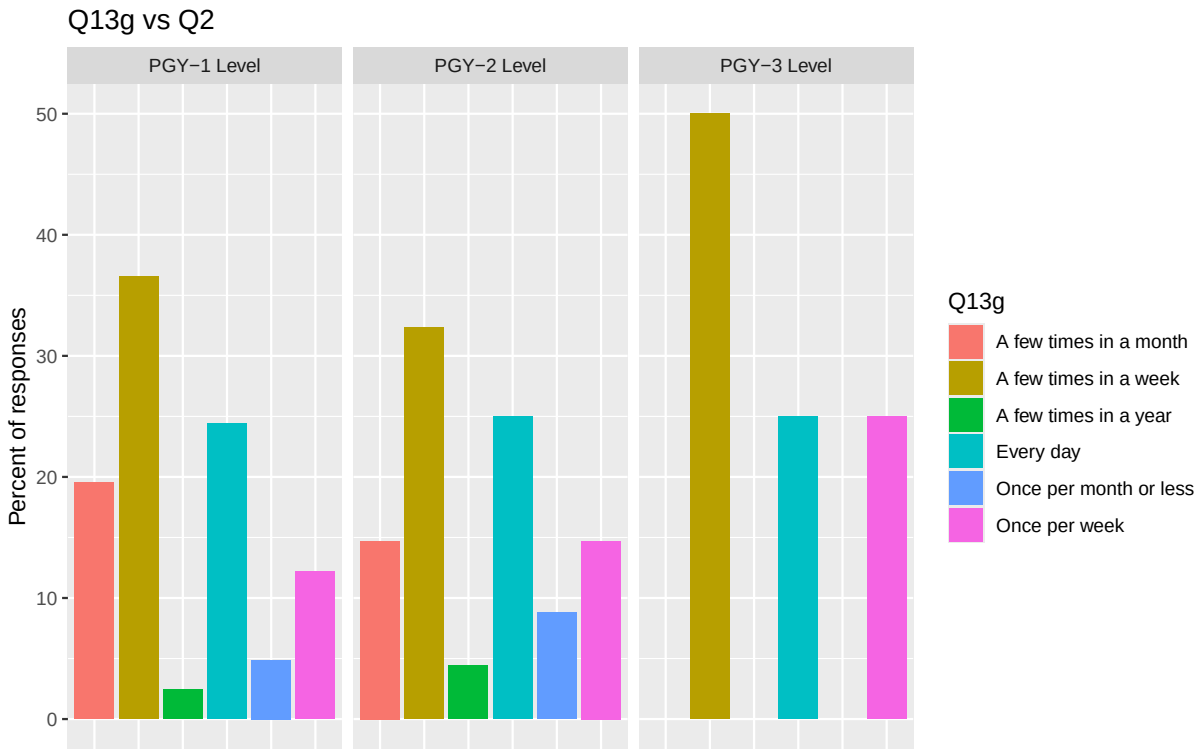

Q13g vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

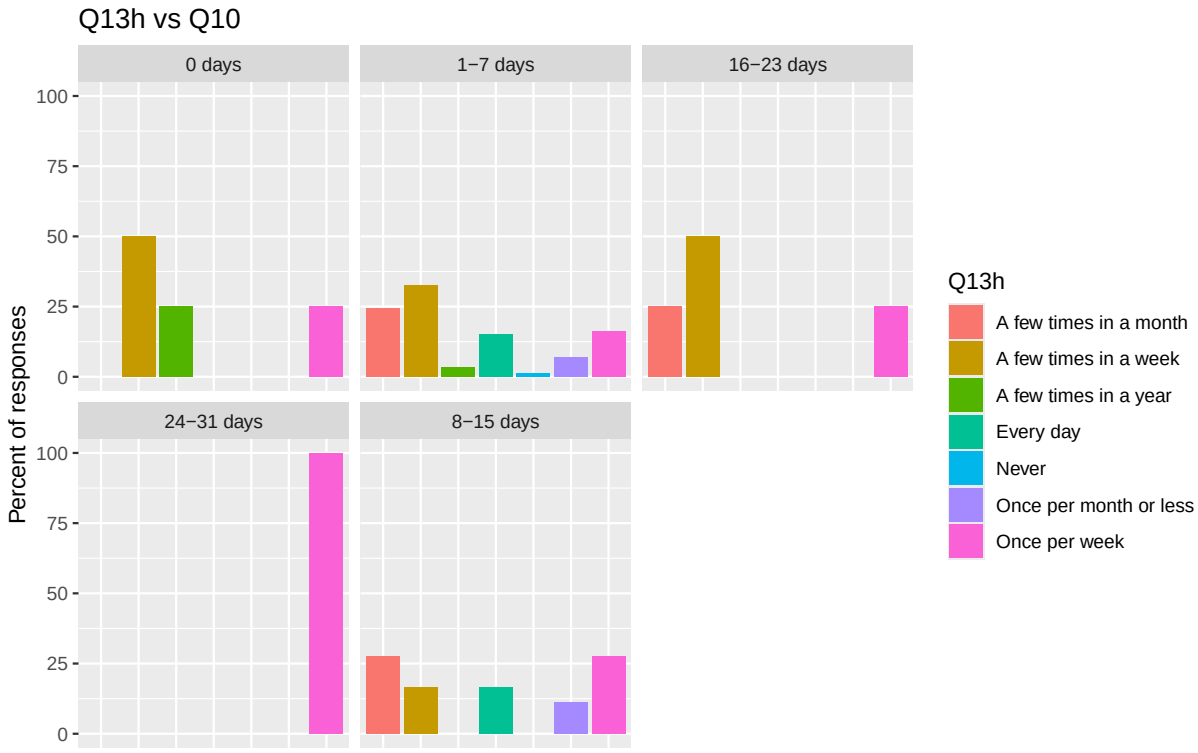

Q13g vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

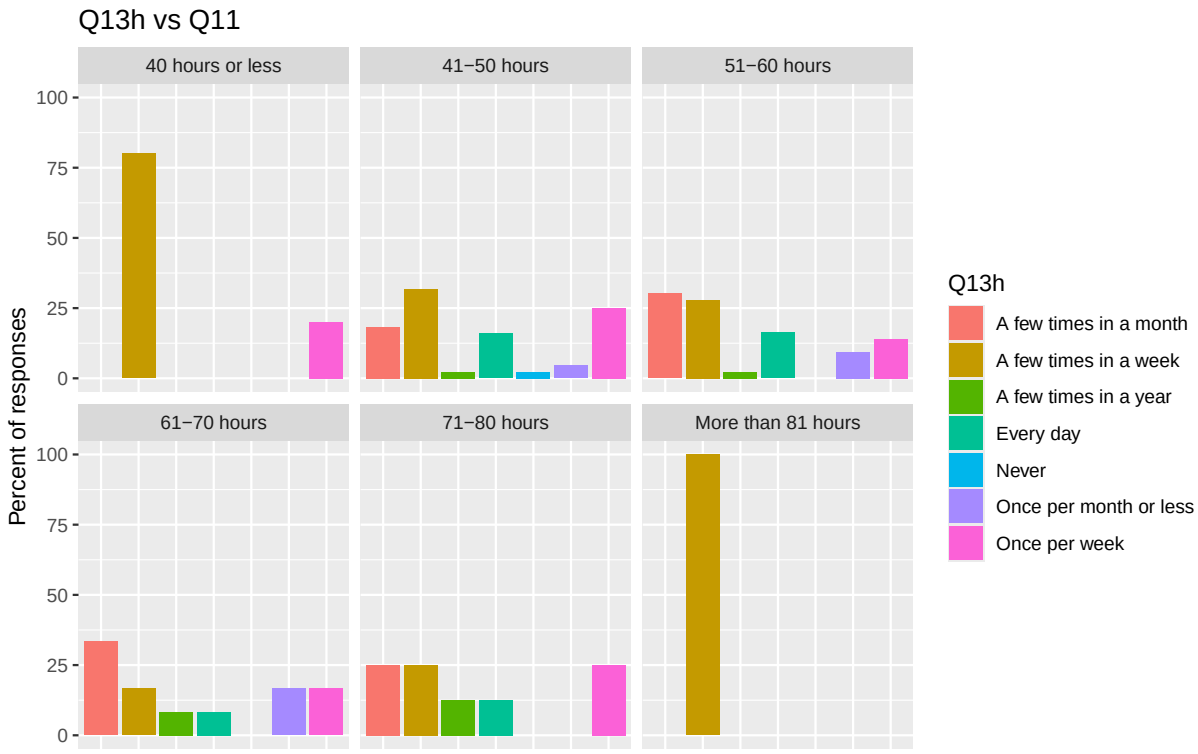

Q13g vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

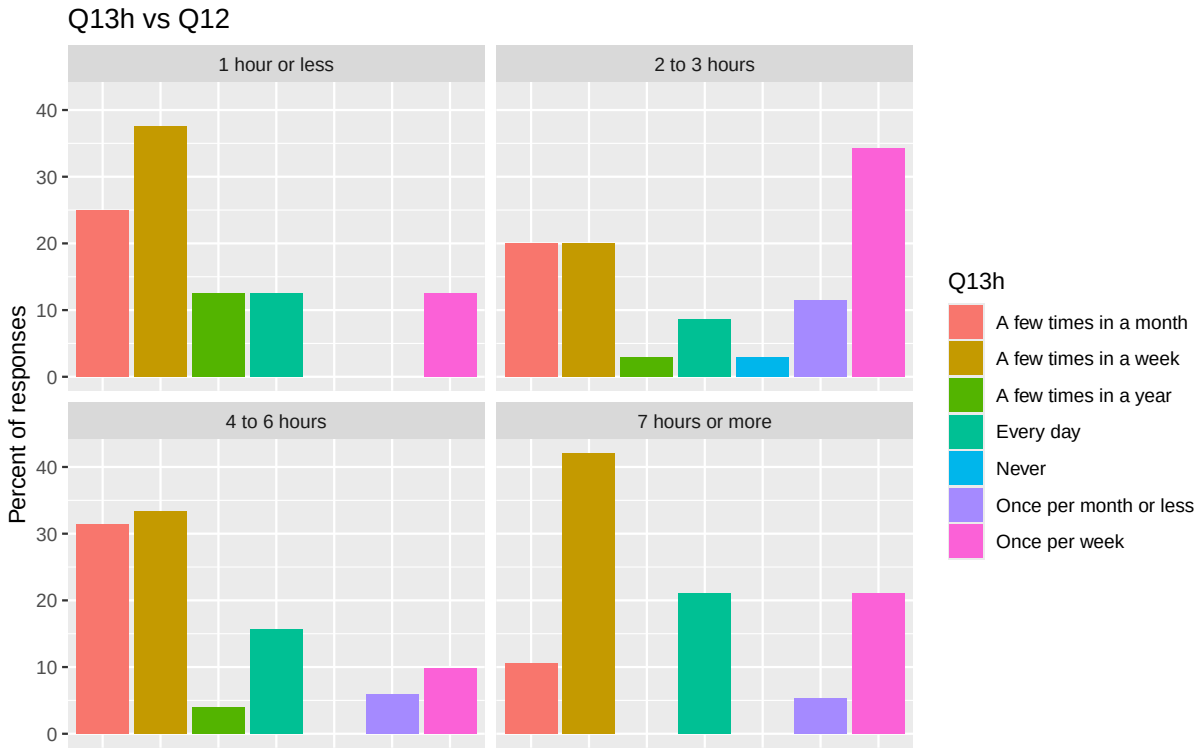

Q13g vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
40
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

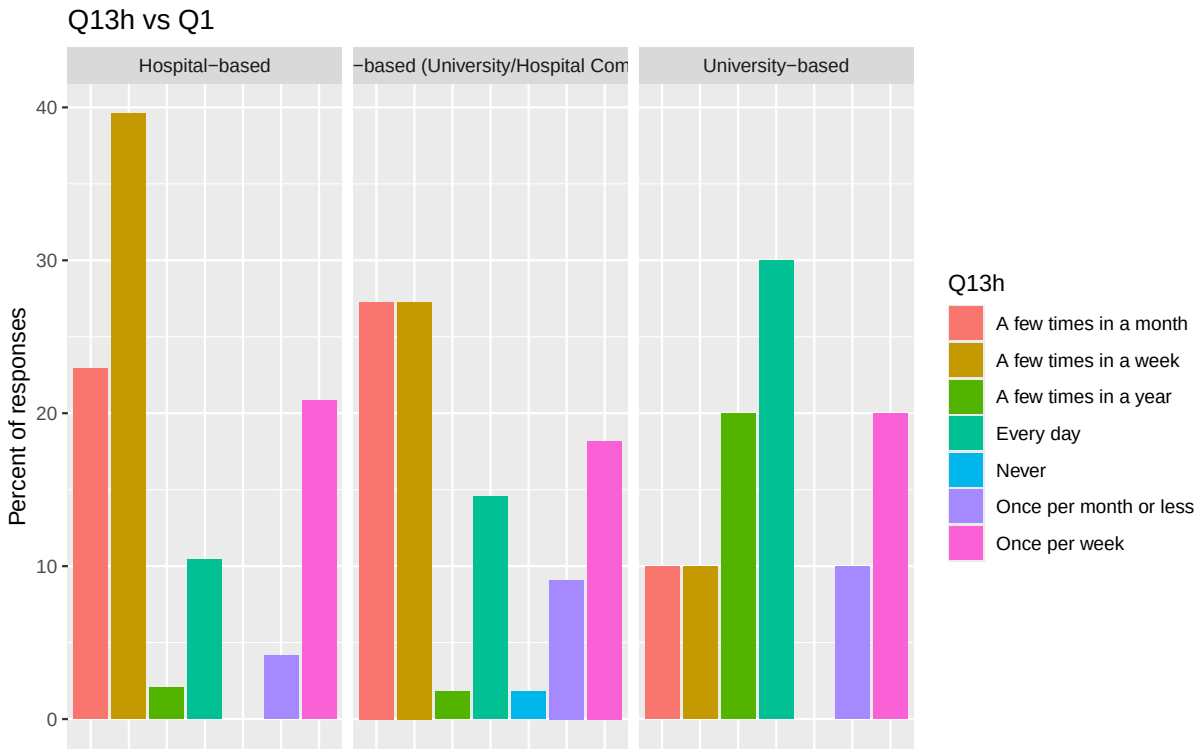

Q13g vs Q5
Divorced
Married
Prefer not to answer
nmarried and not in a relationshi
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

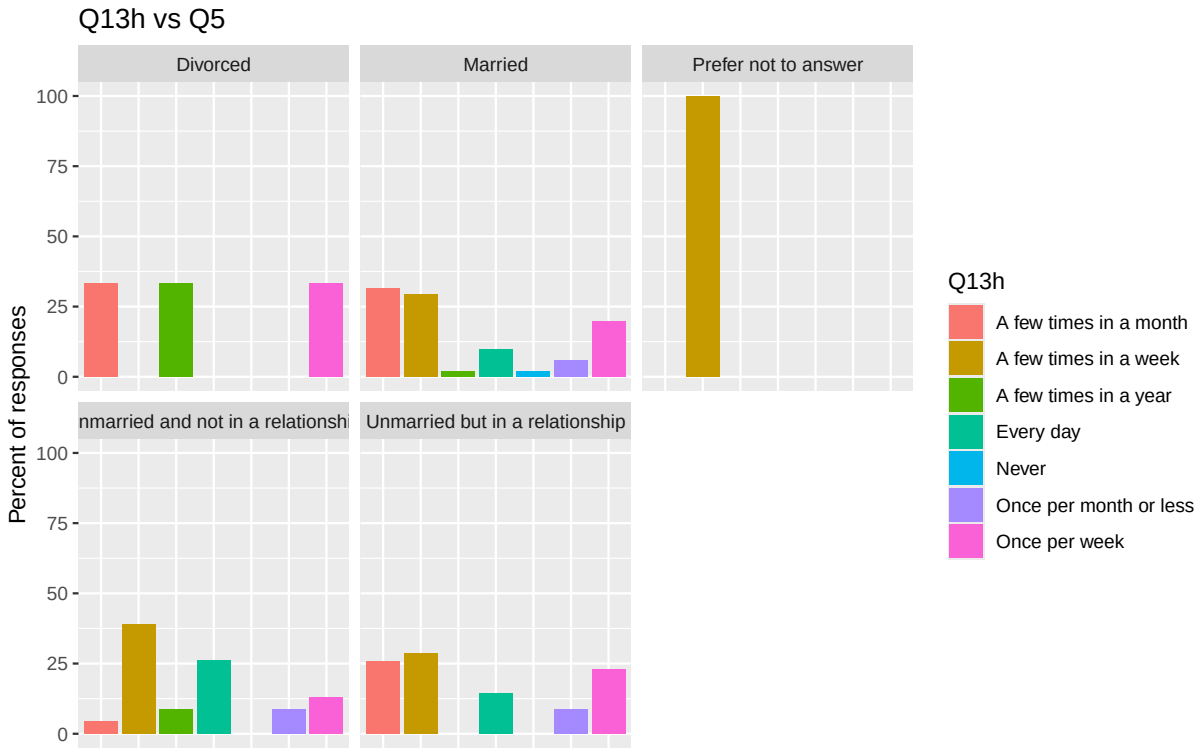

Q13g vs Q2
PGY−1 Level
PGY−2 Level
PGY−3 Level
Percent of responses
50
40
30
20
10
0
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

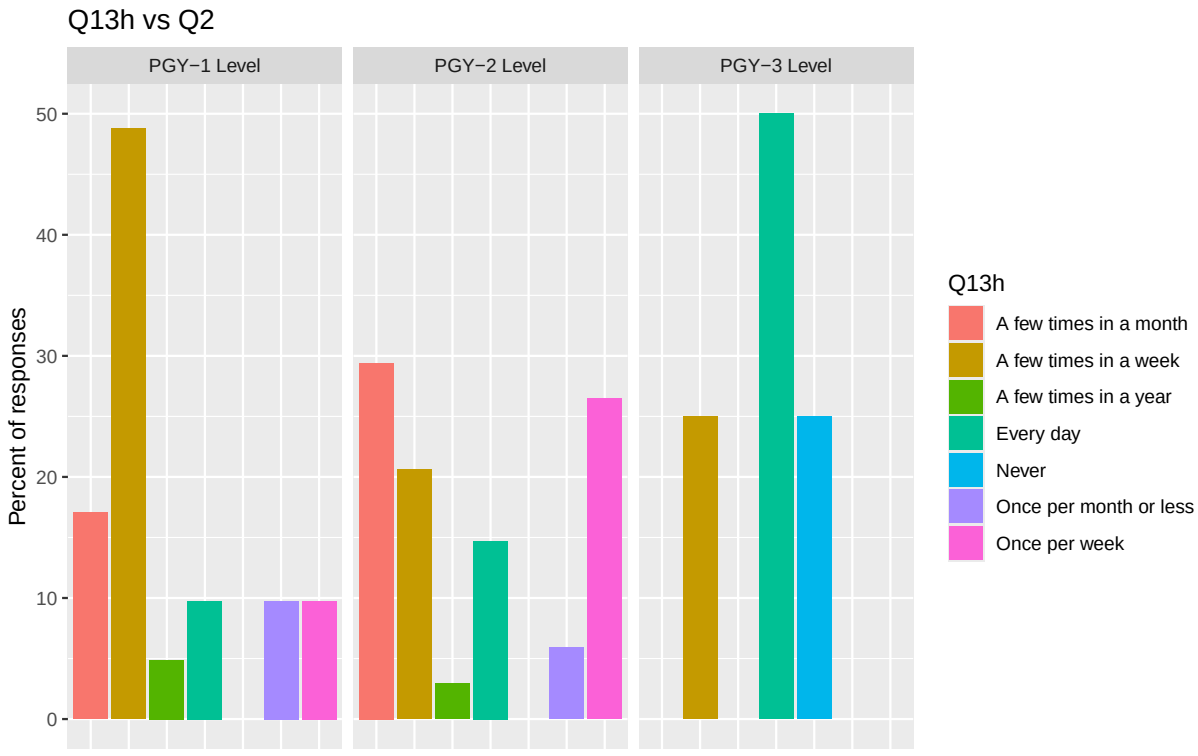

Q13h vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

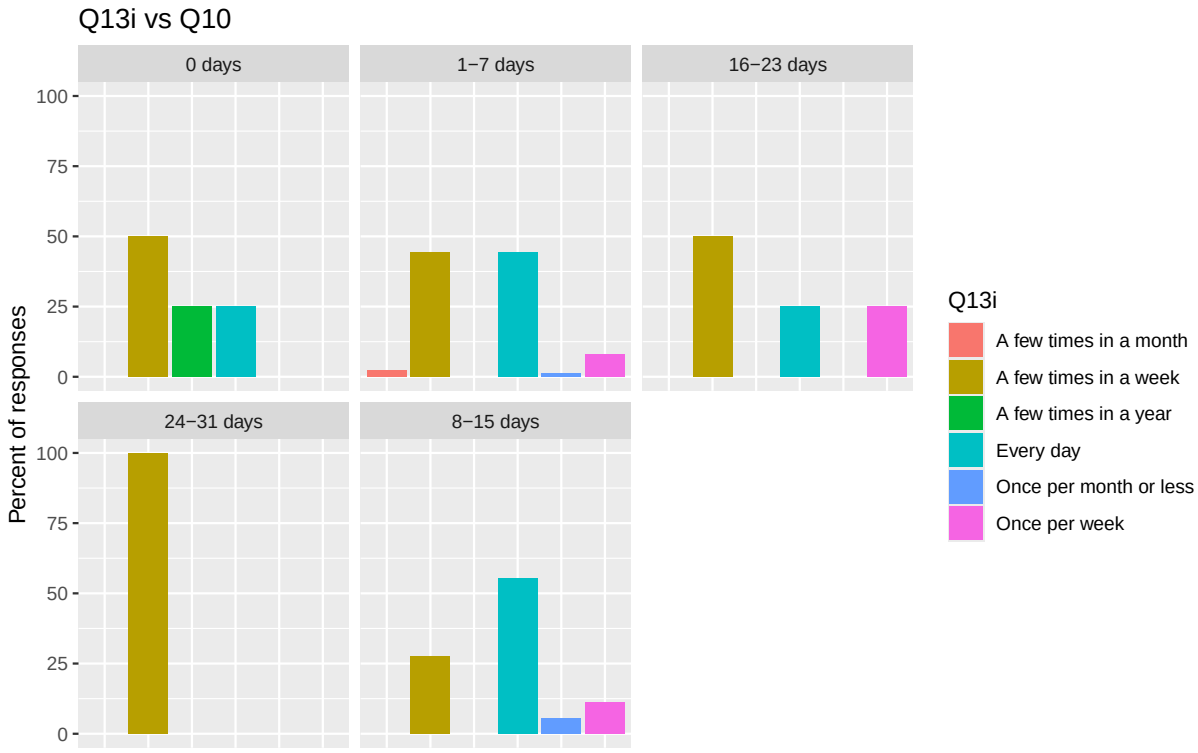

Q13h vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

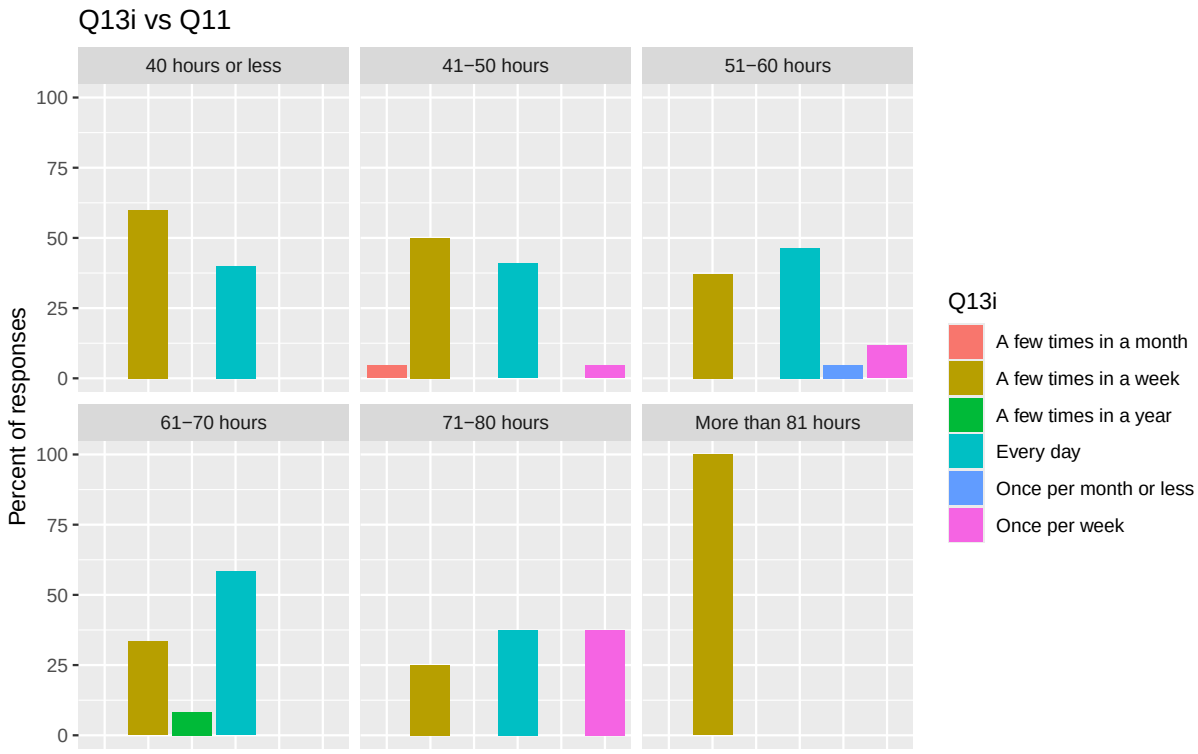

Q13h vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
10
20
30
40
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

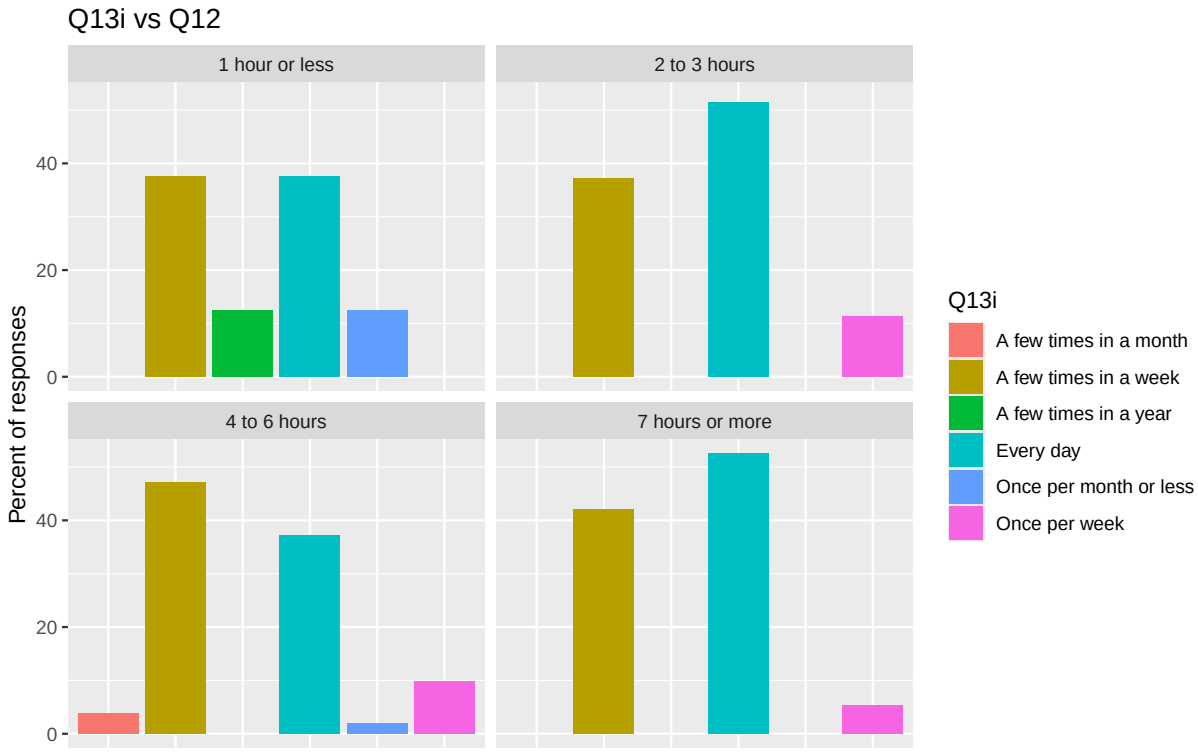

Q13h vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
40
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

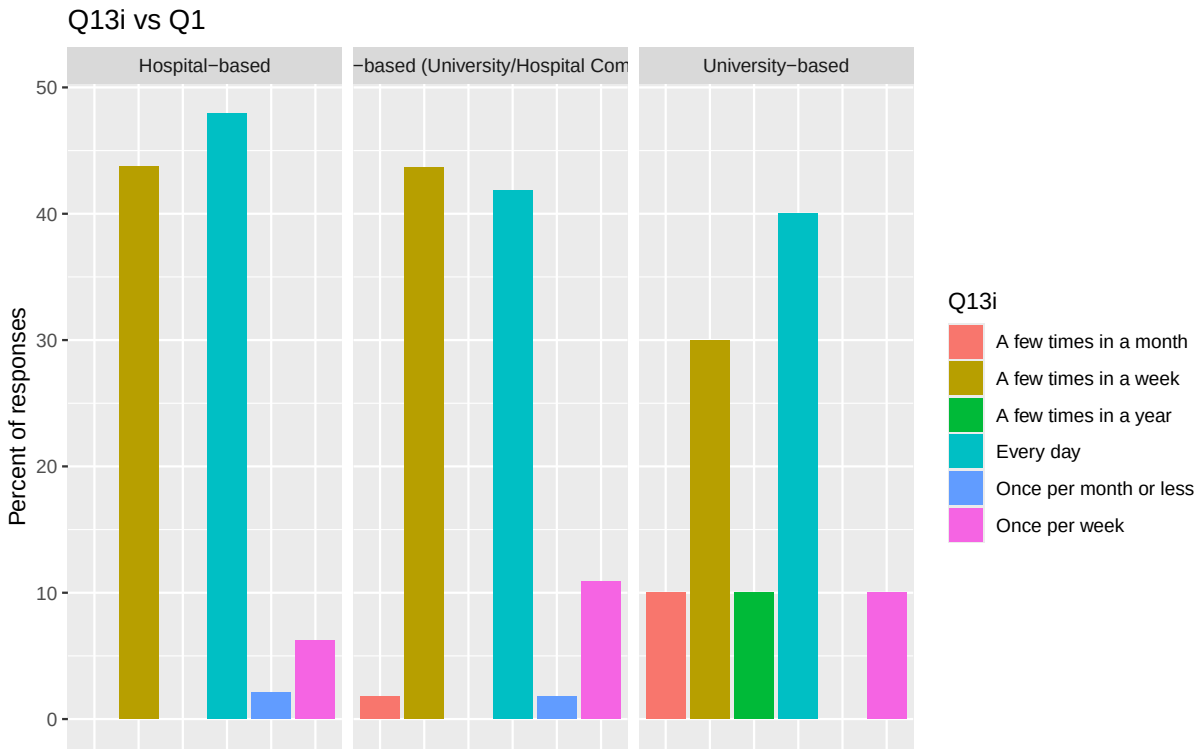

Q13h vs Q5
Divorced
Married
Prefer not to answer
nmarried and not in a relationshi
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

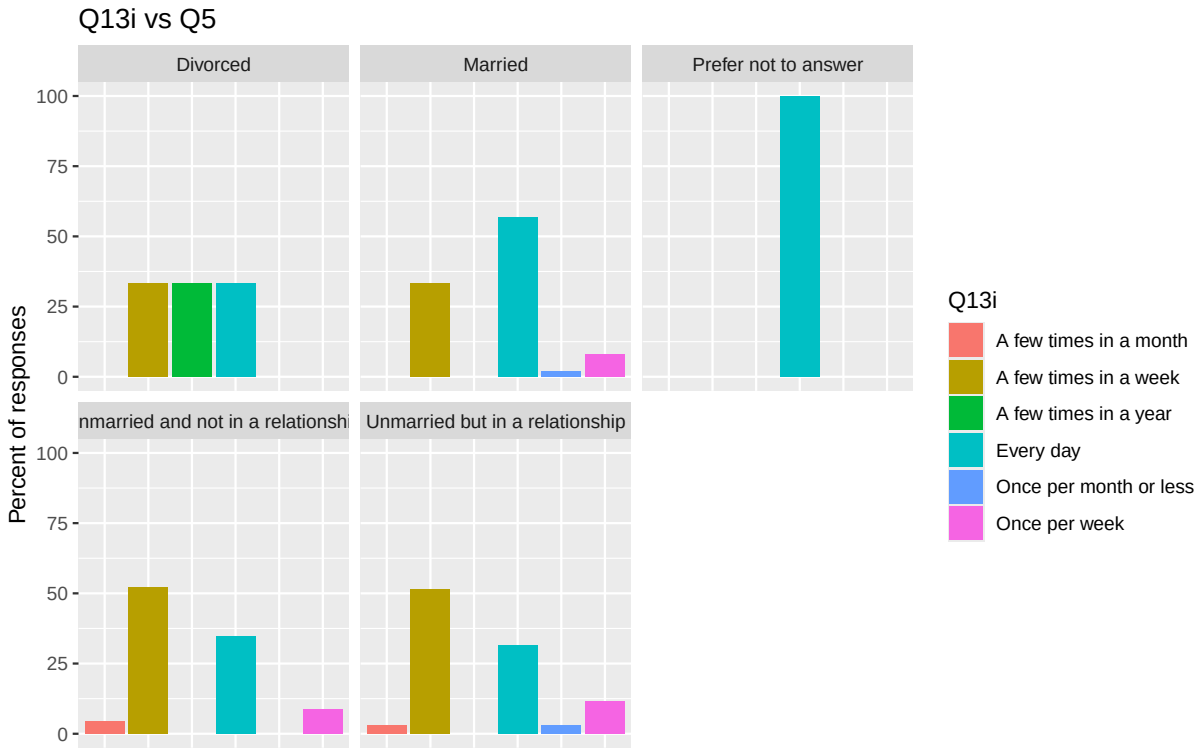

Q13h vs Q2
PGY−1 Level
PGY−2 Level
PGY−3 Level
Percent of responses
50
40
30
20
10
0
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
Never
Once per month or less
Once per week

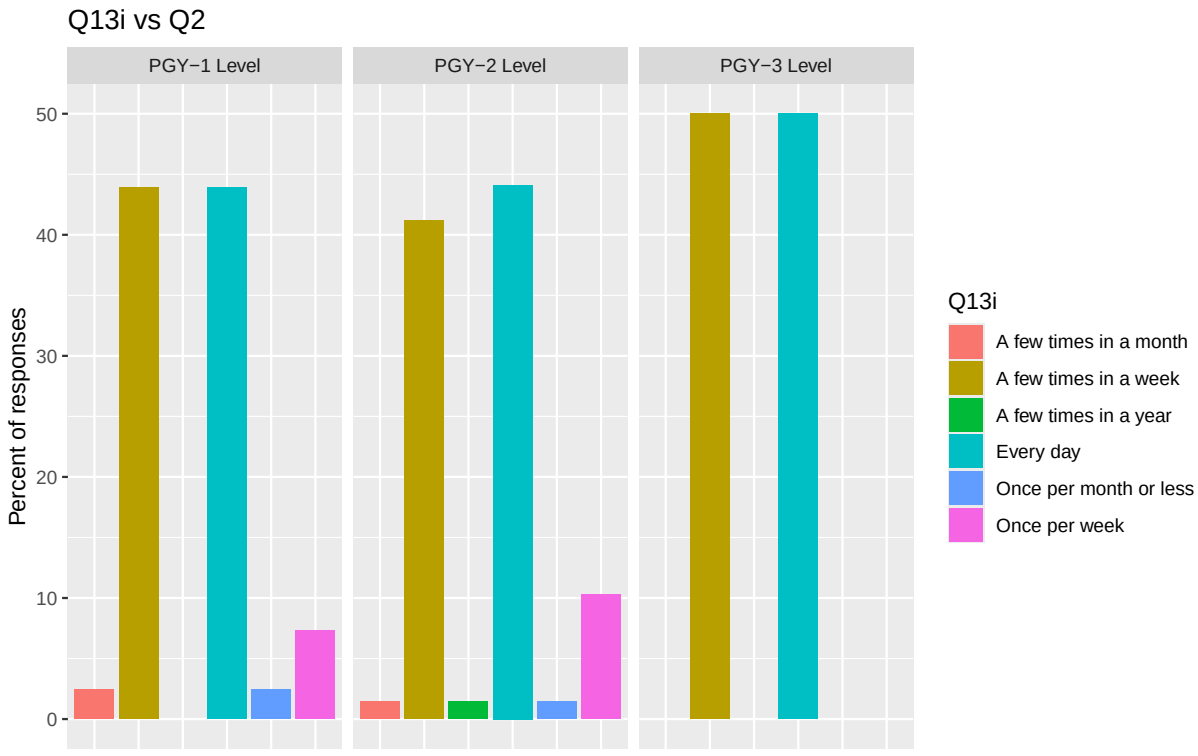

Q13i vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

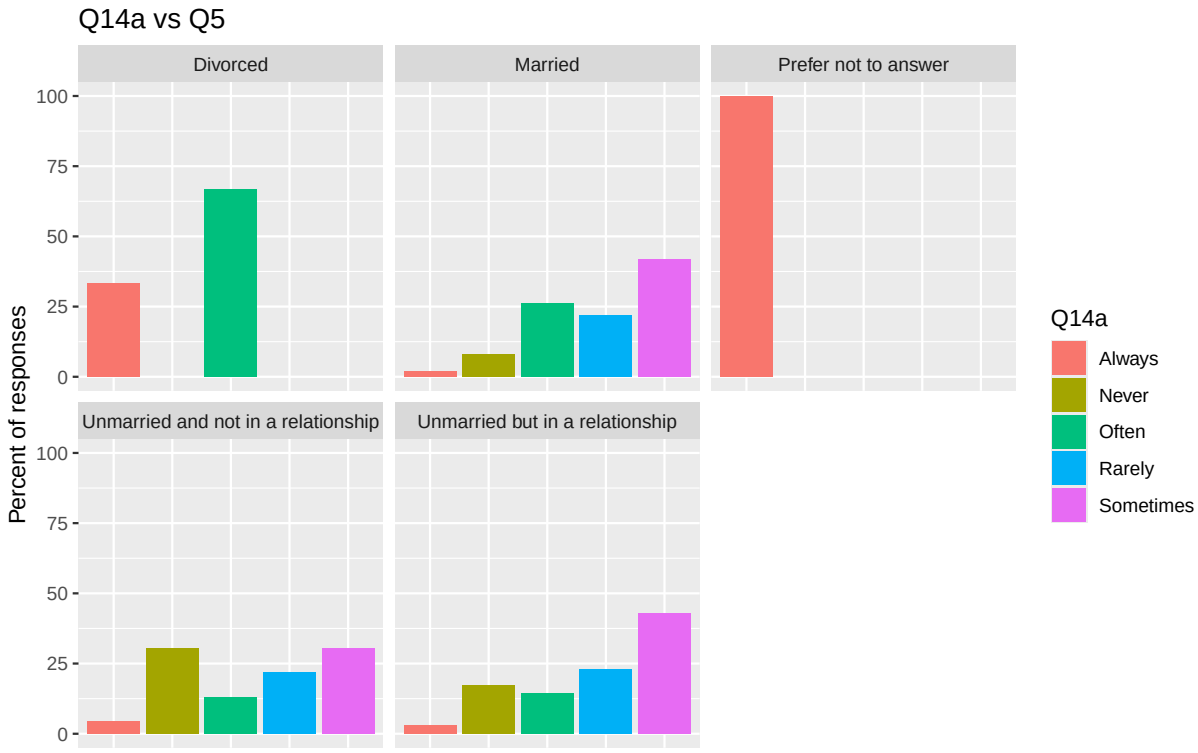

Q13i vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

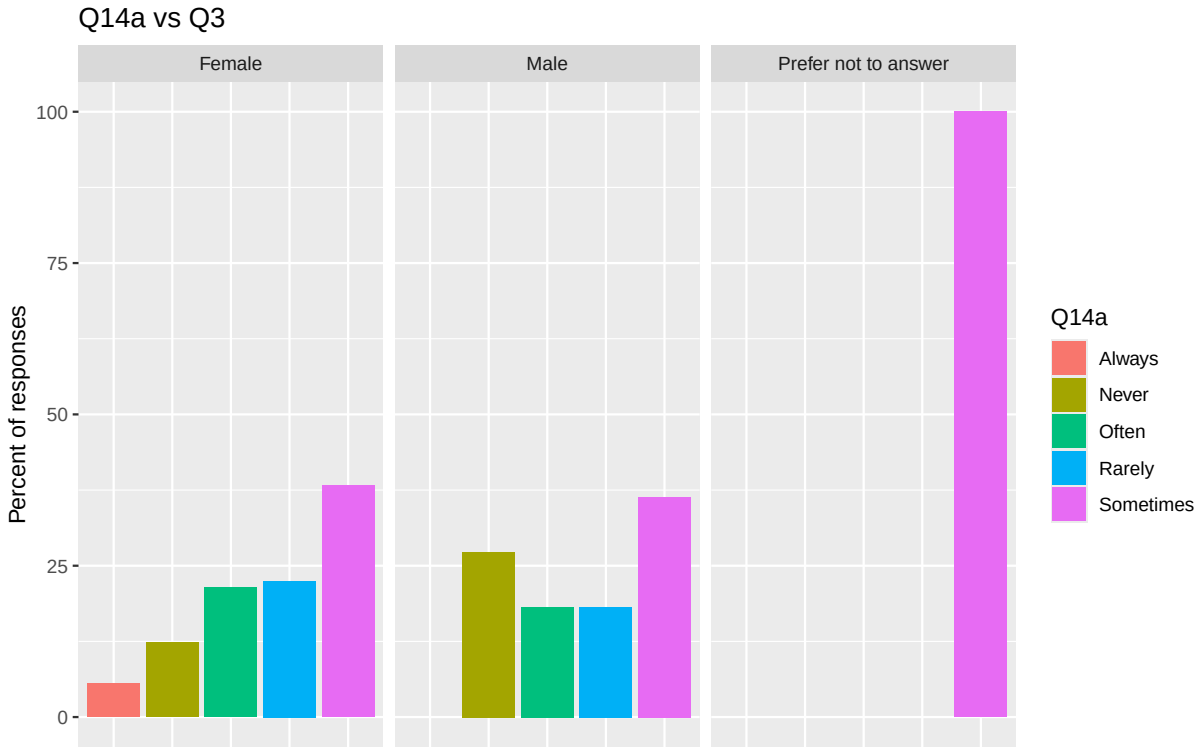

Q13i vs Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
Percent of responses
0
20
40
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

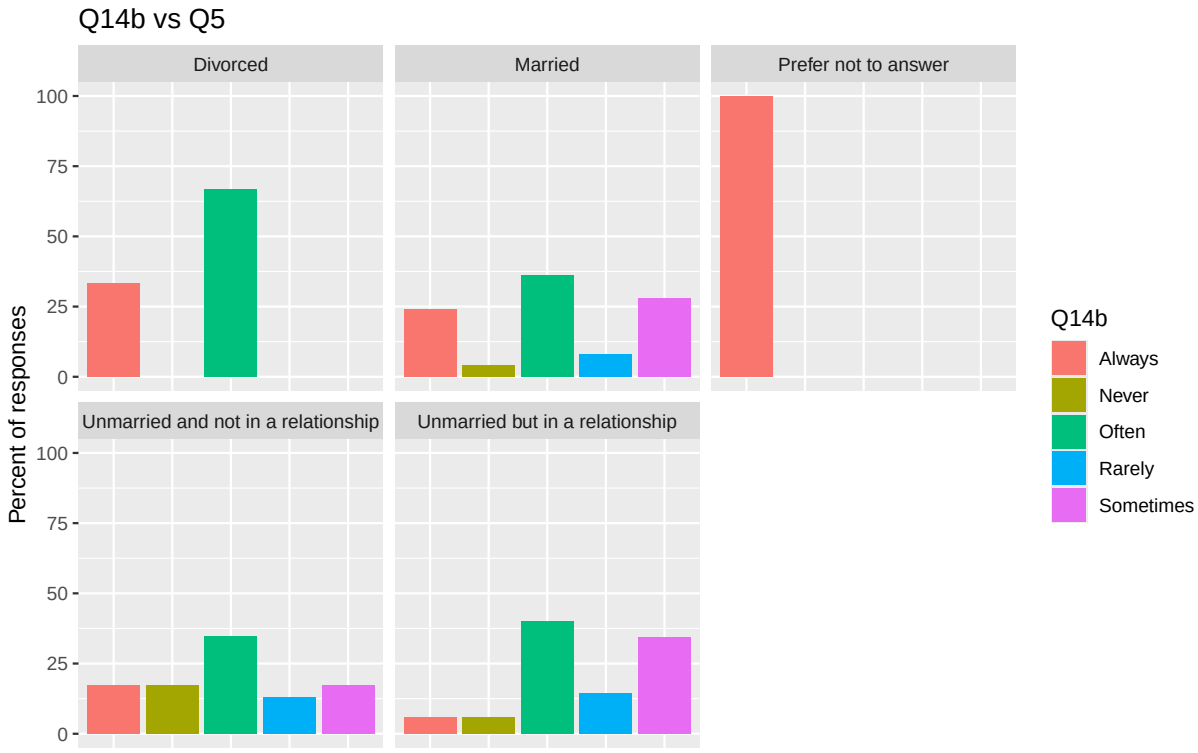

Q13i vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
10
20
30
40
50
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

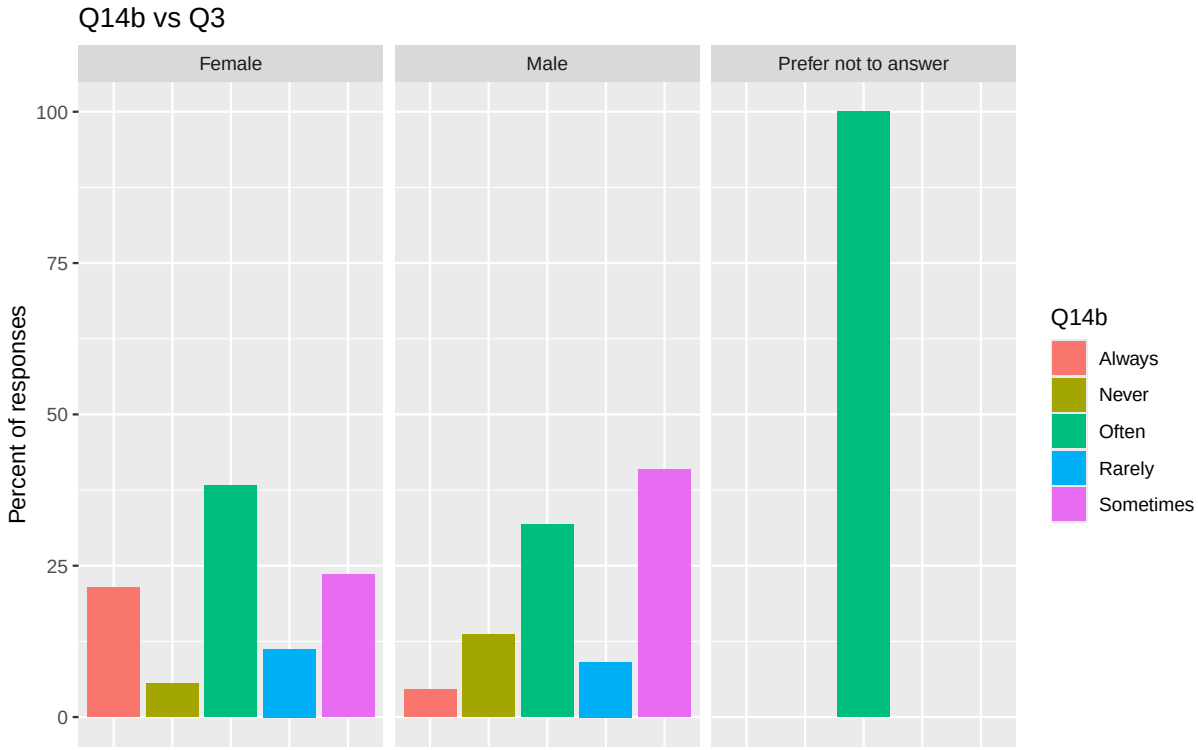

Q13i vs Q5
Divorced
Married
Prefer not to answer
nmarried and not in a relationshi
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

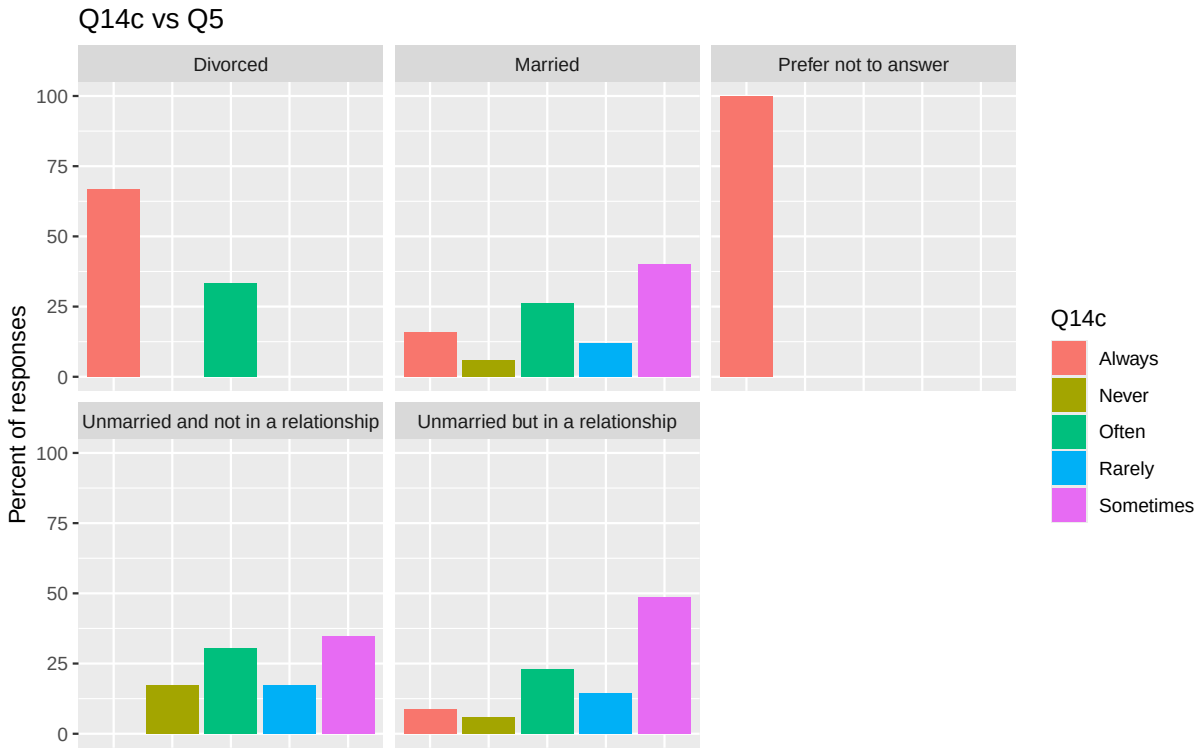

Q13i vs Q2
PGY−1 Level
PGY−2 Level
PGY−3 Level
Percent of responses
50
40
30
20
10
0
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
Once per month or less
Once per week

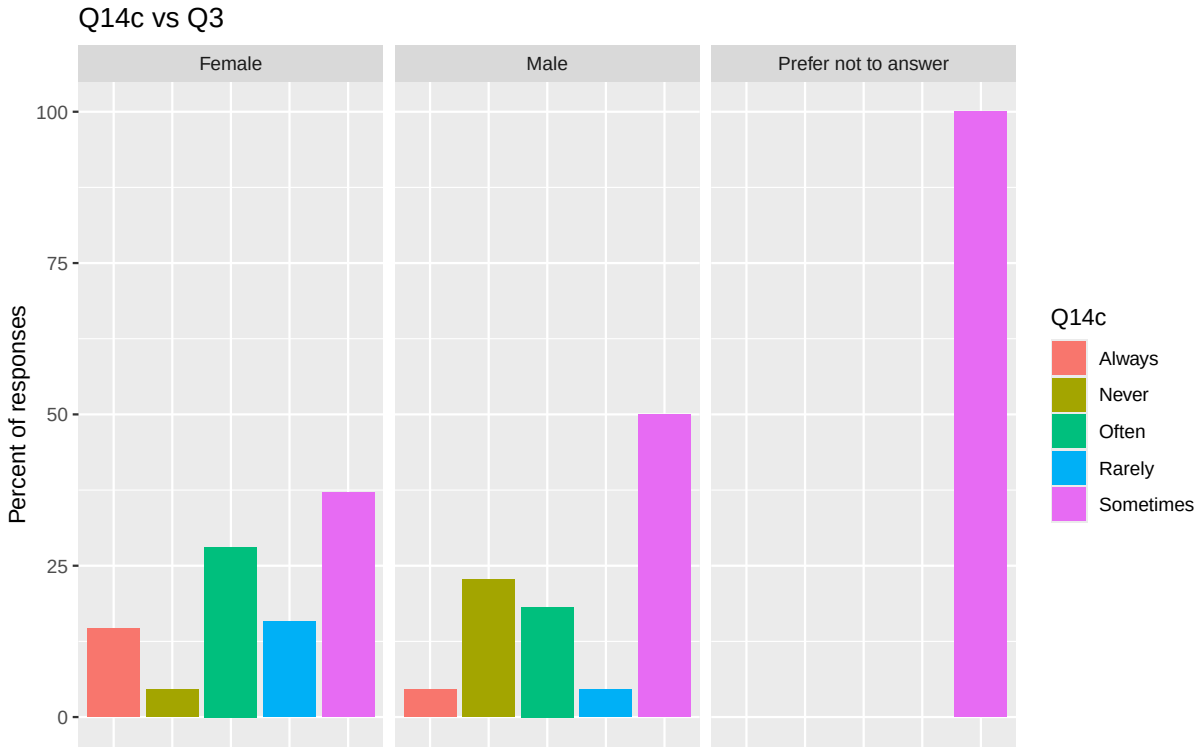

Q14a vs Q5
Divorced
Married
Prefer not to answer
Unmarried and not in a relationship
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q14a
Always
Never
Often
Rarely
Sometimes

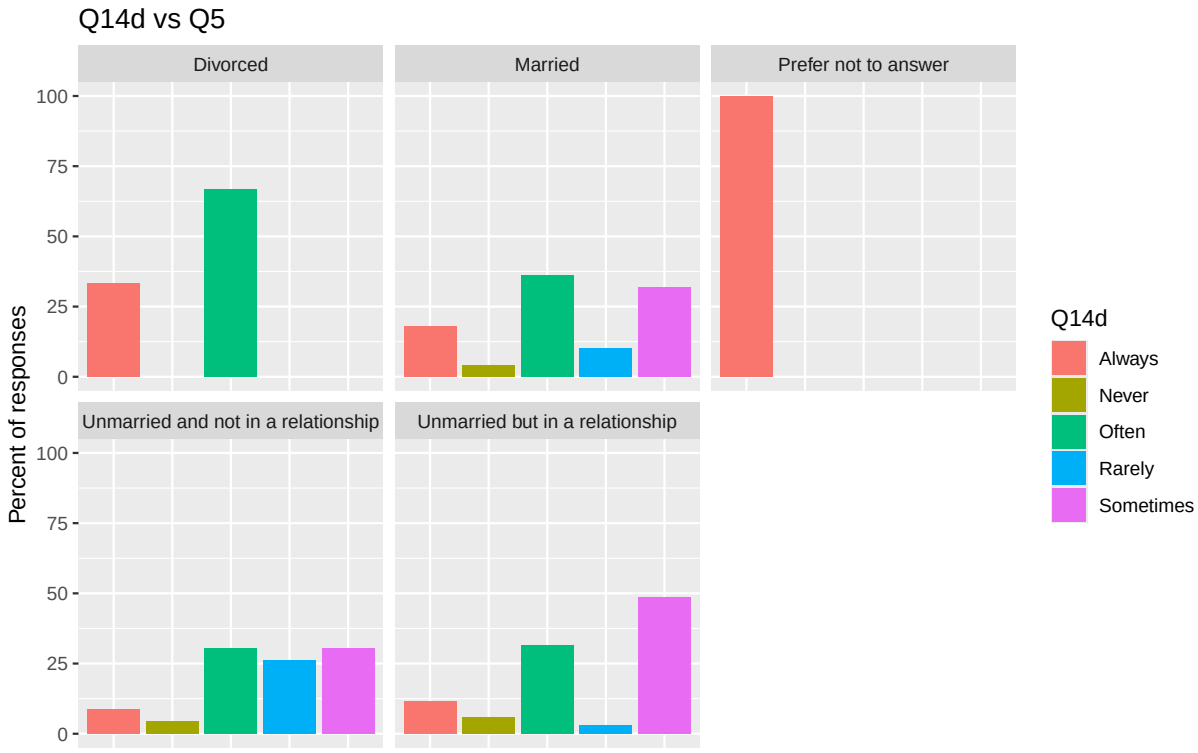

Q14a vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q14a
Always
Never
Often
Rarely
Sometimes

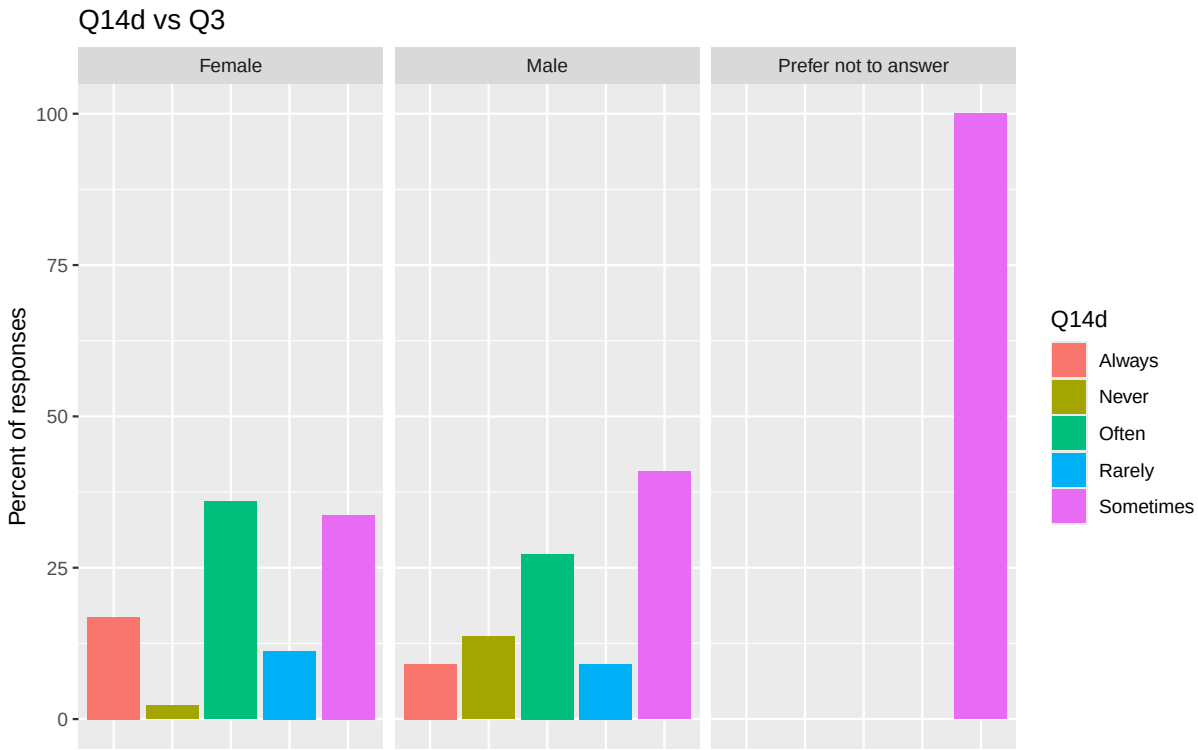

Q14b vs Q5
Divorced
Married
Prefer not to answer
Unmarried and not in a relationship
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q14b
Always
Never
Often
Rarely
Sometimes

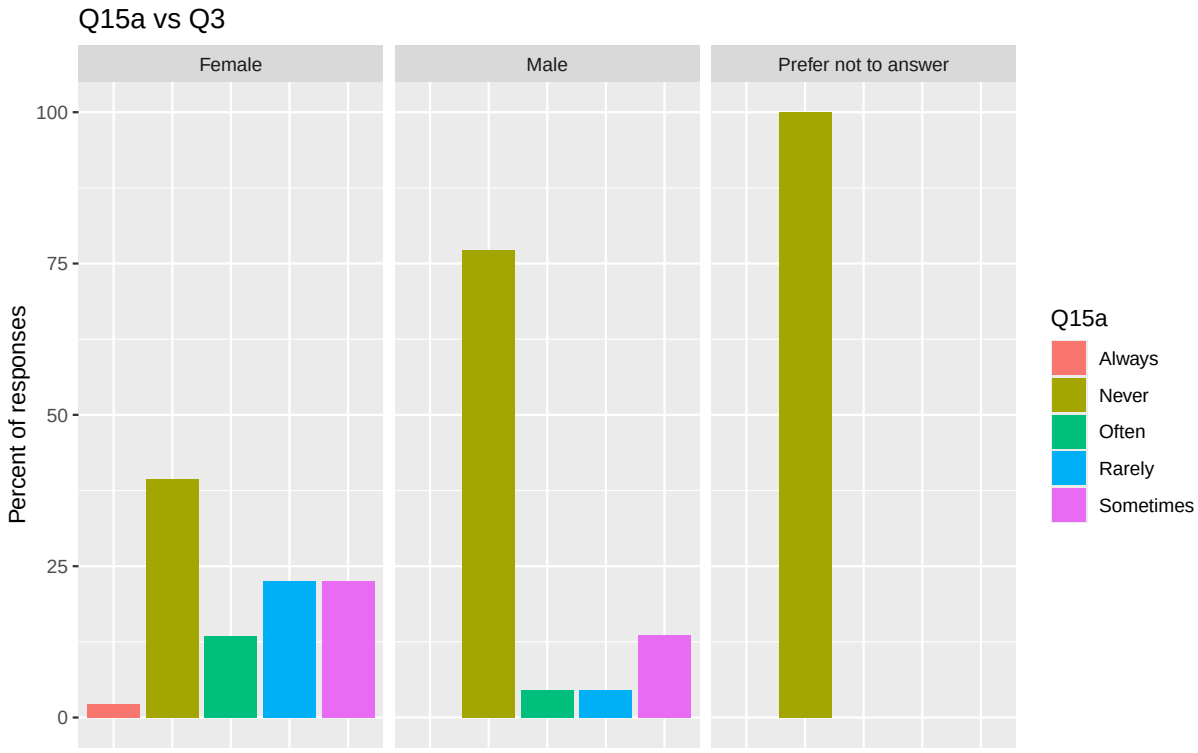

Q14b vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q14b
Always
Never
Often
Rarely
Sometimes

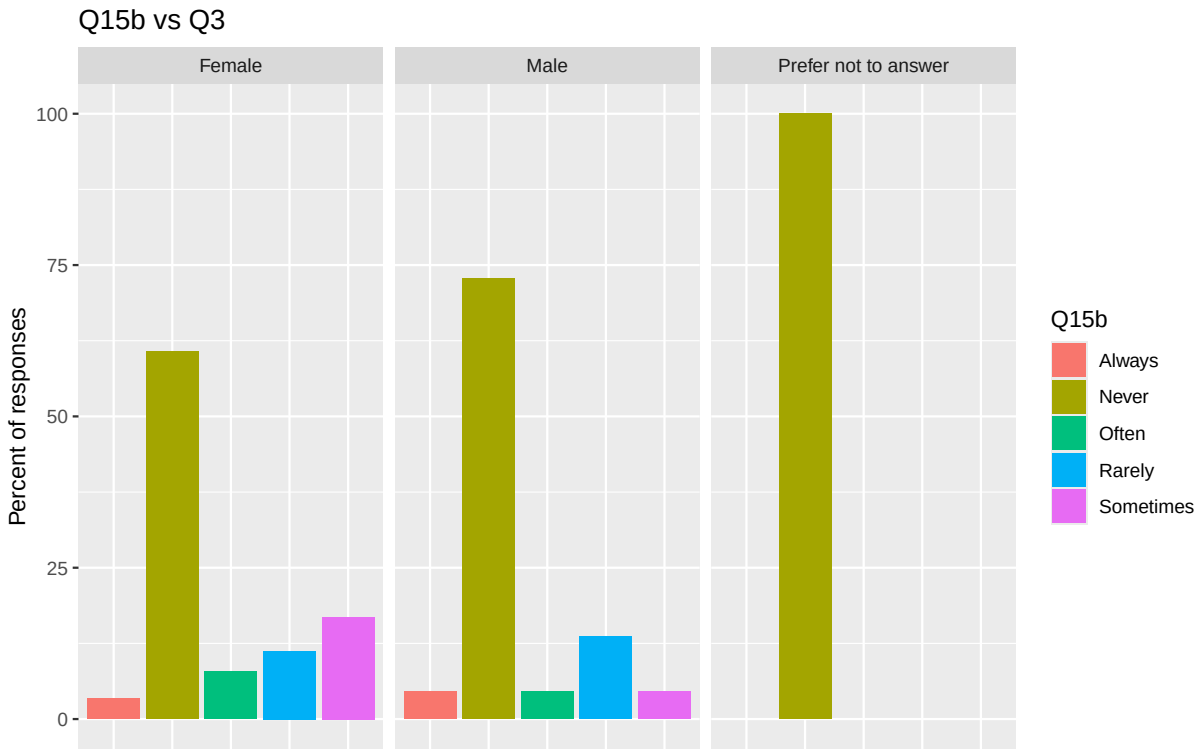

Q14c vs Q5
Divorced
Married
Prefer not to answer
Unmarried and not in a relationship
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q14c
Always
Never
Often
Rarely
Sometimes

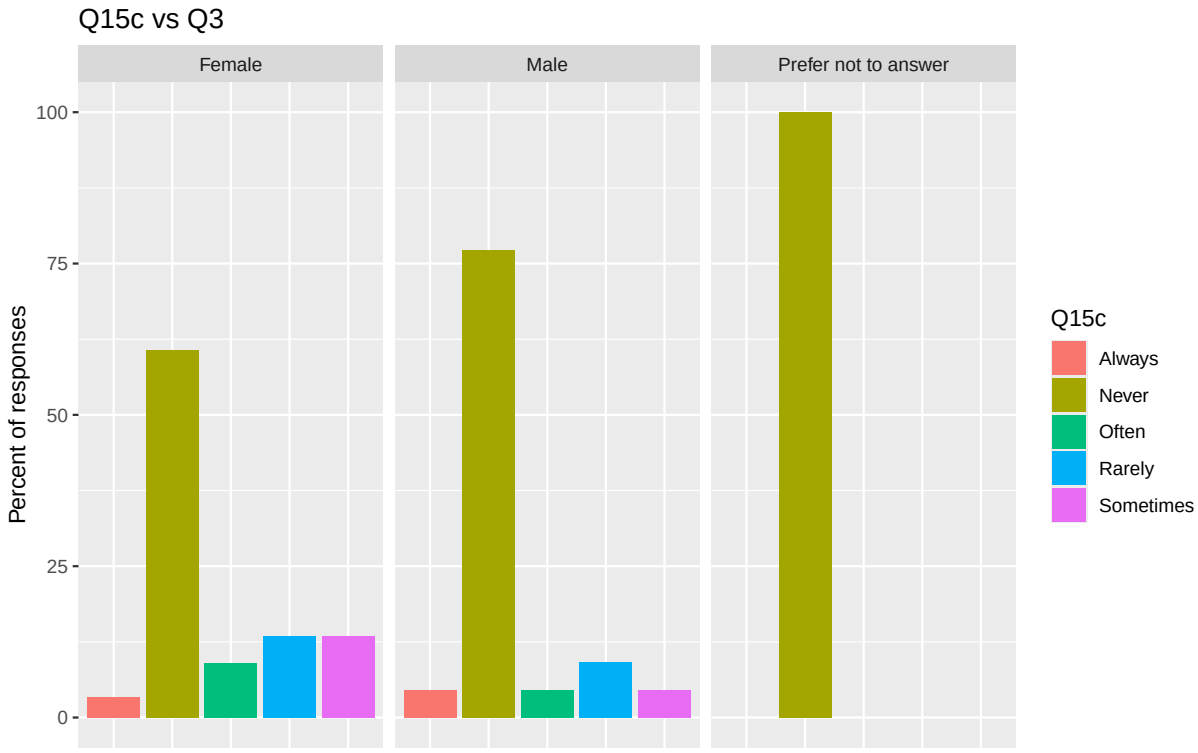

Q14c vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q14c
Always
Never
Often
Rarely
Sometimes

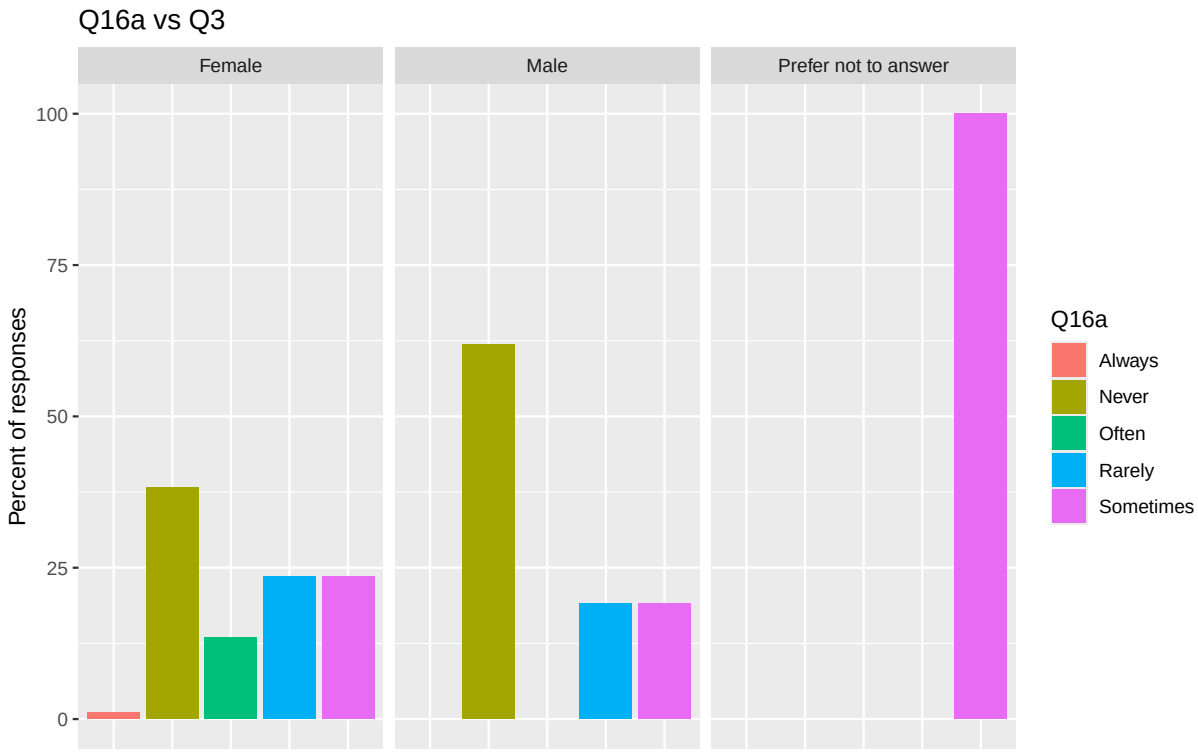

Q14d vs Q5
Divorced
Married
Prefer not to answer
Unmarried and not in a relationship
Unmarried but in a relationship
Percent of responses
100
75
50
25
0
Q14d
Always
Never
Often
Rarely
Sometimes

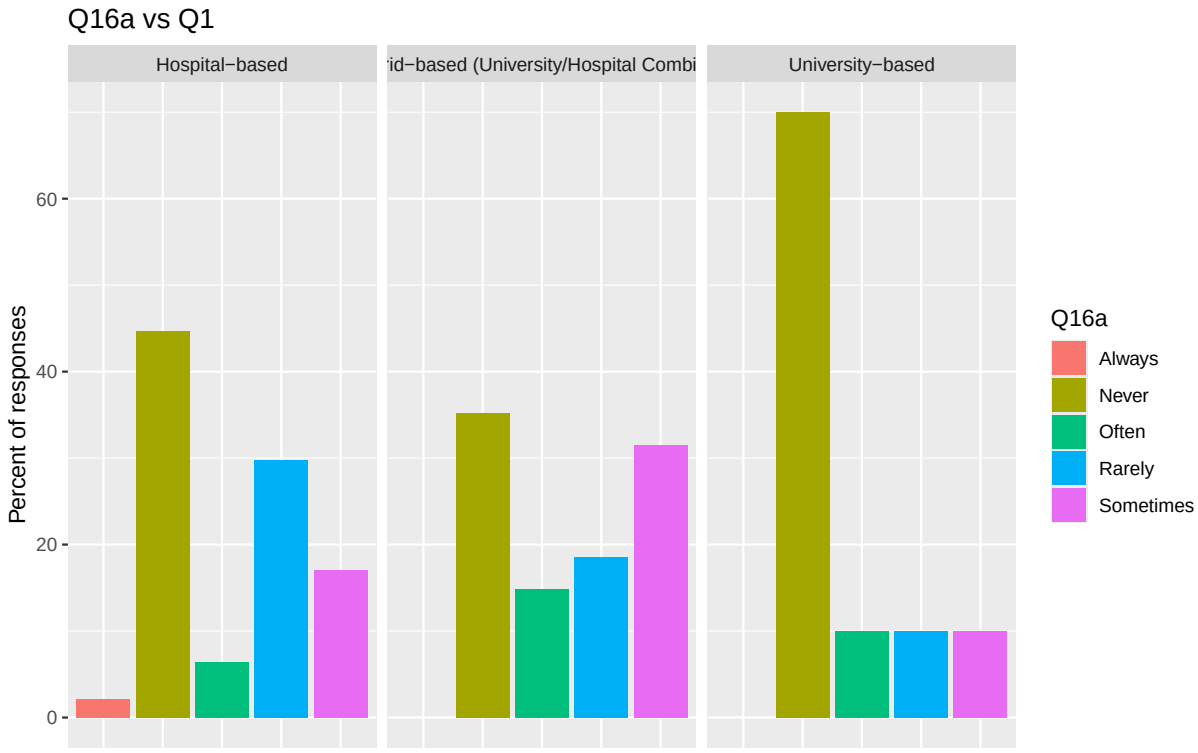

Q14d vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q14d
Always
Never
Often
Rarely
Sometimes

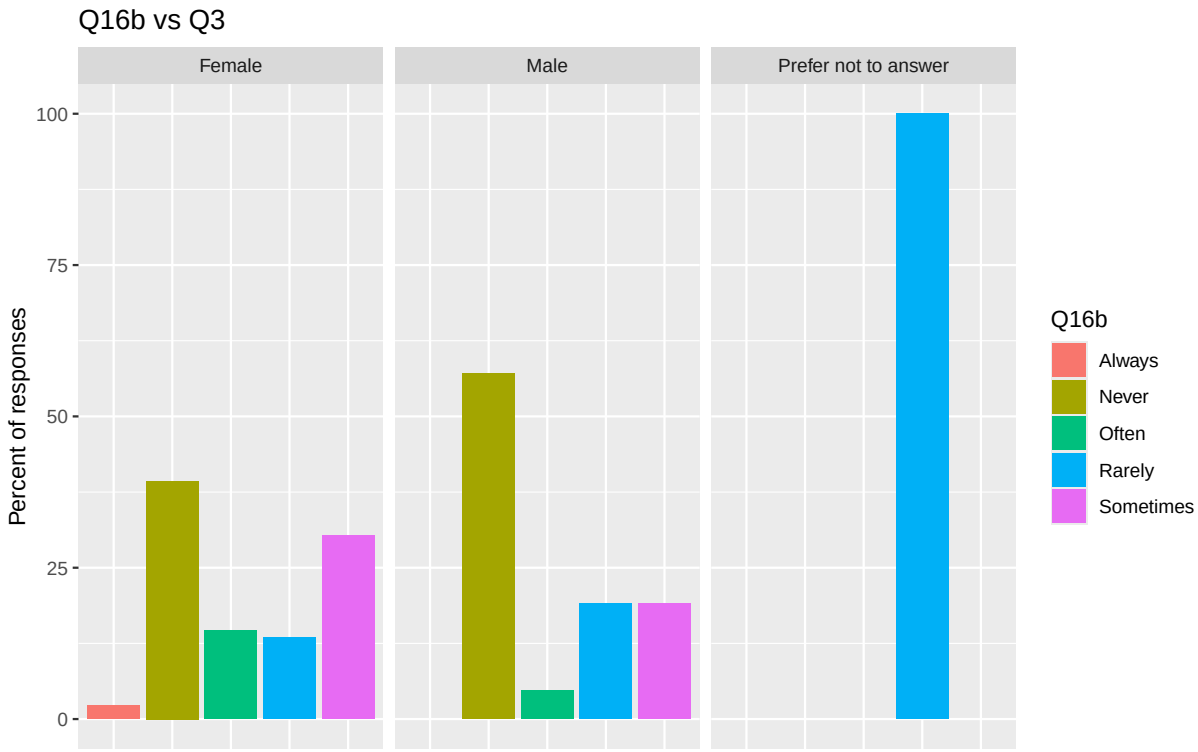

Q15a vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q15a
Always
Never
Often
Rarely
Sometimes

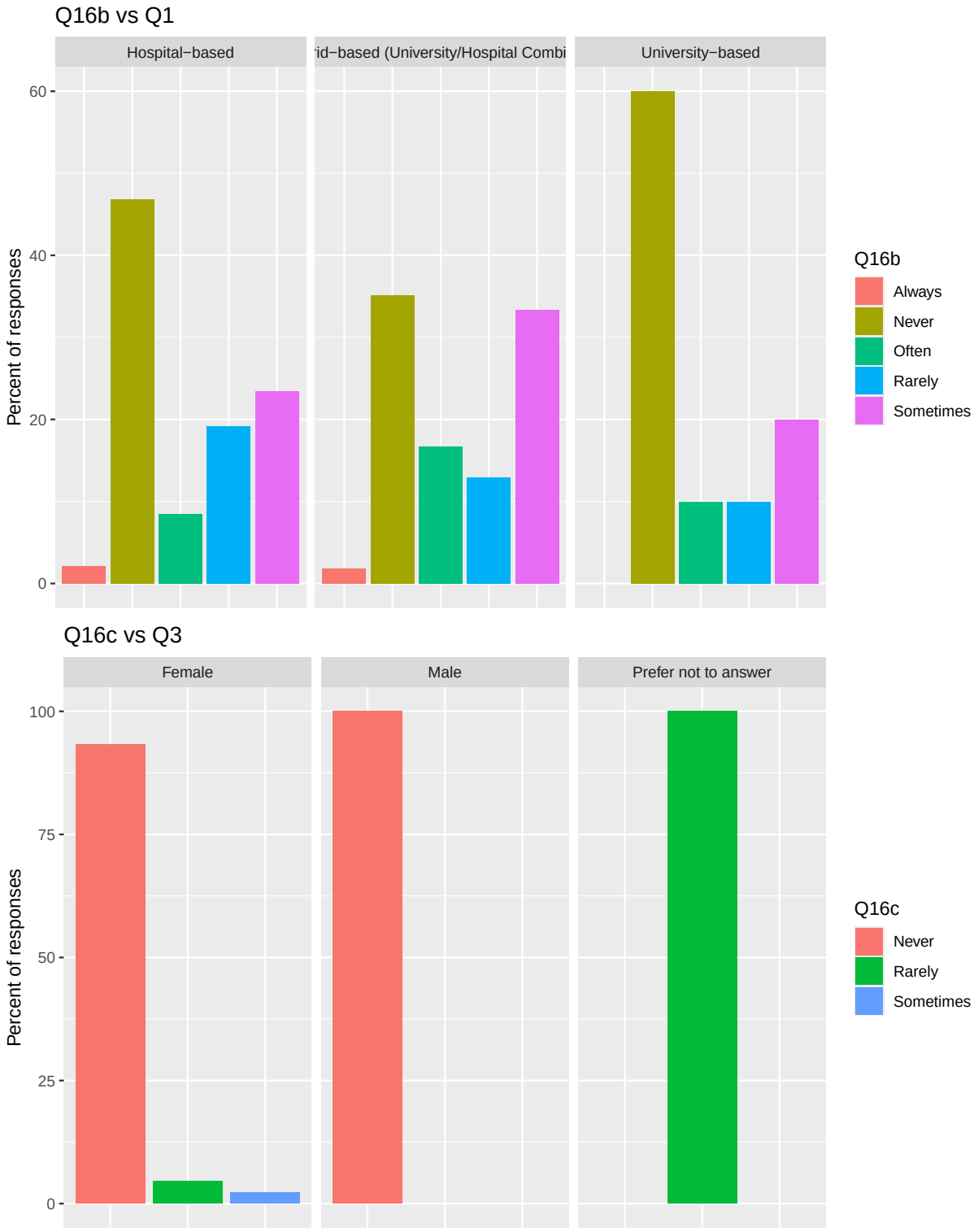

Q15b vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q15b
Always
Never
Often
Rarely
Sometimes

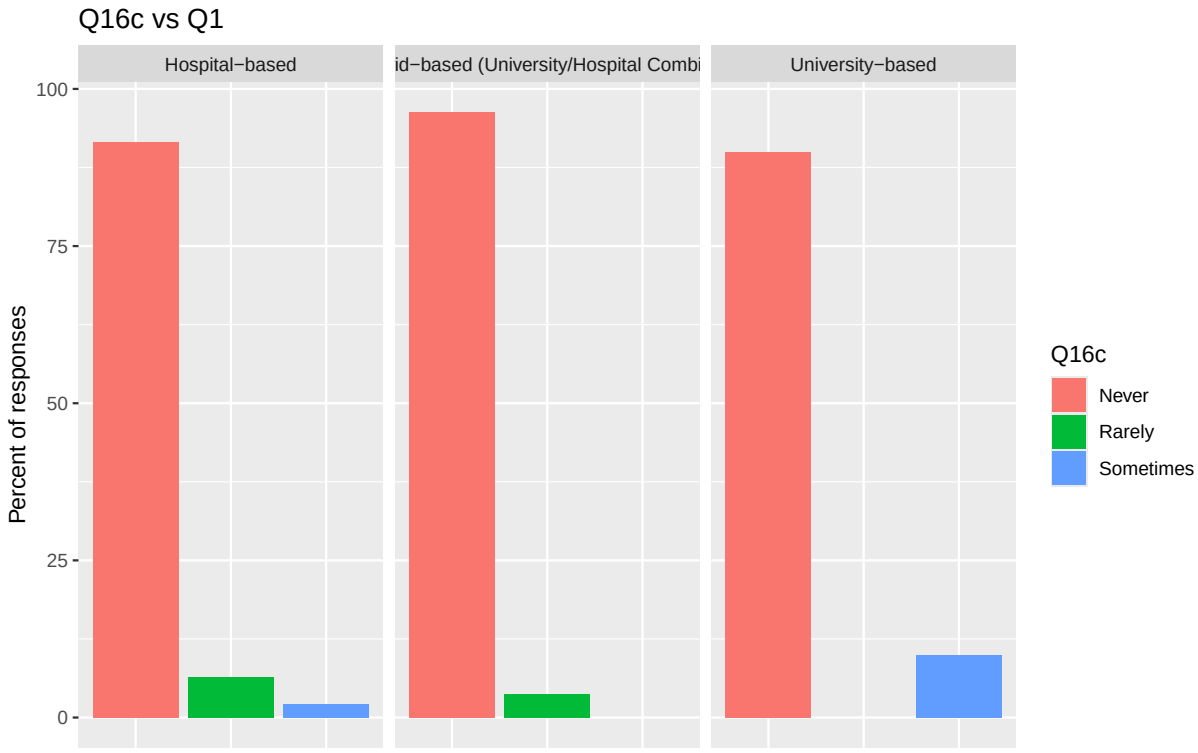

Q15c vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q15c
Always
Never
Often
Rarely
Sometimes

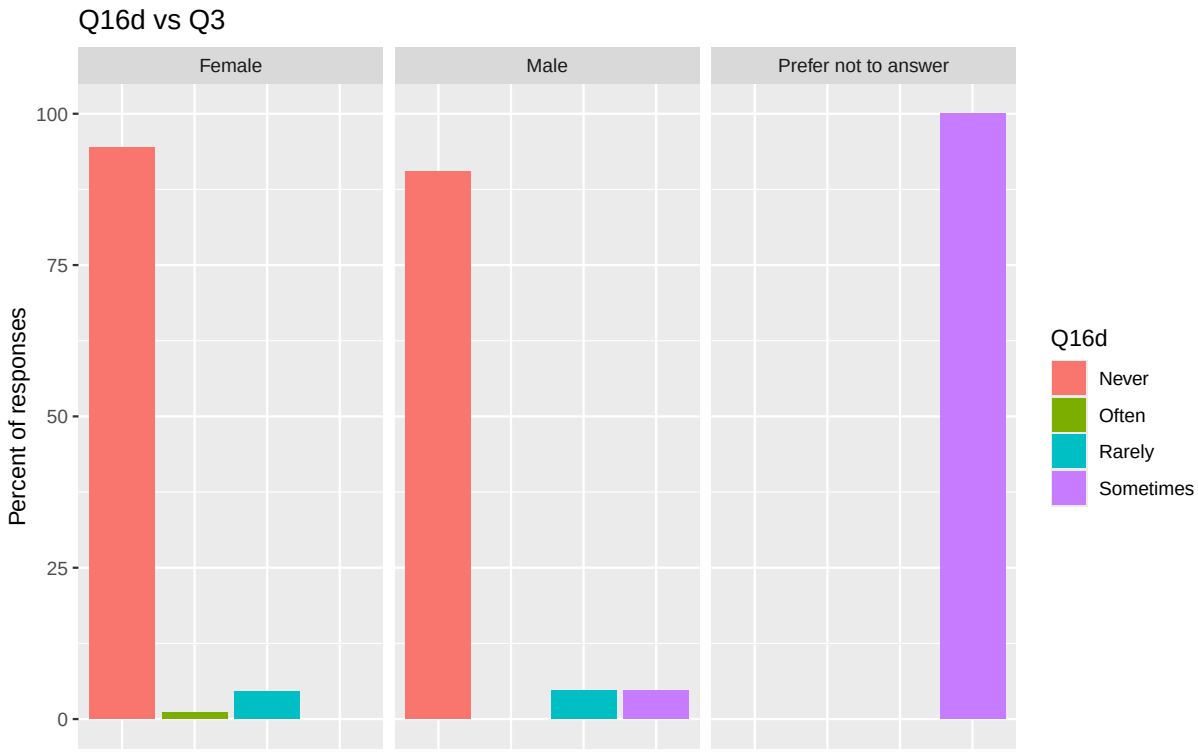

Q16a vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q16a
Always
Never
Often
Rarely
Sometimes

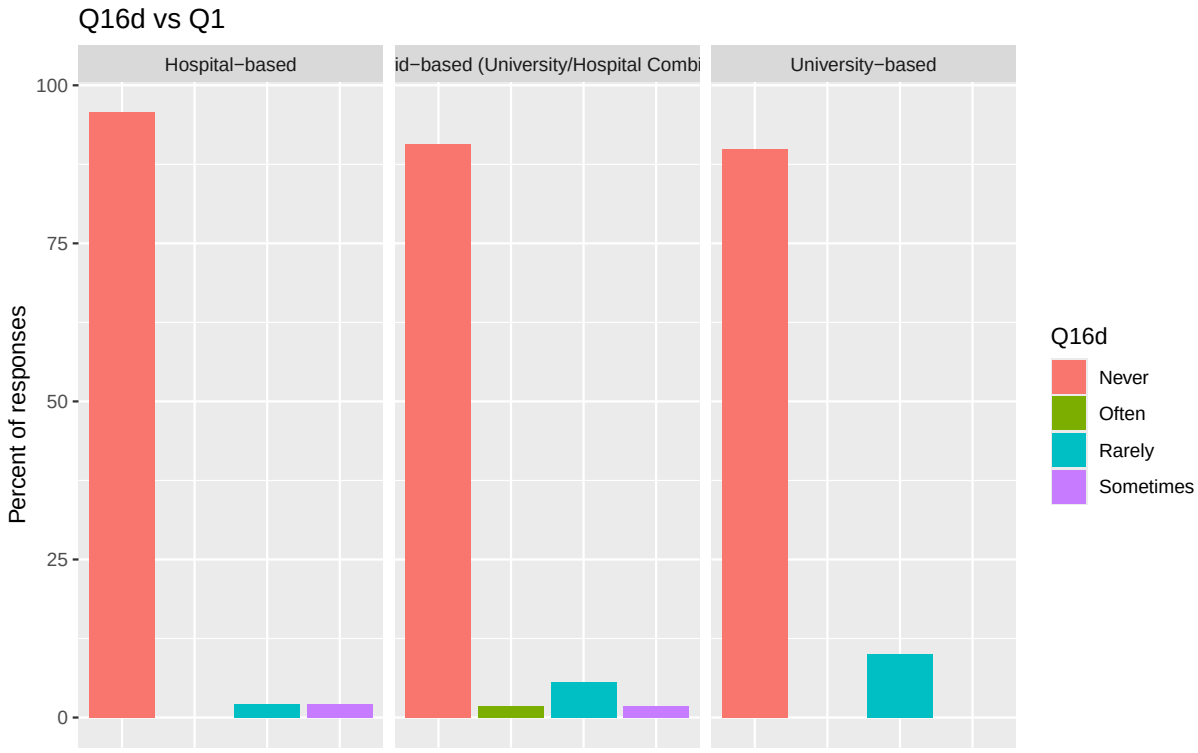

Q16a vs Q1
Hospital−based
id−based (University/Hospital Combi
University−based
Percent of responses
0
20
40
60
Q16a
Always
Never
Often
Rarely
Sometimes

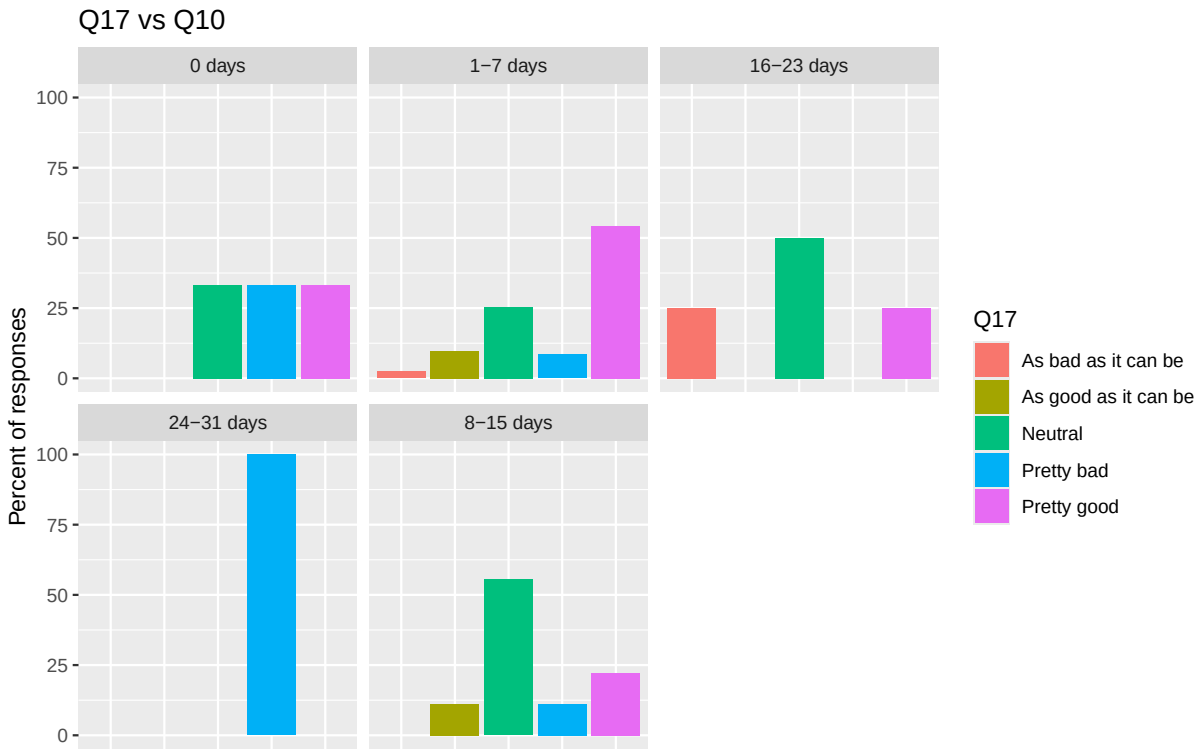

Q16b vs Q3
Female
Male
Prefer not to answer
Percent of responses
0
25
50
75
100
Q16b
Always
Never
Often
Rarely
Sometimes

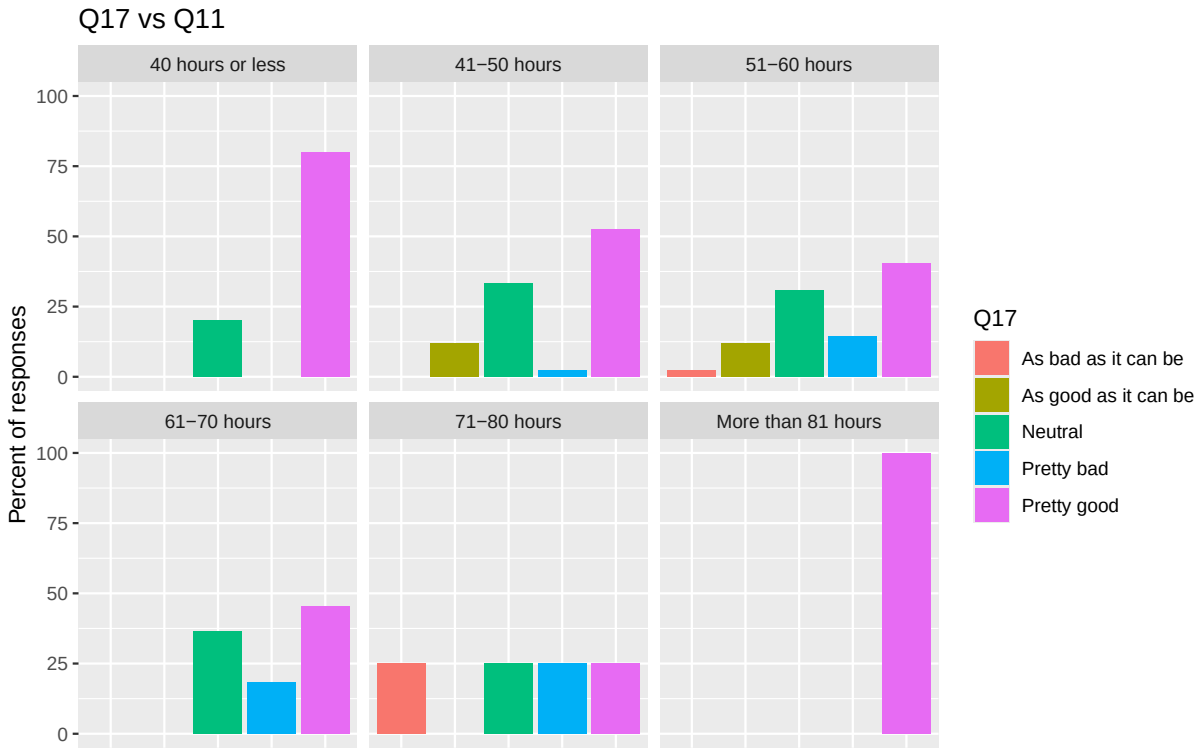

Q16b vs Q1
Hospital−based
id−based (University/Hospital Combi
University−based
Percent of responses
60
40
20
0
Q16b
Always
Never
Often
Rarely
Sometimes
Q16c vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q16c
Never
Rarely
Sometimes

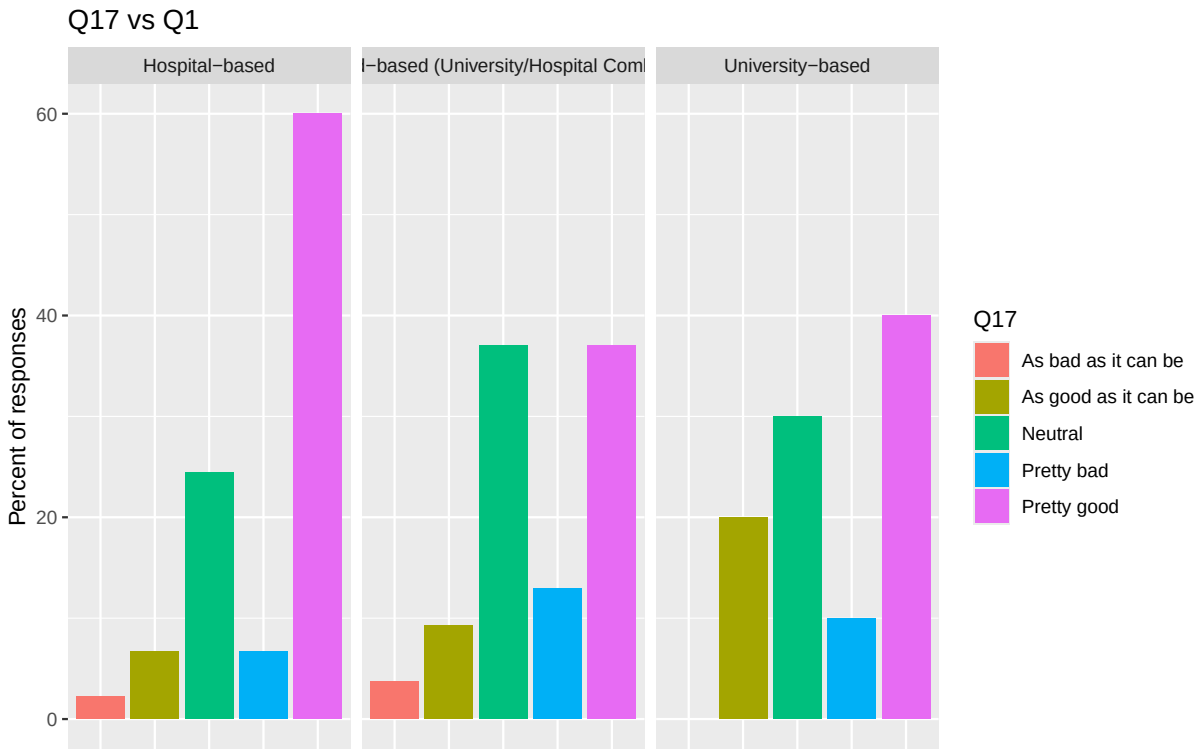

Q16c vs Q1
Hospital−based
id−based (University/Hospital Combi
University−based
Percent of responses
100
75
50
25
0
Q16c
Never
Rarely
Sometimes

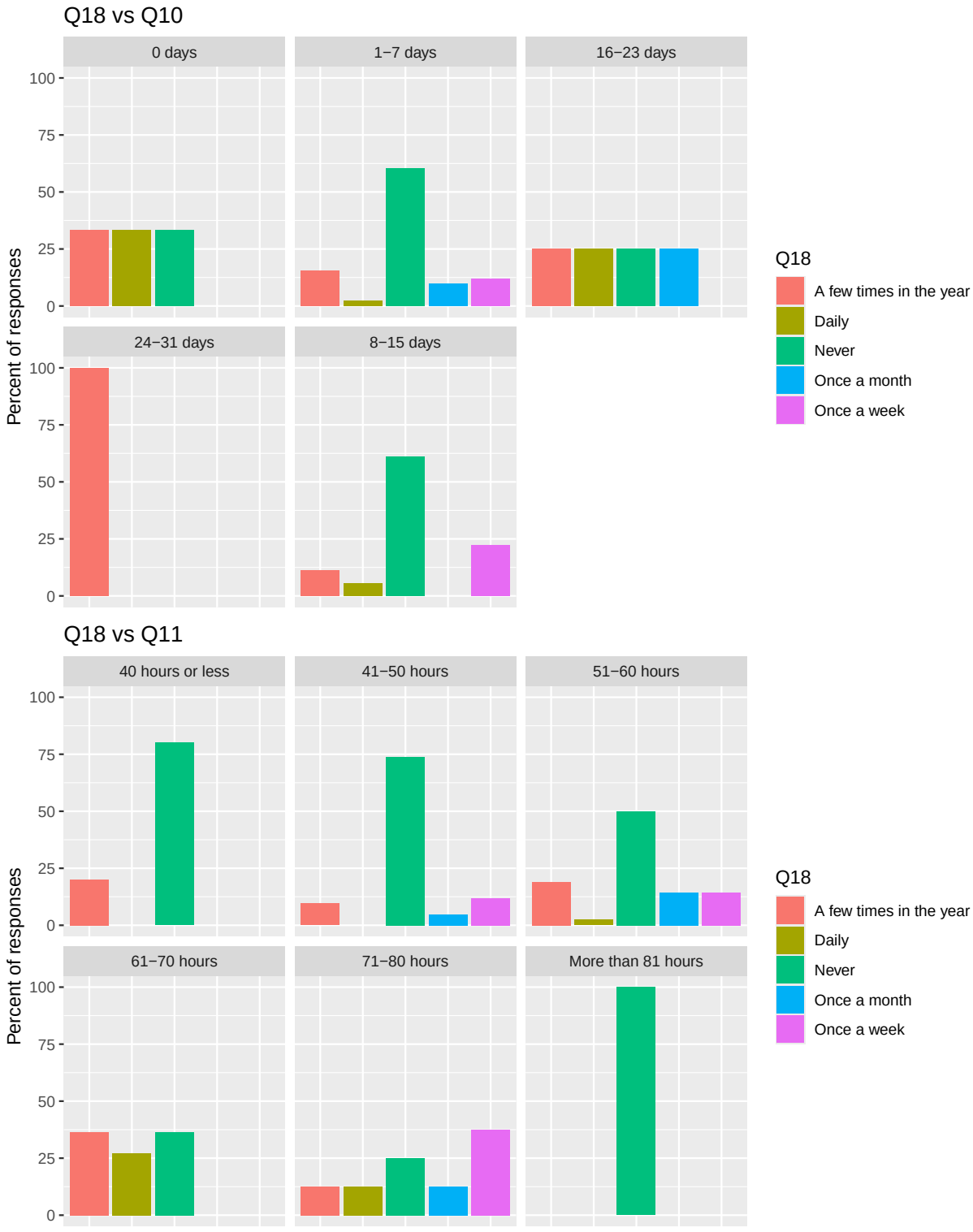

Q16d vs Q3
Female
Male
Prefer not to answer
Percent of responses
100
75
50
25
0
Q16d
Never
Often
Rarely
Sometimes

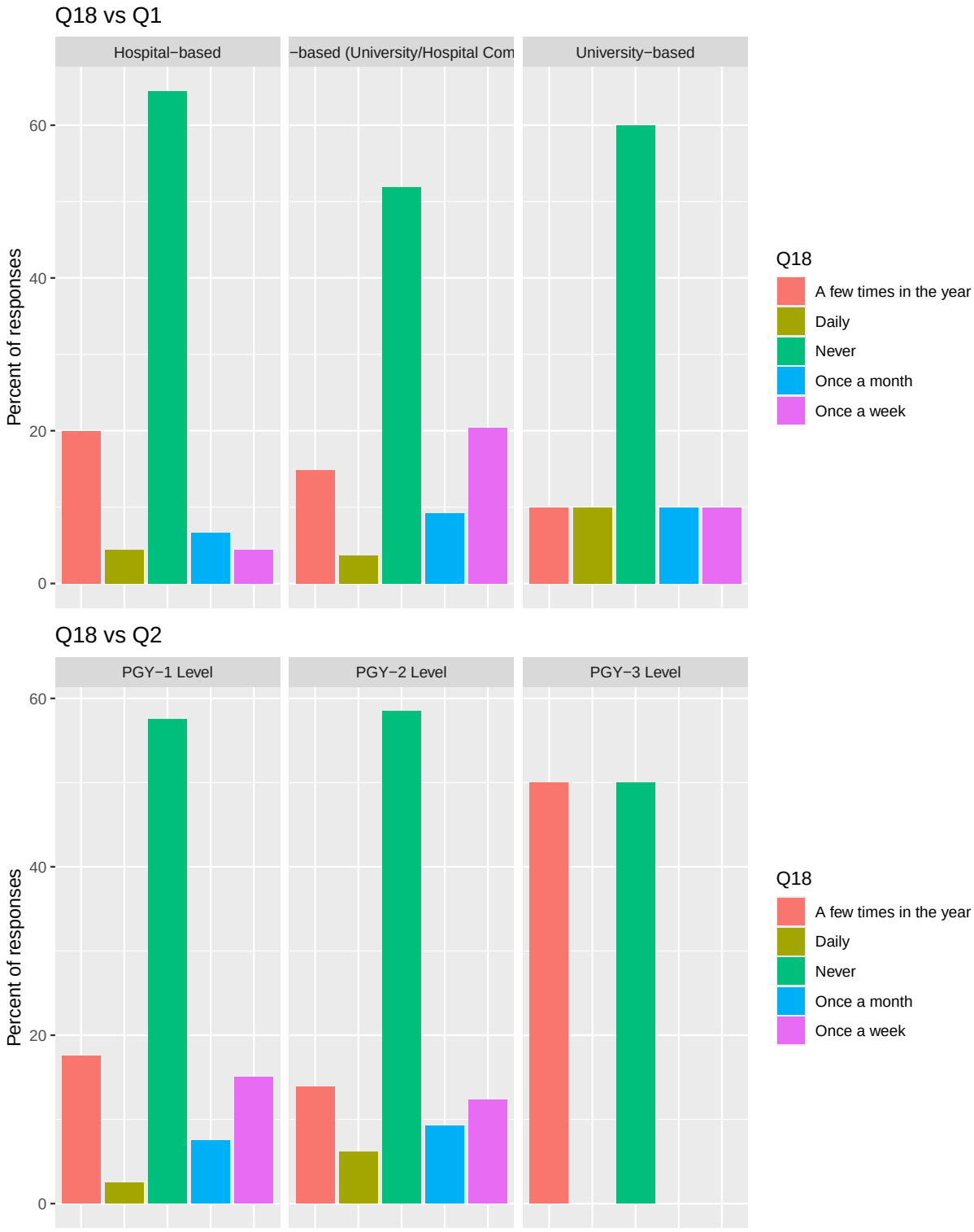

Q16d vs Q1
Hospital−based
id−based (University/Hospital Combi
University−based
Percent of responses
100
75
50
25
0
Q16d
Never
Often
Rarely
Sometimes

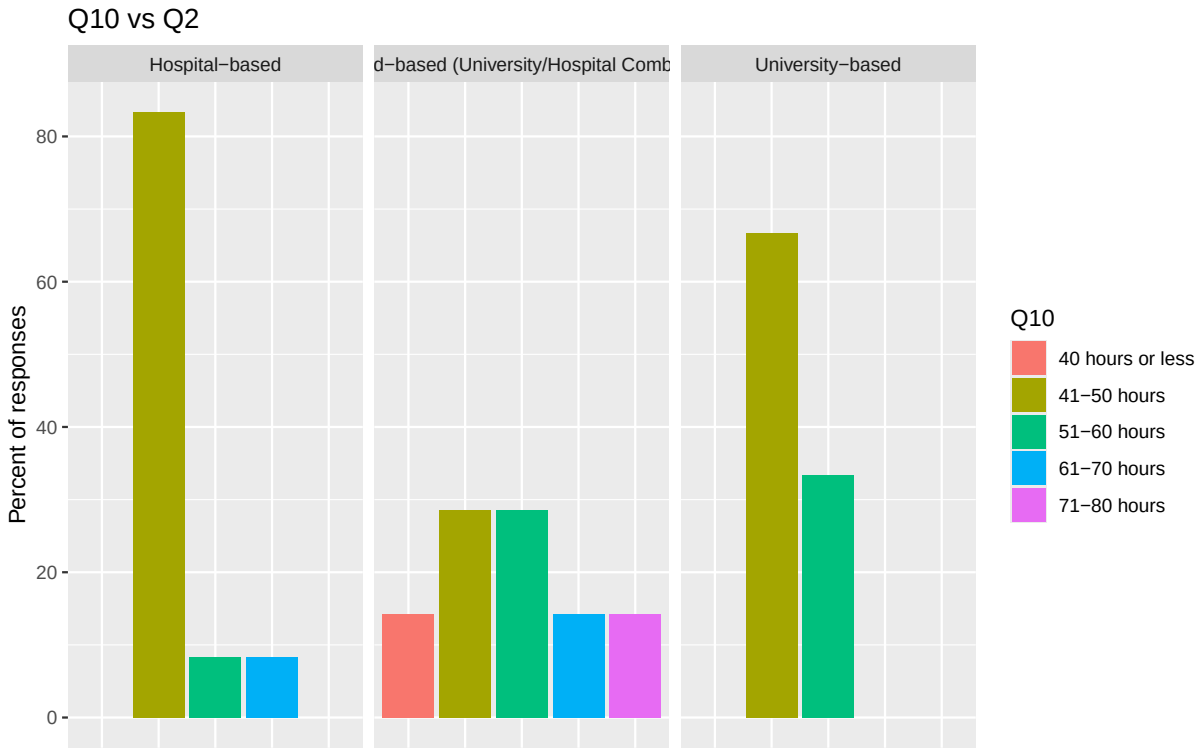

Q17 vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q17
As bad as it can be
As good as it can be
Neutral
Pretty bad
Pretty good

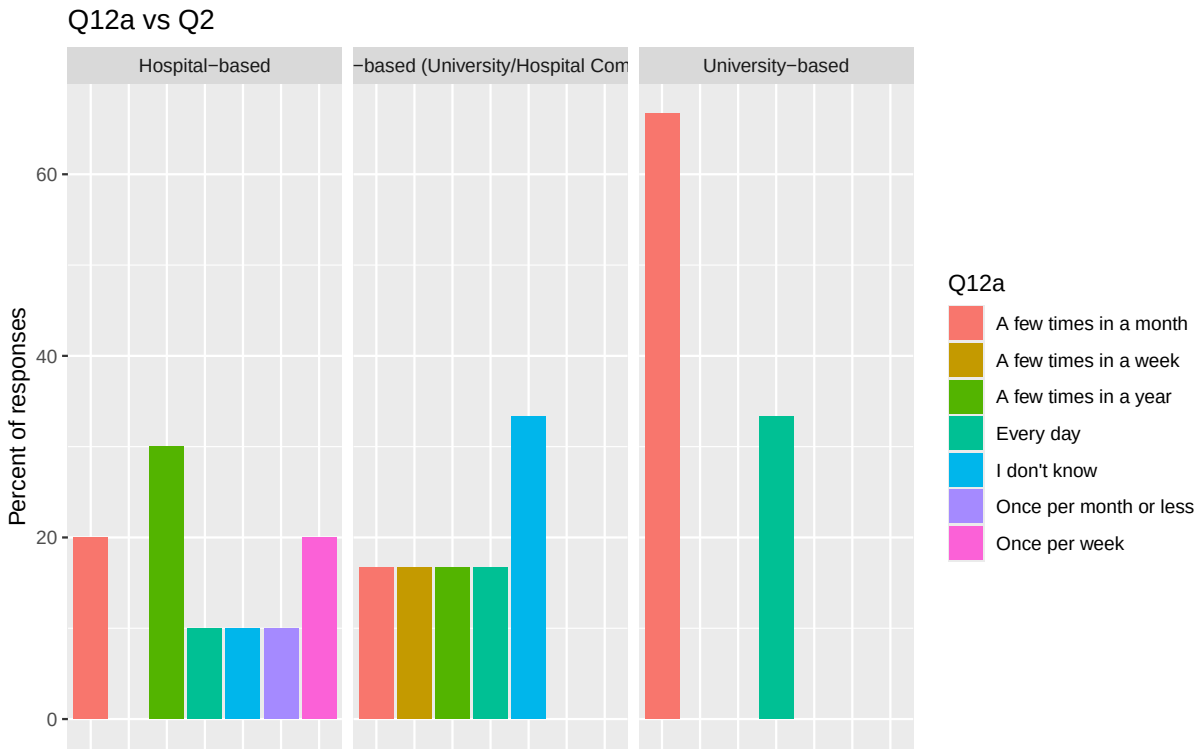

Q17 vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours
Percent of responses
100
75
50
25
0
Q17
As bad as it can be
As good as it can be
Neutral
Pretty bad
Pretty good

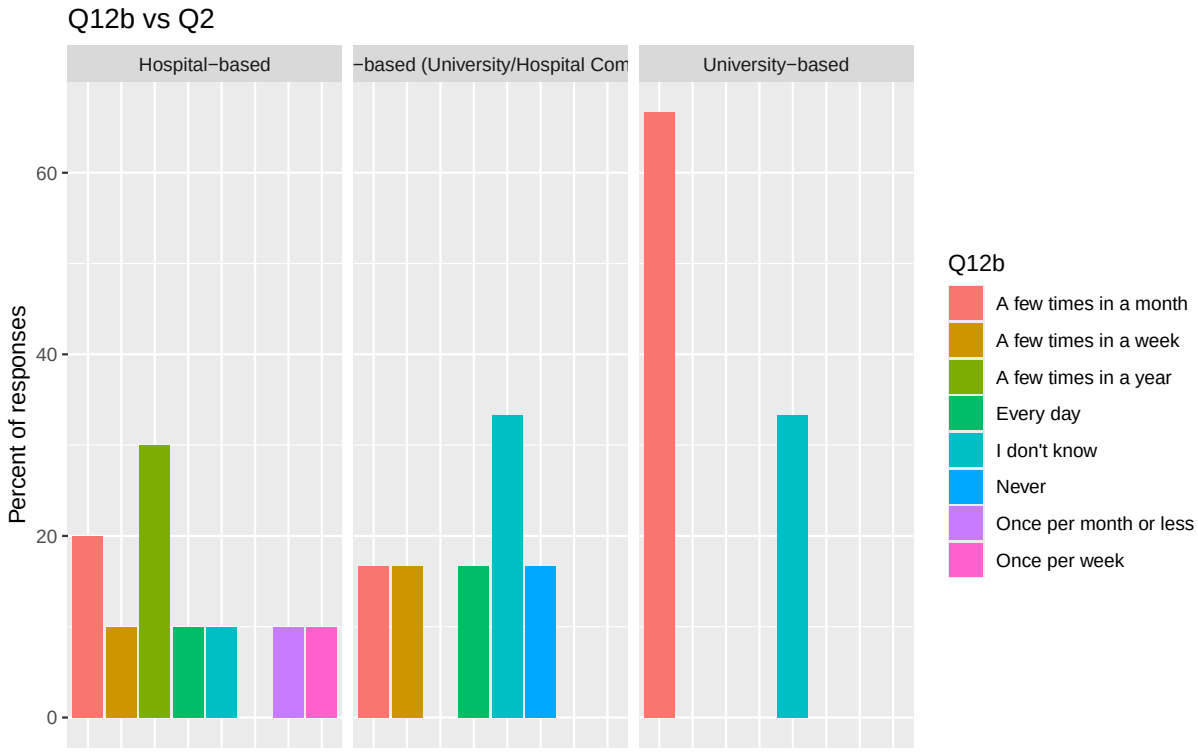

Q17 vs Q1
Hospital−based
I−based (University/Hospital Com
University−based
Percent of responses
60
40
20
0
Q17
As bad as it can be
As good as it can be
Neutral
Pretty bad
Pretty good

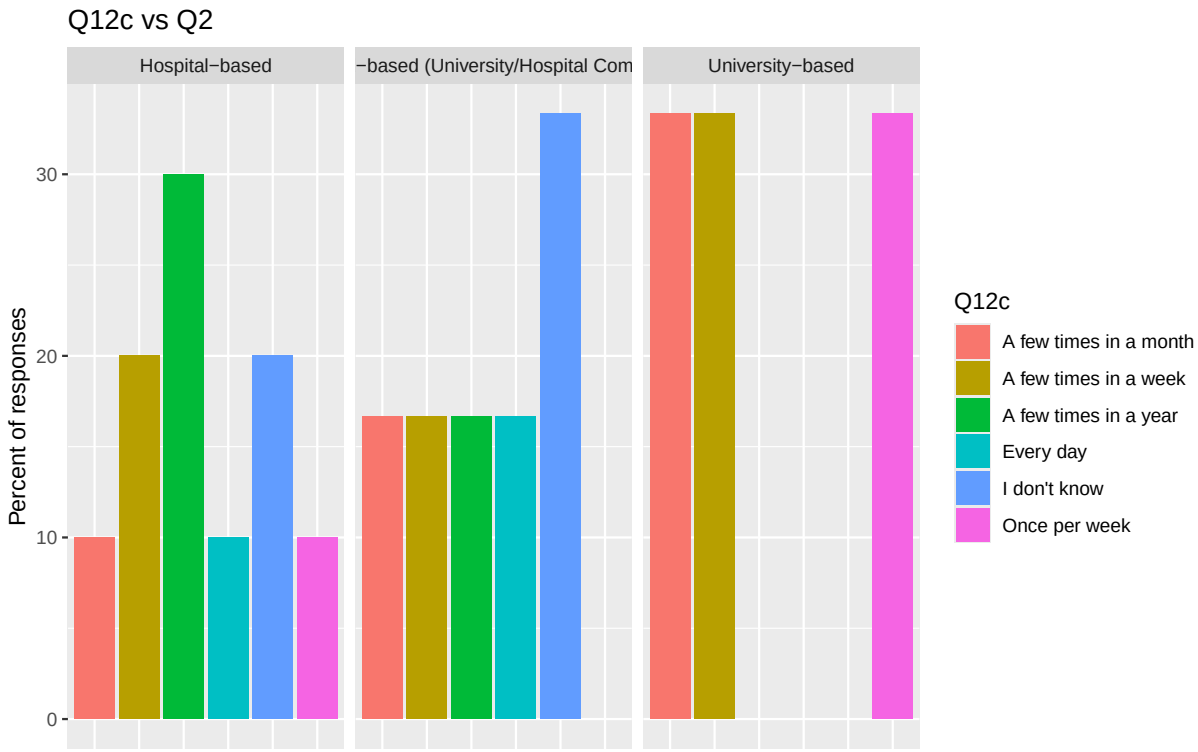

Q18 vs Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days
Percent of responses
100
75
50
25
0
Q18
A few times in the year
Daily
Never
Once a month
Once a week
Q18 vs Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours

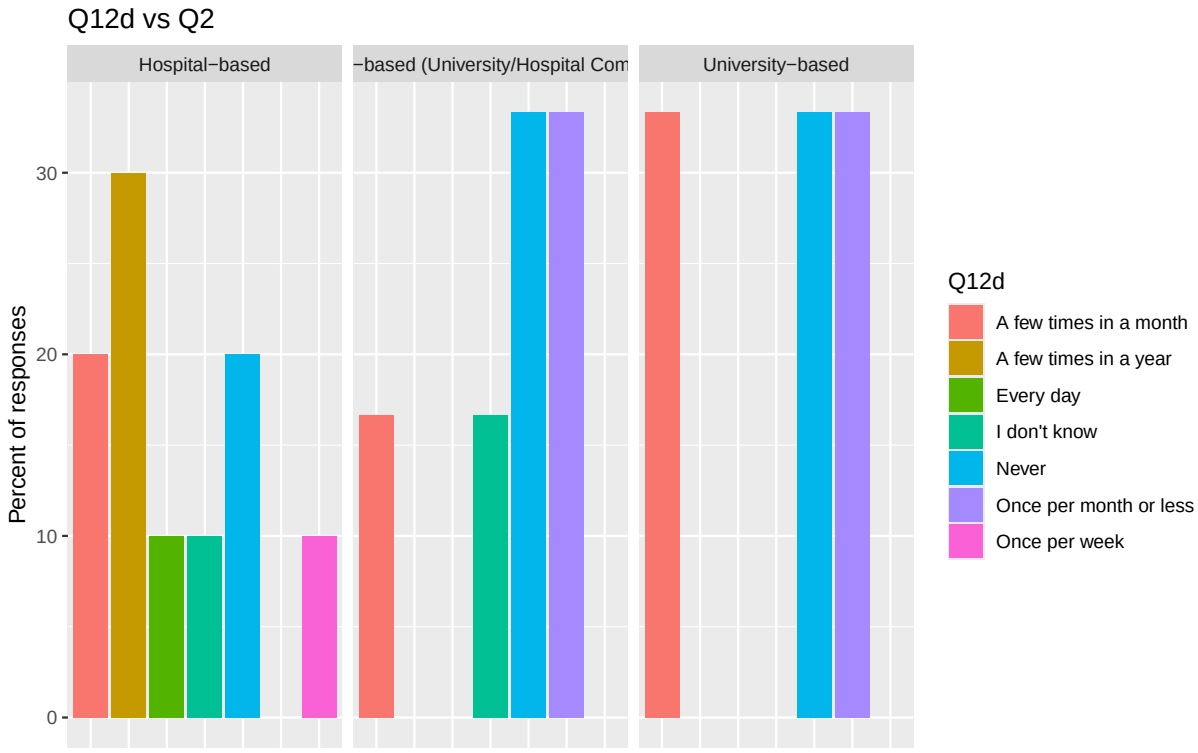

Q18 vs Q1
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
20
40
60
Q18
A few times in the year
Daily
Never
Once a month
Once a week
Q18 vs Q2
PGY−1 Level
PGY−2 Level
PGY−3 Level
Percent of responses
0
20
40
60
Q18
A few times in the year
Daily
Never
Once a month
Once a week

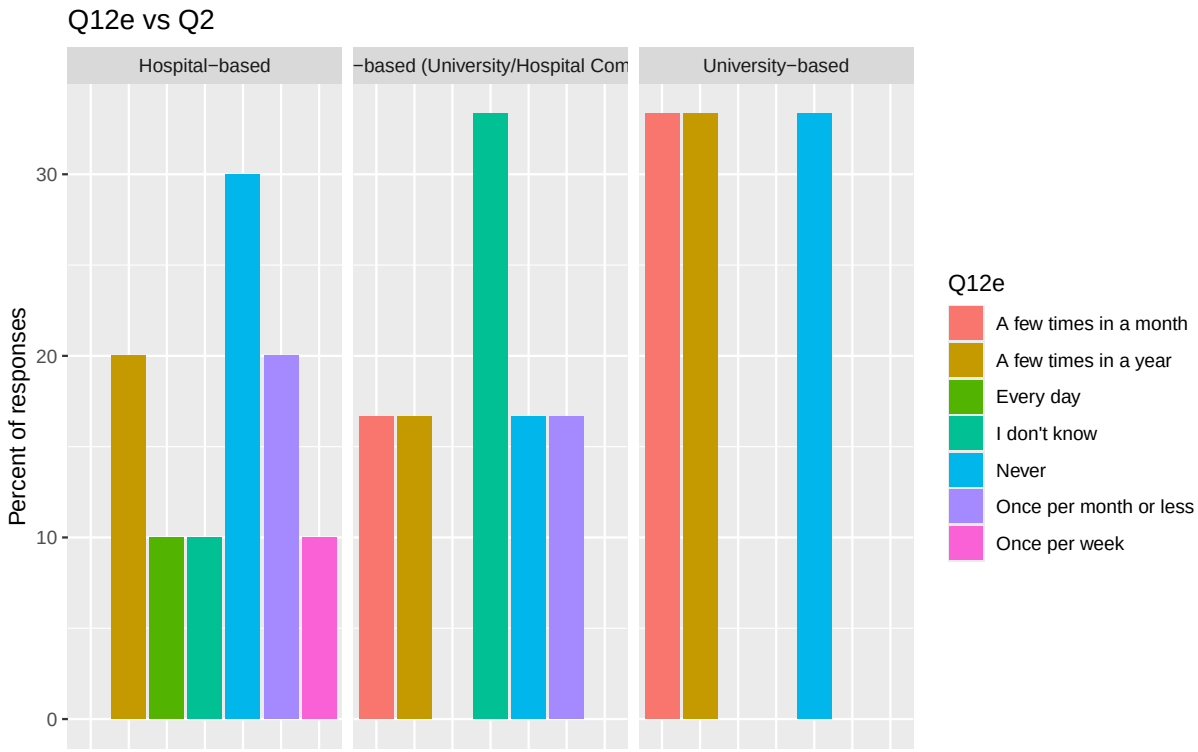

Q10 vs Q2
Hospital−based
d−based (University/Hospital Comb
University−based
Percent of responses
0
20
40
60
80
Q10
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours

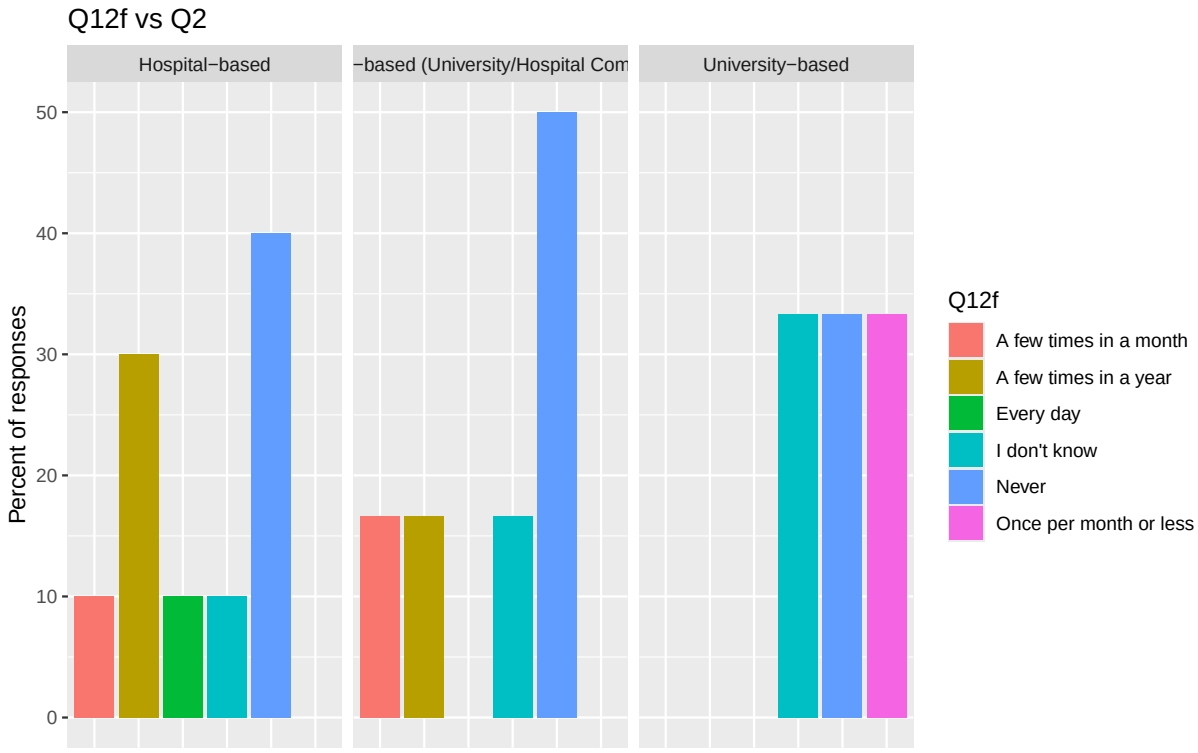

Q12a vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
0
20
40
60
Q12a
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Once per month or less
Once per week

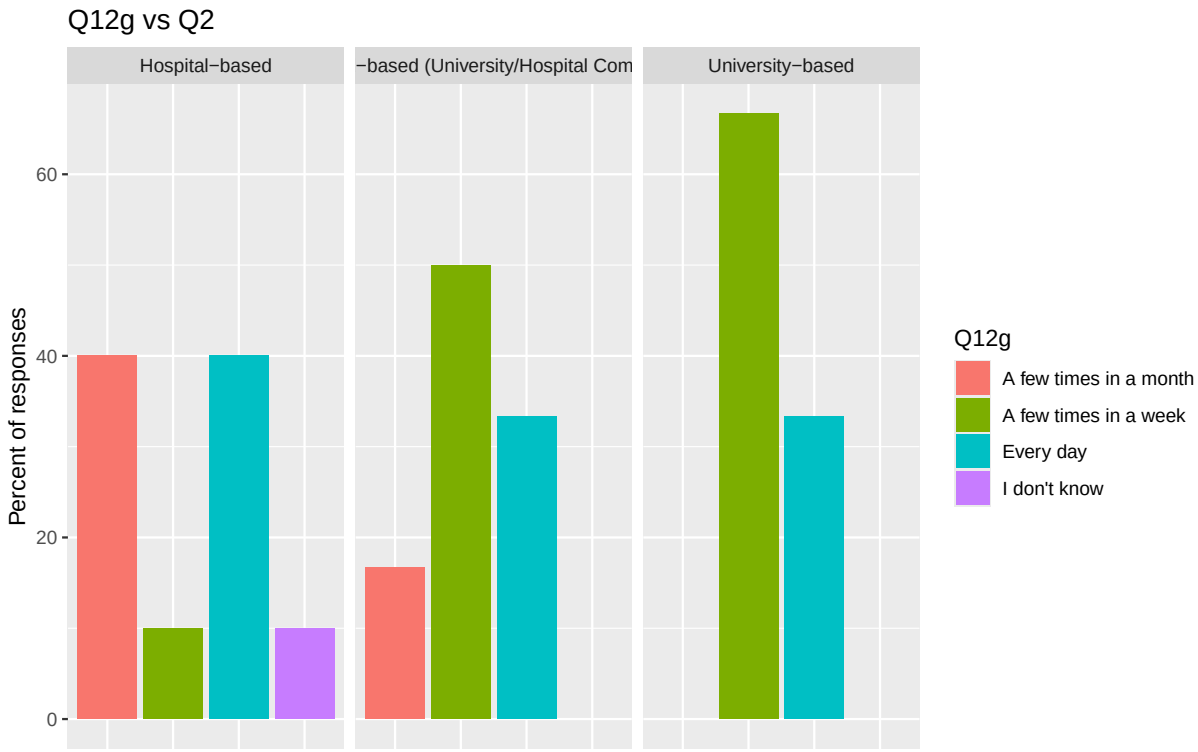

Q12b vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
60
40
20
0
Q12b
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

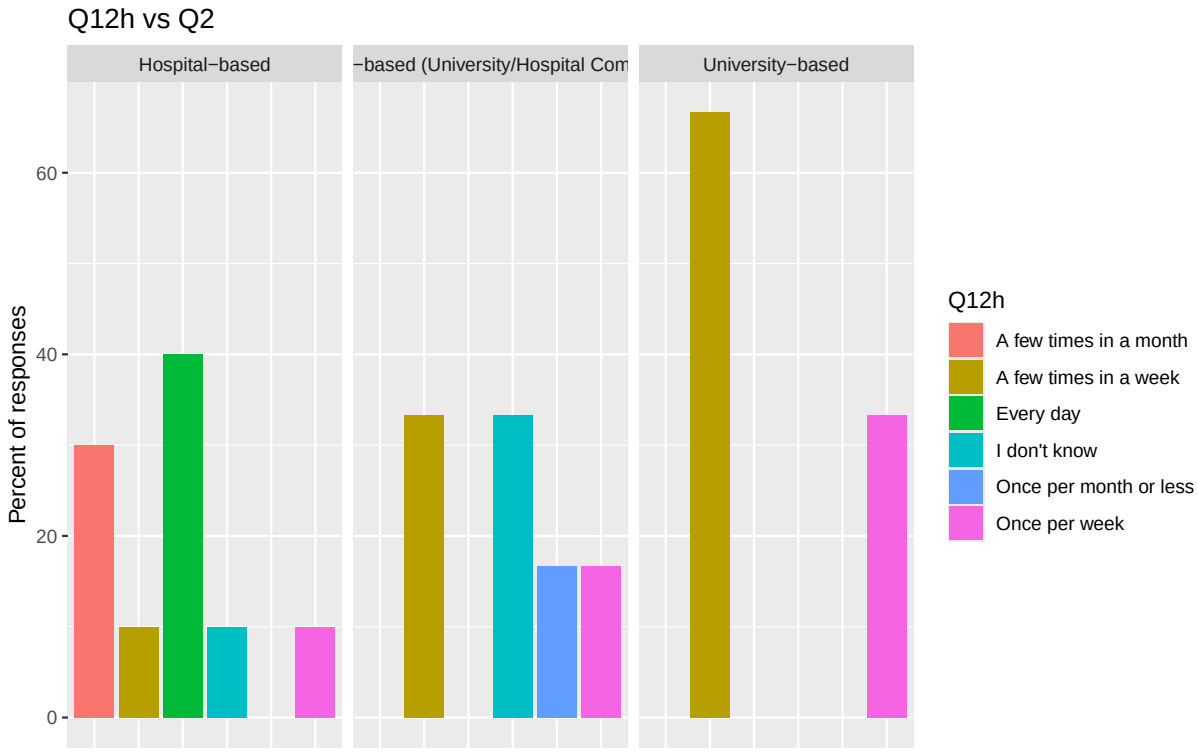

Q12c vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
30
20
10
0
Q12c
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Once per week

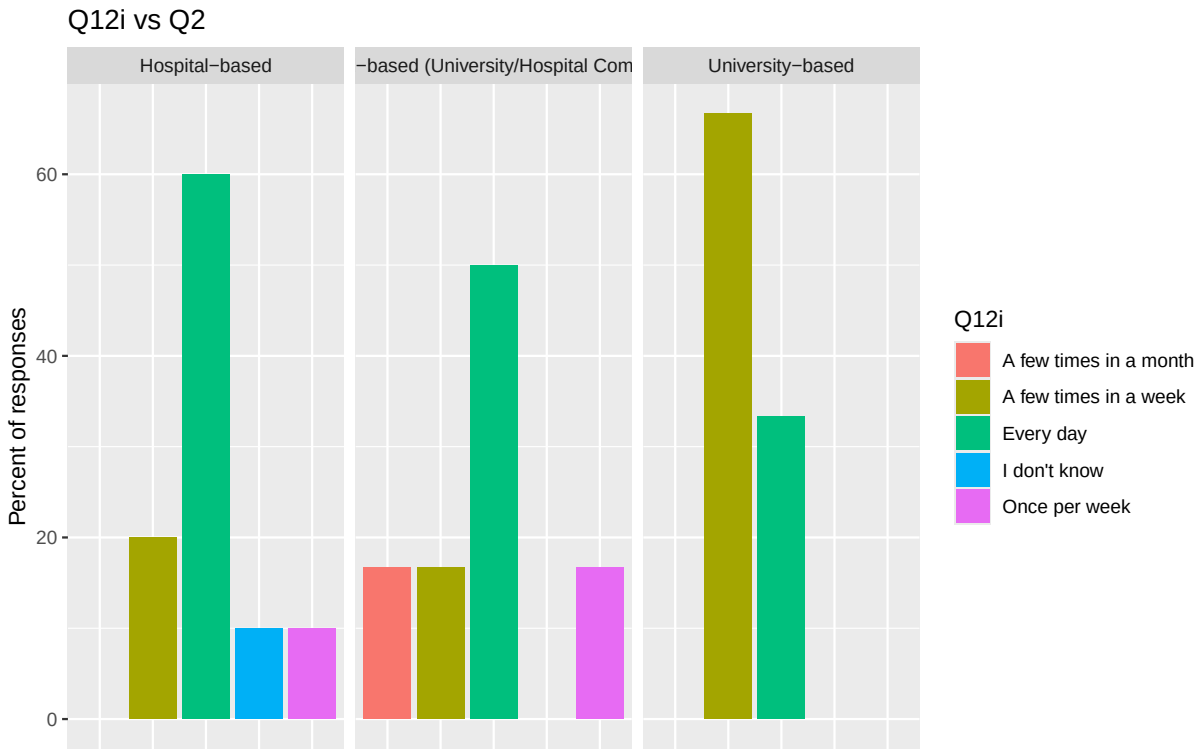

Q12d vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
30
20
10
0
Q12d
A few times in a month
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

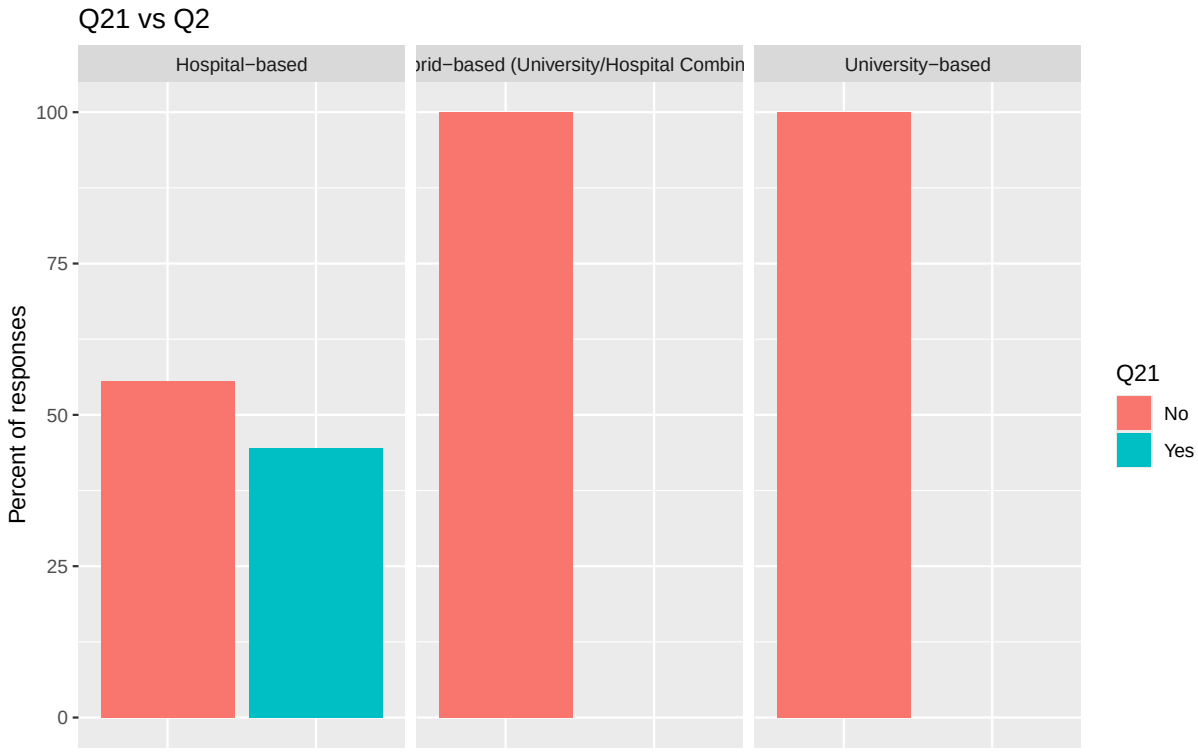

Q12e vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
30
20
10
0
Q12e
A few times in a month
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

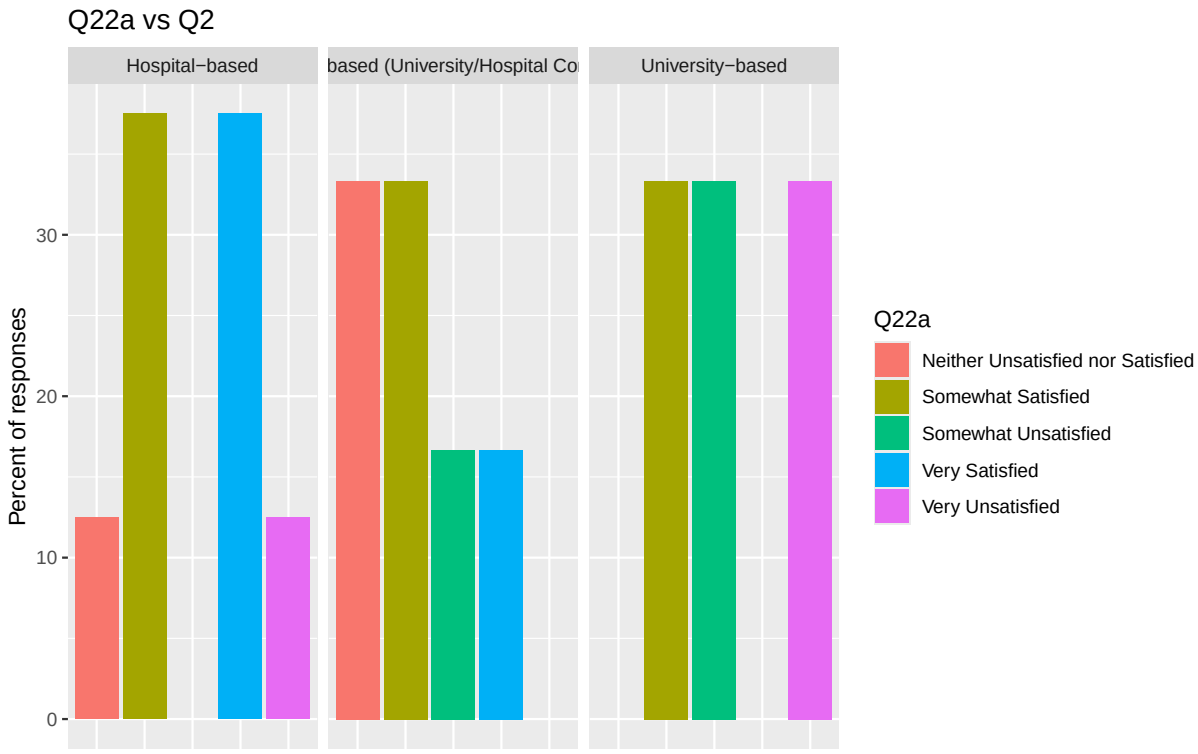

Q12f vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
50
40
30
20
10
0
Q12f
A few times in a month
A few times in a year
Every day
I don't know
Never
Once per month or less

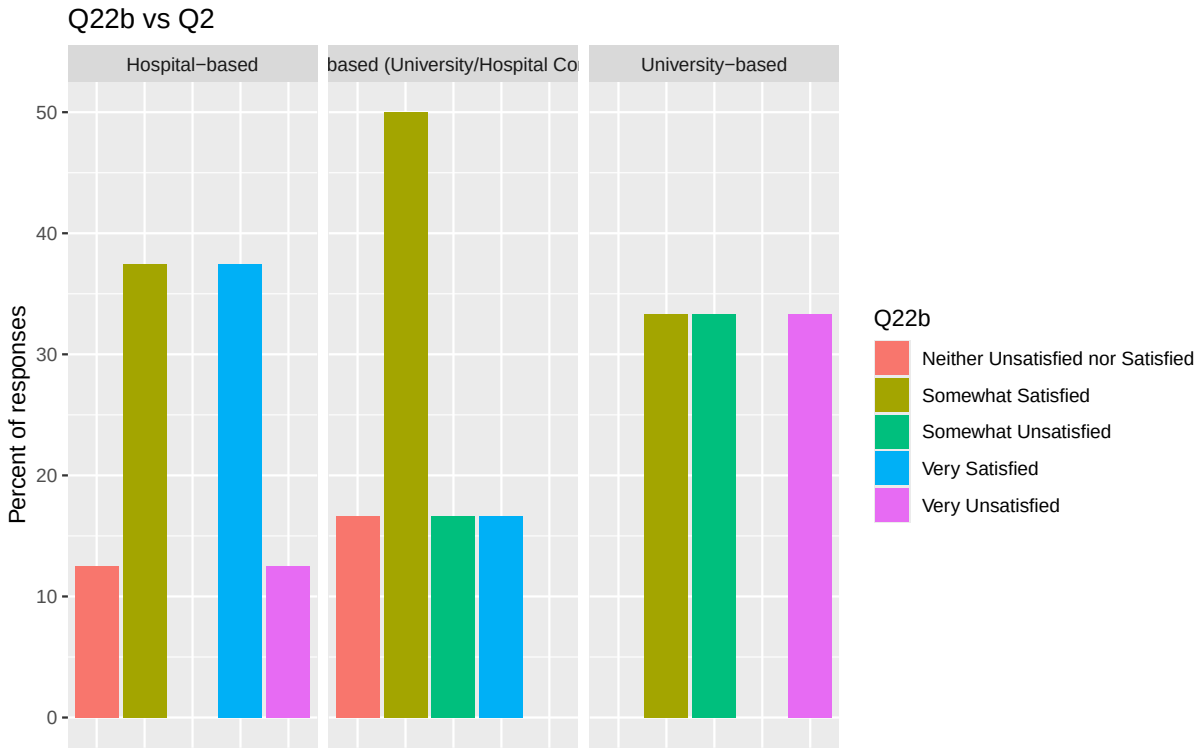

Q12g vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
60
40
20
0
Q12g
A few times in a month
A few times in a week
Every day
I don't know

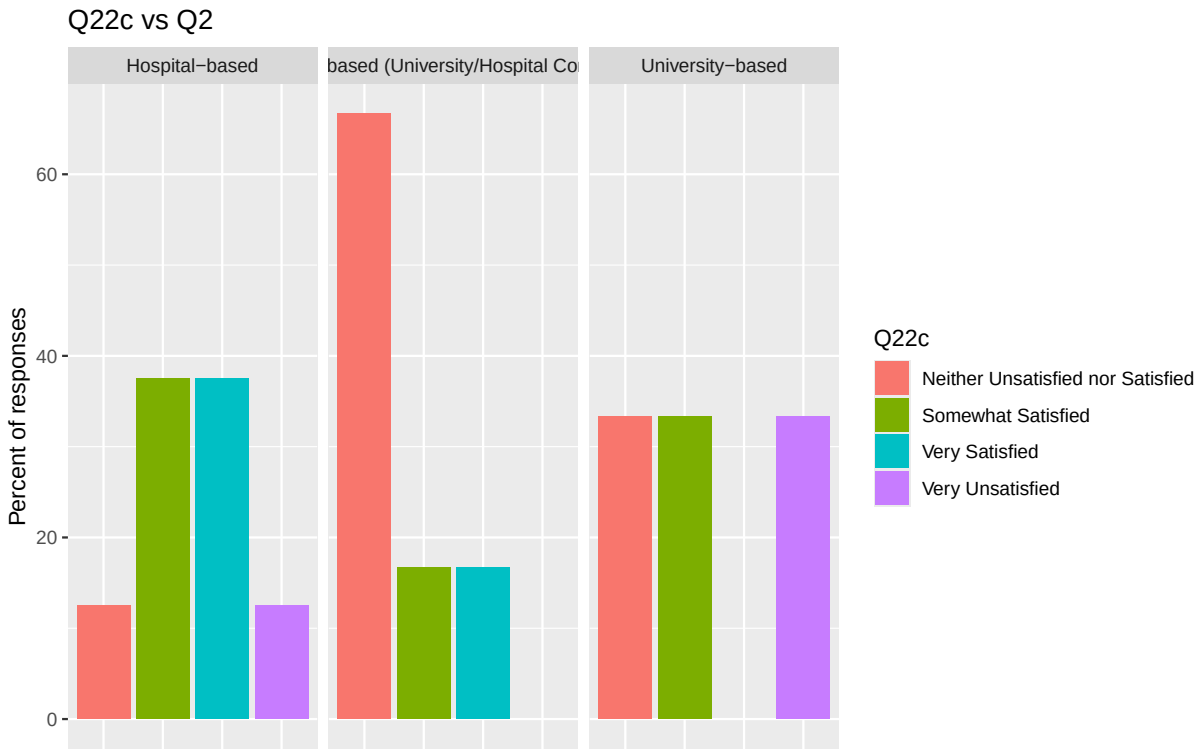

Q12h vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
60
40
20
0
Q12h
A few times in a month
A few times in a week
Every day
I don't know
Once per month or less
Once per week

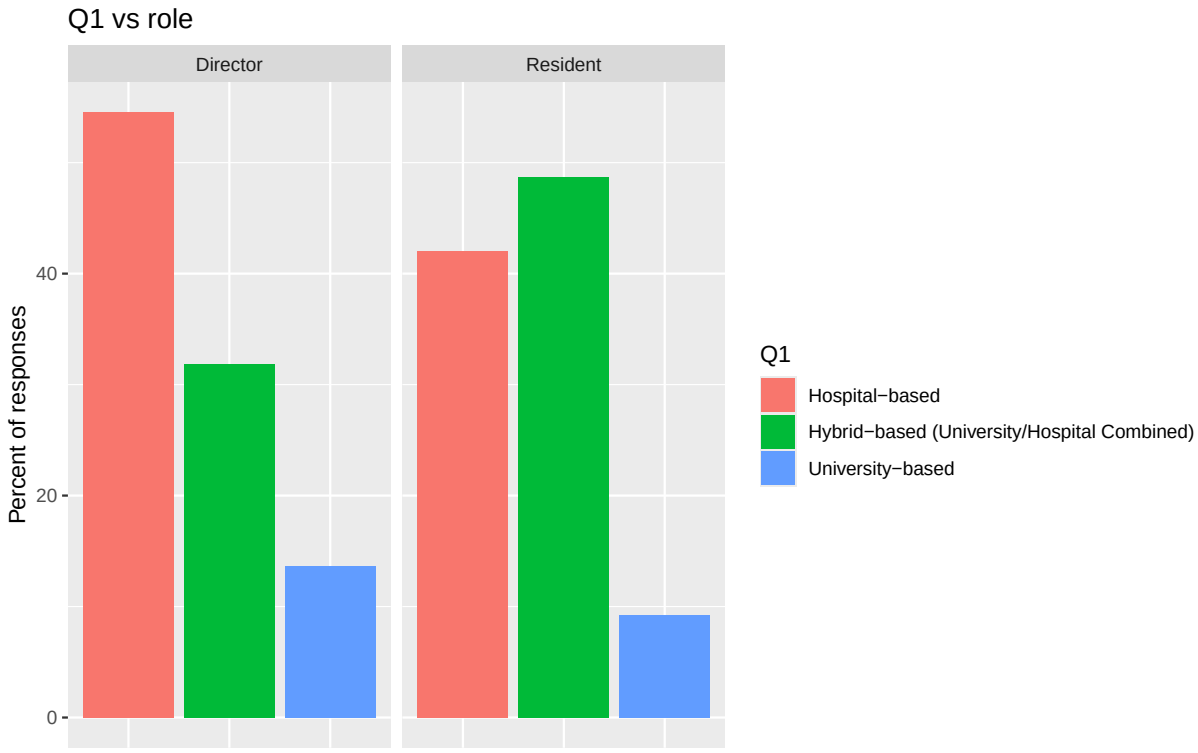

Q12i vs Q2
Hospital−based
−based (University/Hospital Com
University−based
Percent of responses
60
40
20
0
Q12i
A few times in a month
A few times in a week
Every day
I don't know
Once per week

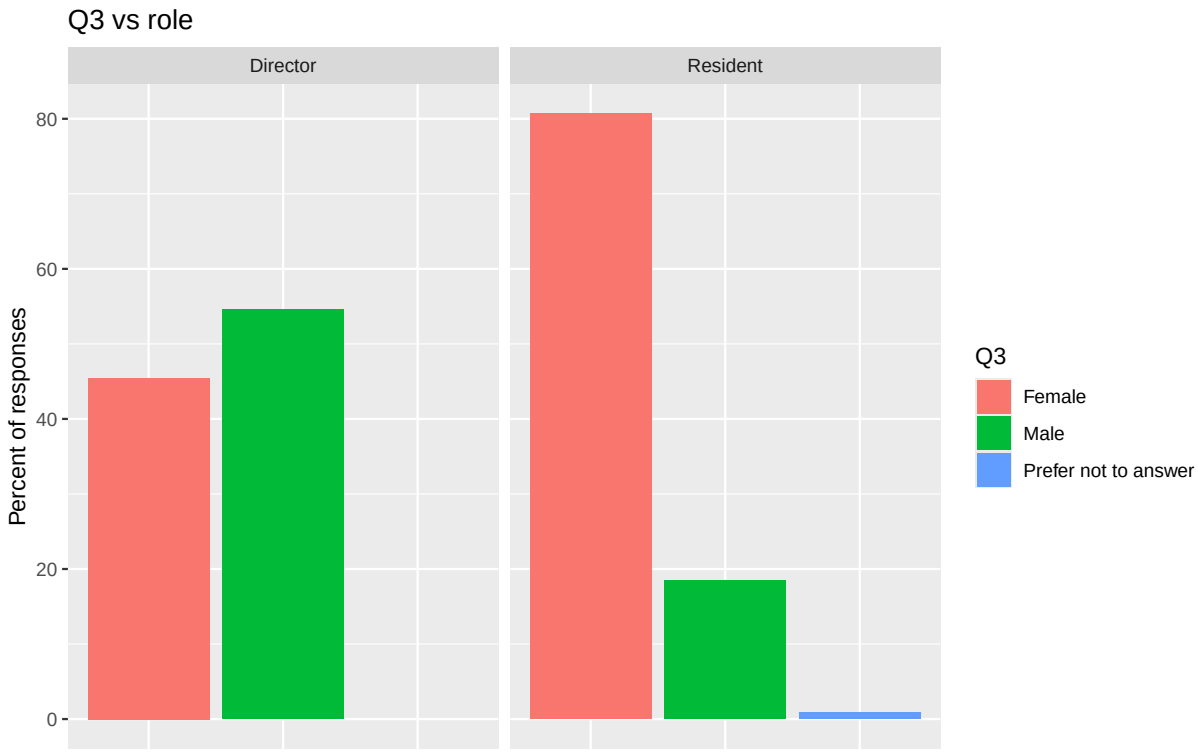

Q21 vs Q2
Hospital−based
rid−based (University/Hospital Combin
University−based
100
75
50
25
0
Percent of responses
Q21
No
Yes


Q22a vs Q2
Hospital−based
based (University/Hospital Co
University−based
30
20
10
0
Percent of responses
Q22a
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Somewhat Unsatisfied
Very Satisfied
Very Unsatisfied


Q22b vs Q2
Hospital−based
based (University/Hospital Co
University−based
Percent of responses
0
10
20
30
40
50
Q22b
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Somewhat Unsatisfied
Very Satisfied
Very Unsatisfied


Q22c vs Q2
Hospital−based
based (University/Hospital Co
University−based
Percent of responses
0
20
40
60
Q22c
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Very Satisfied
Very Unsatisfied


Q1 vs role
Director
Resident
Percent of responses
0
20
40
Q1
Hospital−based
Hybrid−based (University/Hospital Combined)
University−based


Q3 vs role
Director
Resident
Percent of responses
0
20
40
60
80
Q3
Female
Male
Prefer not to answer

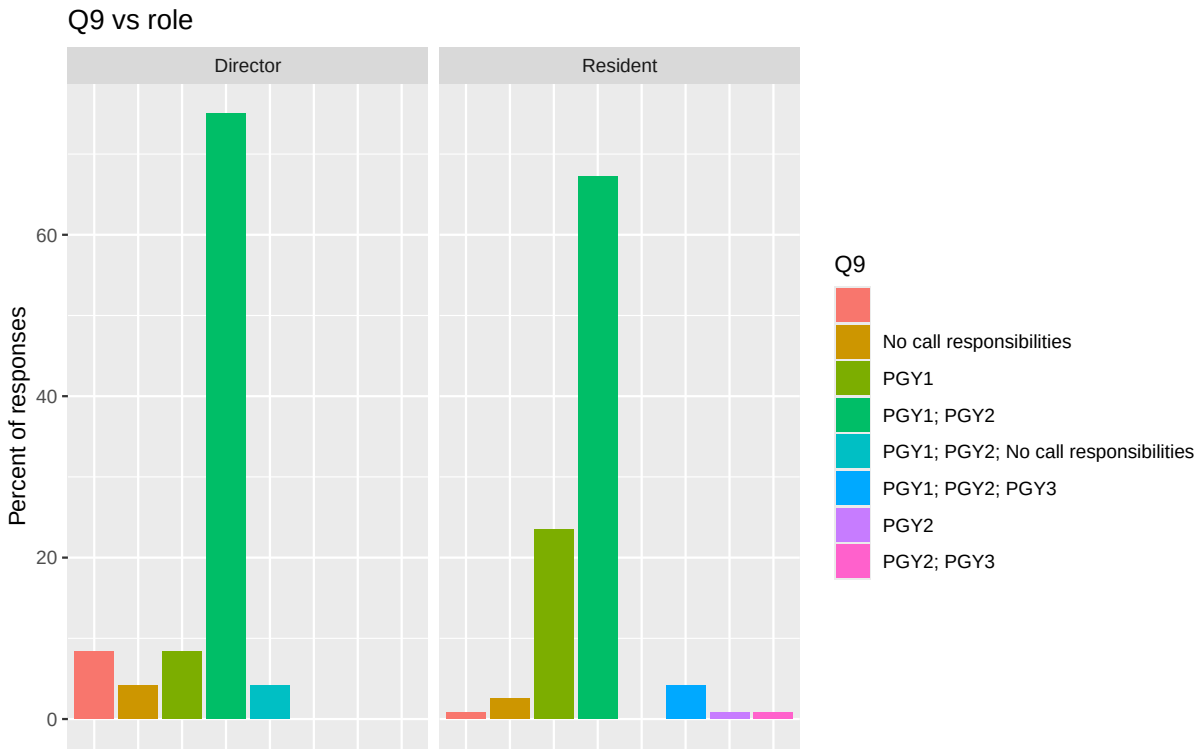
ebraska, North Dakota, Ohio, South Dakota, Wisconsin, and the Canadian provinces of Manitoba and Ontario)

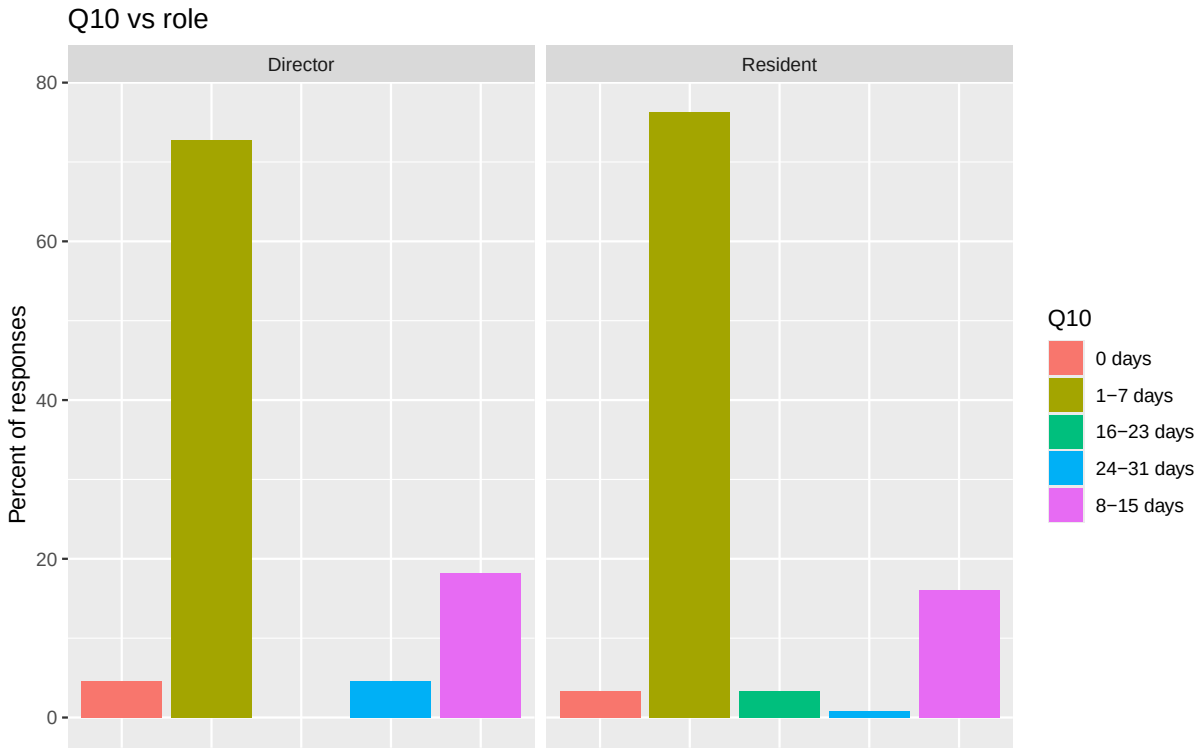
ıe, Maryland, Massachusetts, New Hampshire, New Jersey, New York, Pennsylvania, Rhode Island, Vermont, and the Canadian provinces of New

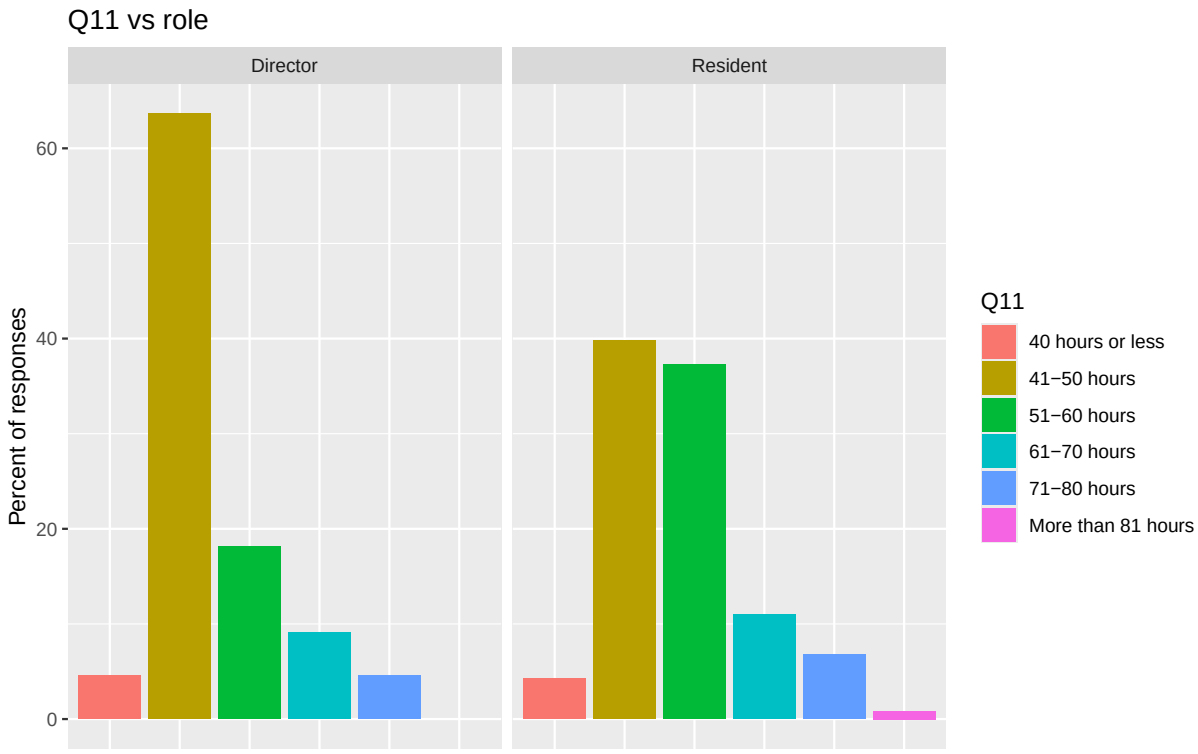
i, North Carolina, South Carolina, Tennessee, Virginia, West Virginia, the Commonwealth of Puerto Rico, and the U.S. Virgin Islands)

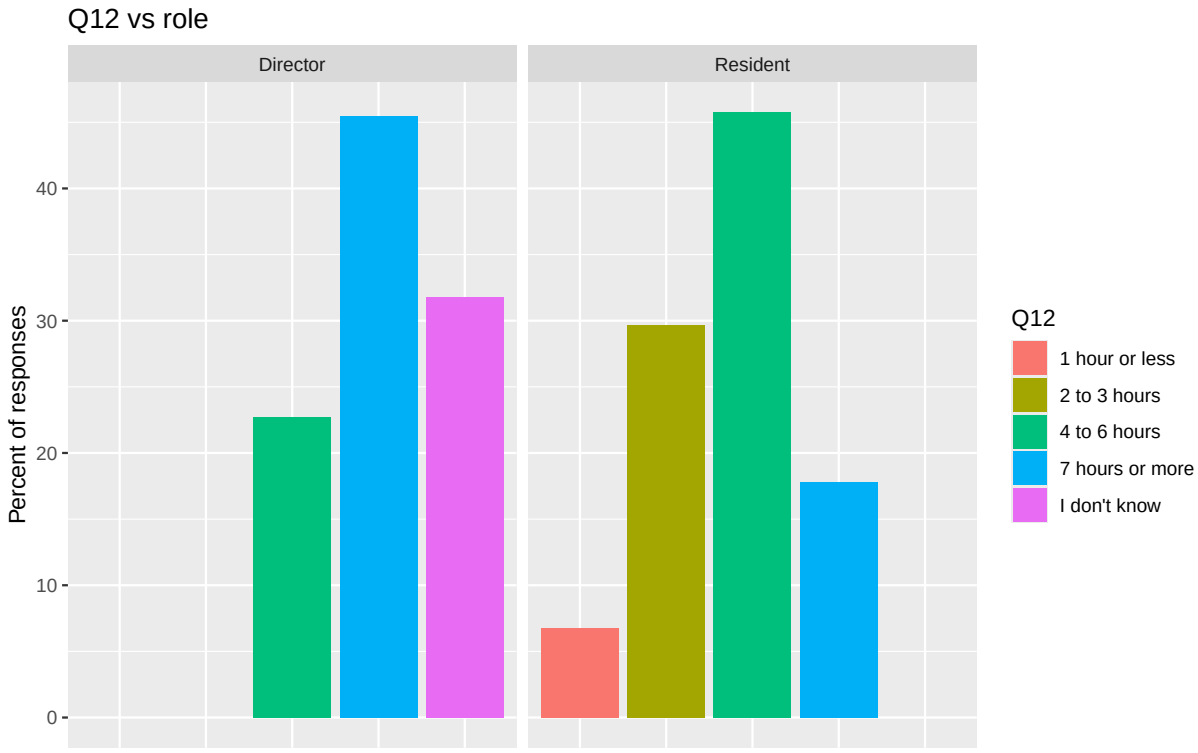
sas, Louisiana, Texas and Mexico

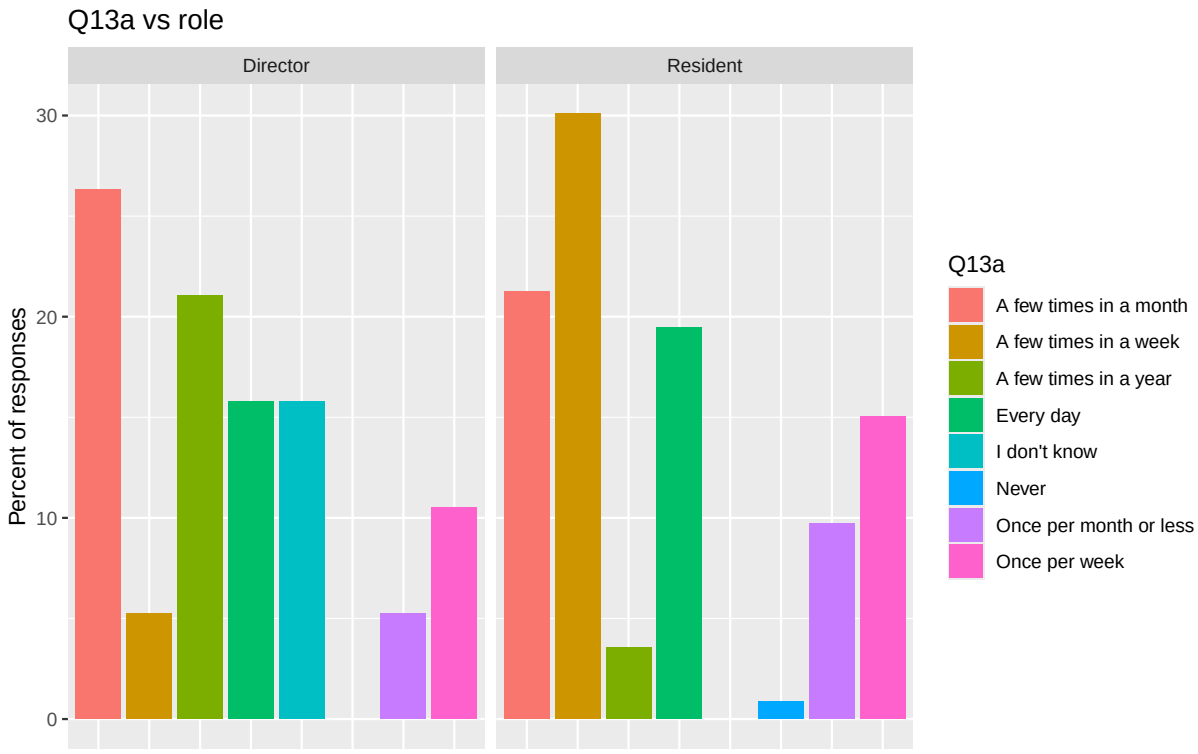
Nevada, Oregon, Utah, Washington, Wyoming, and the Canadian provinces of Saskatchewan, Alberta, British Columbia, Northwest Territories, Nu

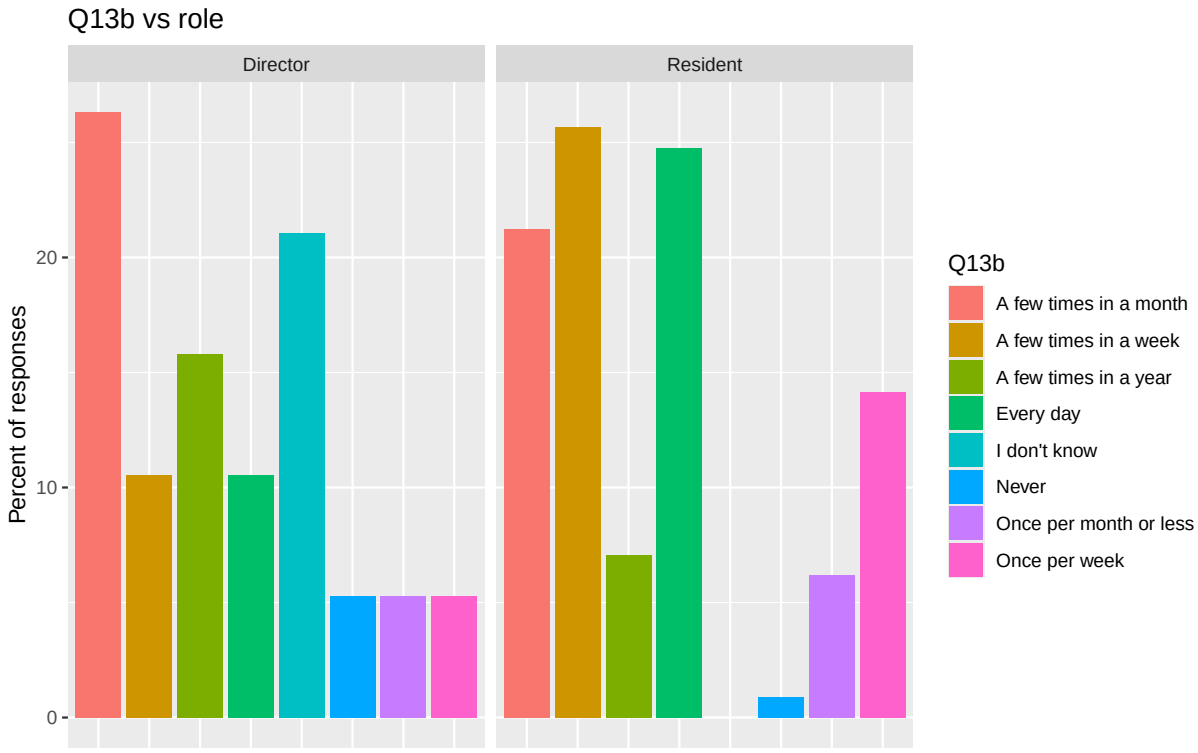


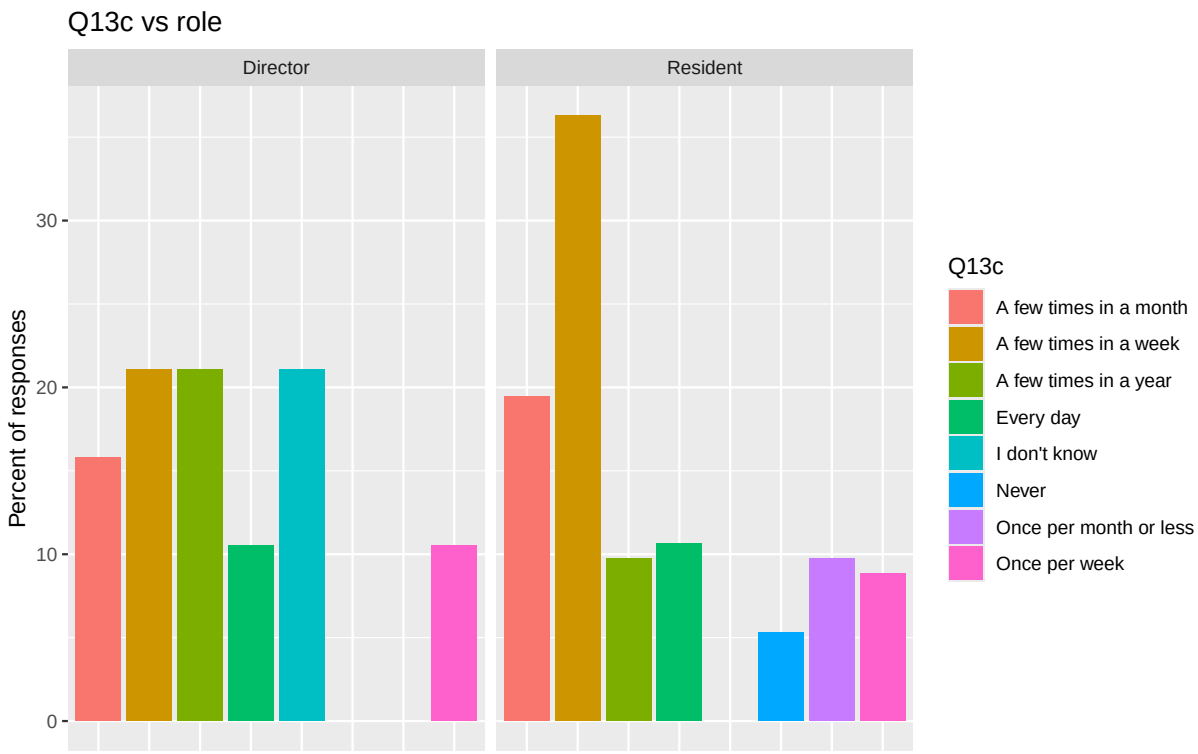

Q10 vs role
Director
Resident
Percent of responses
80
60
40
20
0
Q10
0 days
1−7 days
16−23 days
24−31 days
8−15 days

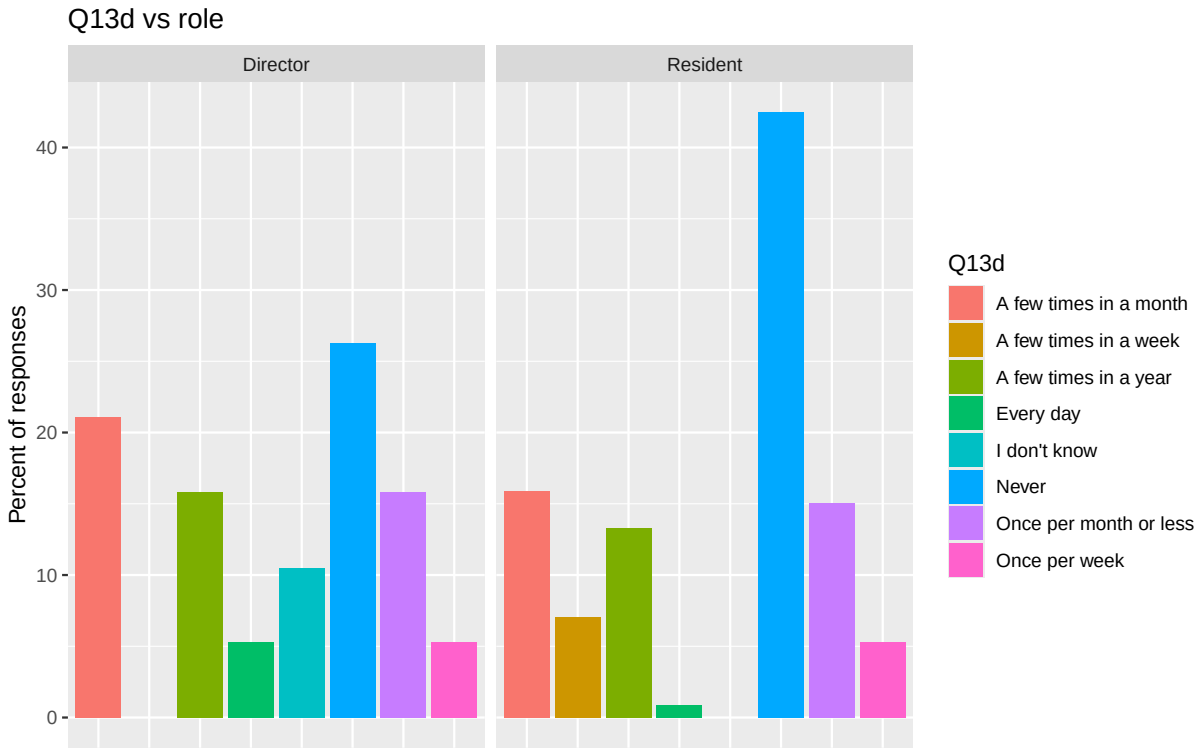

Q11 vs role
Director
Resident
Percent of responses
60
40
20
0
Q11
40 hours or less
41−50 hours
51−60 hours
61−70 hours
71−80 hours
More than 81 hours

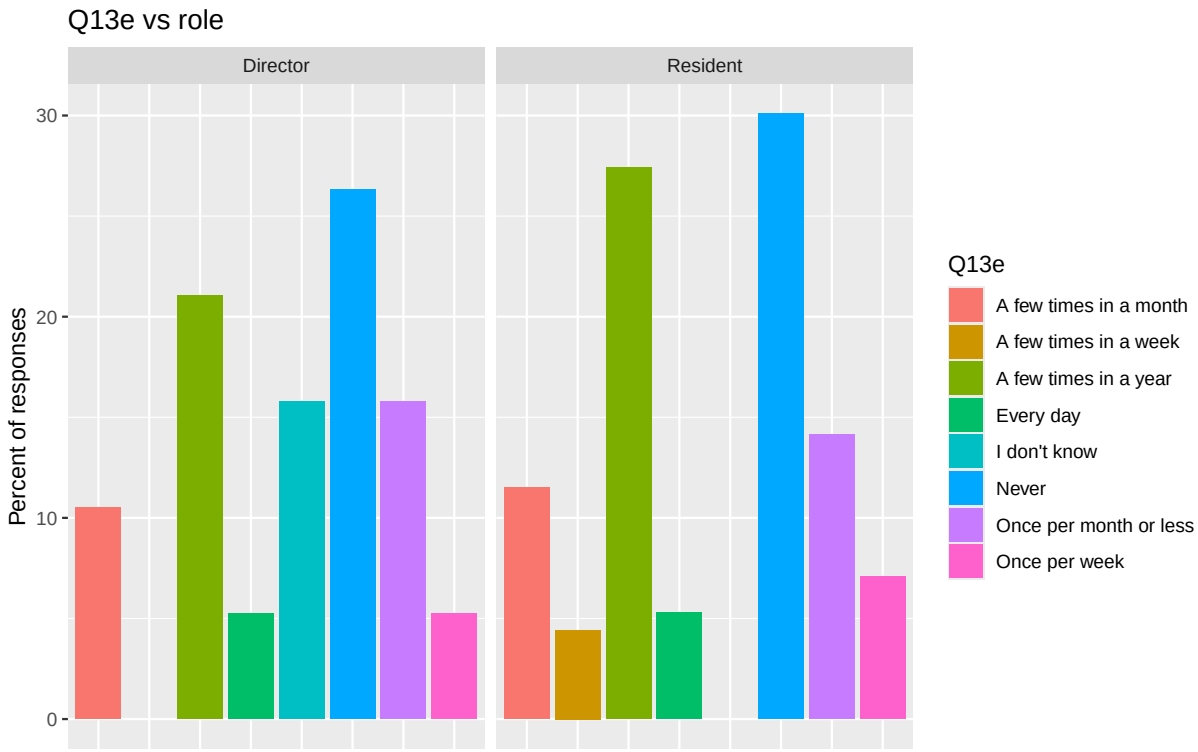

Q12 vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q12
1 hour or less
2 to 3 hours
4 to 6 hours
7 hours or more
I don't know

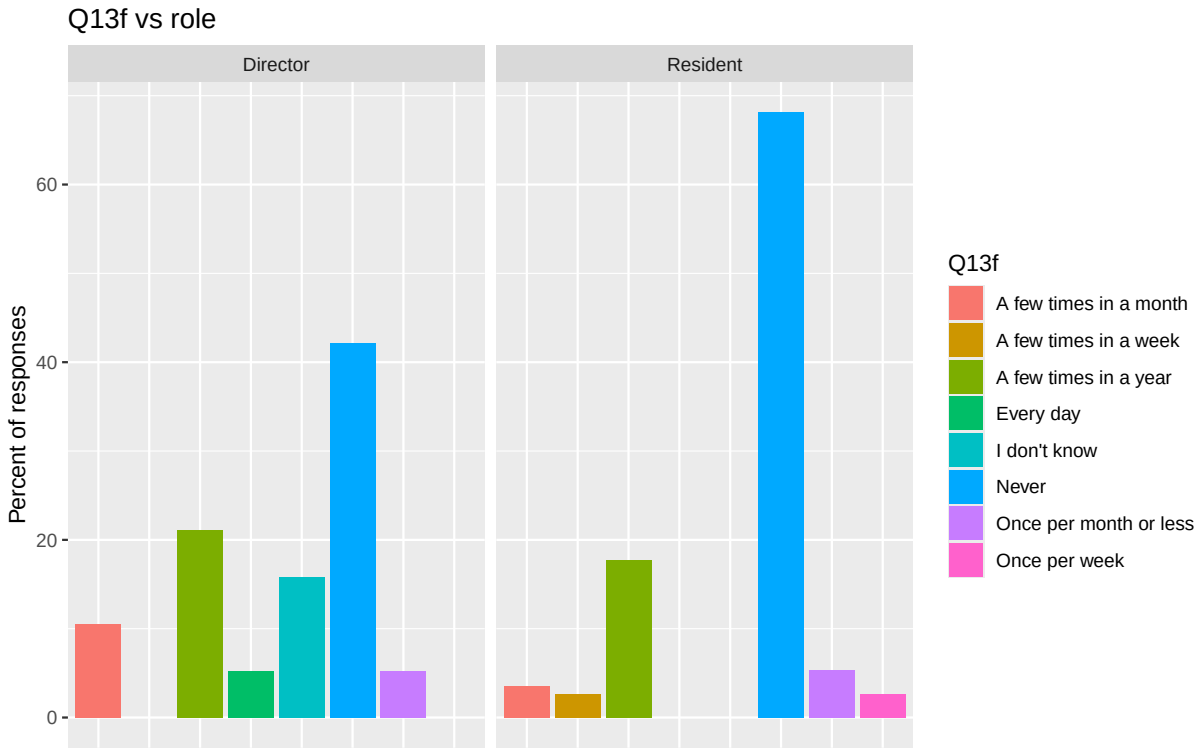

Q13a vs role
Director
Resident
Percent of responses
0
10
20
30
Q13a
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

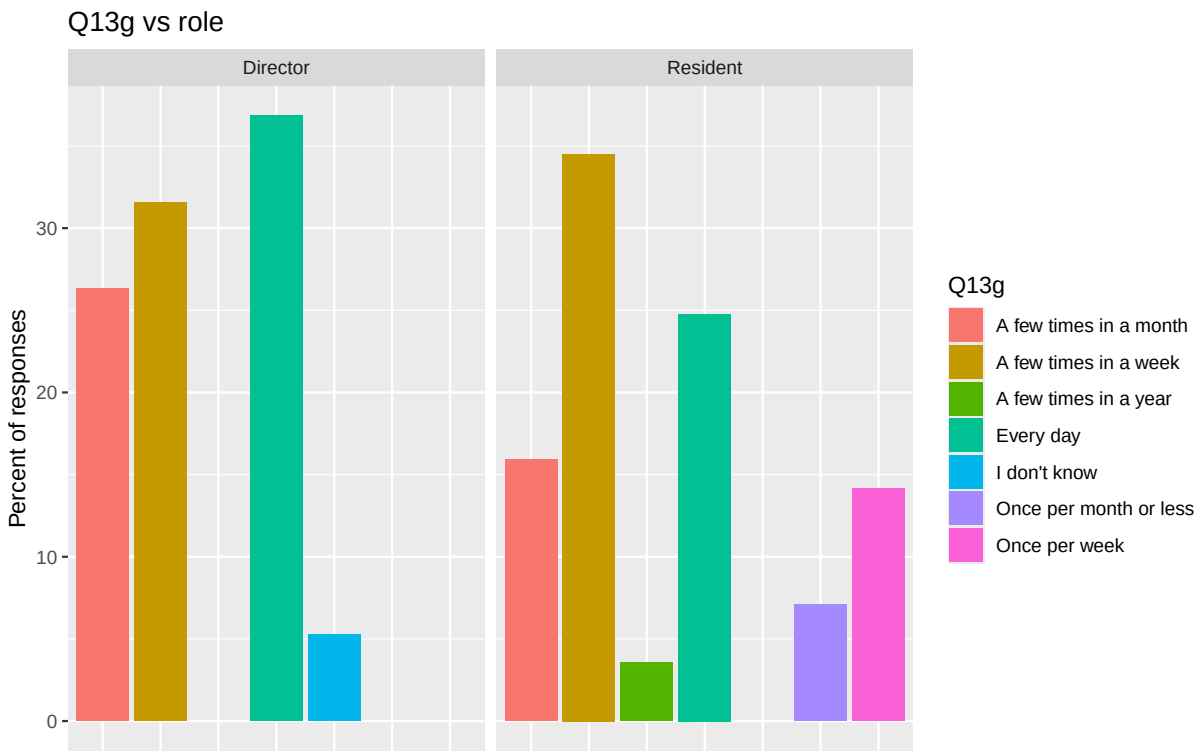

Q13b vs role
Director
Resident
Percent of responses
20
10
0
Q13b
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

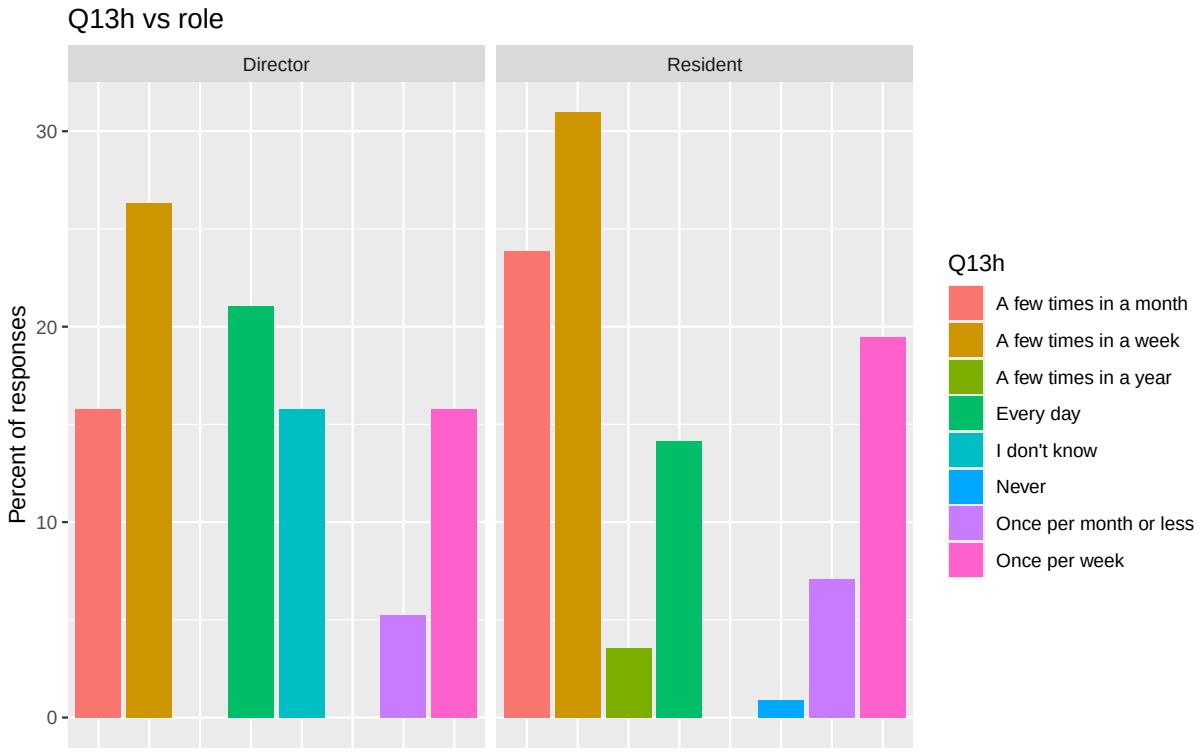

Q13c vs role
Director
Resident
Percent of responses
30
20
10
0
Q13c
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

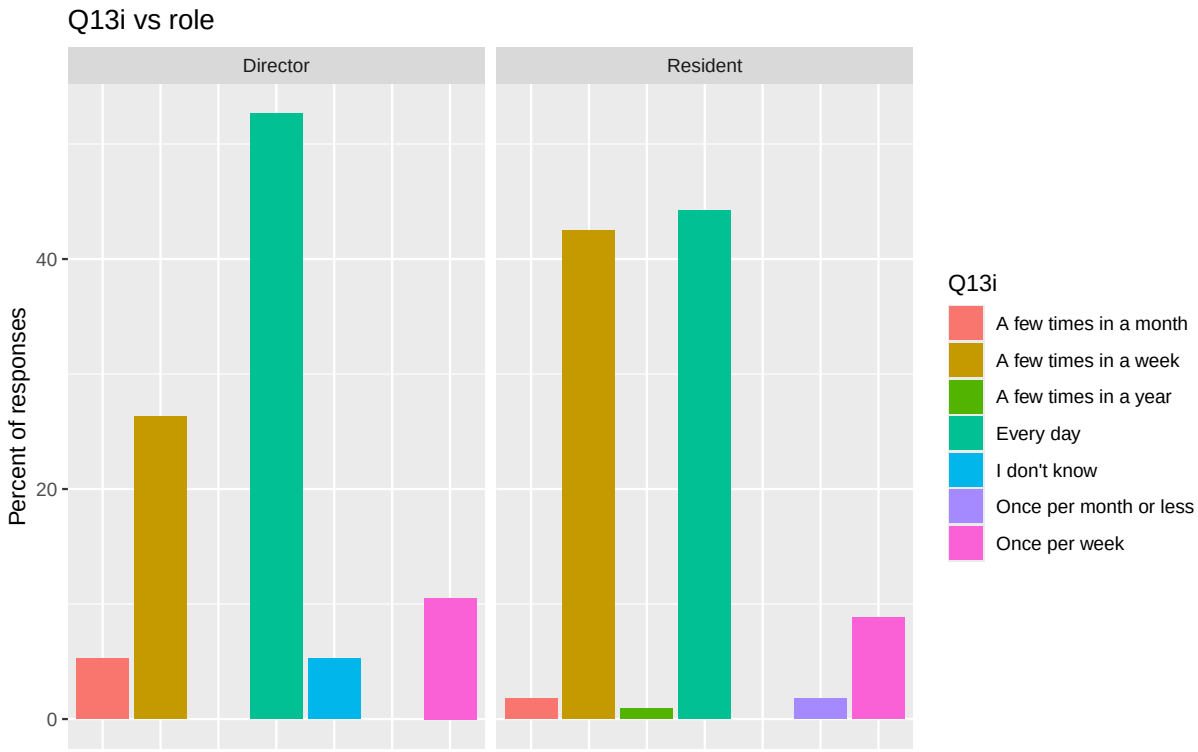

Q13d vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q13d
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

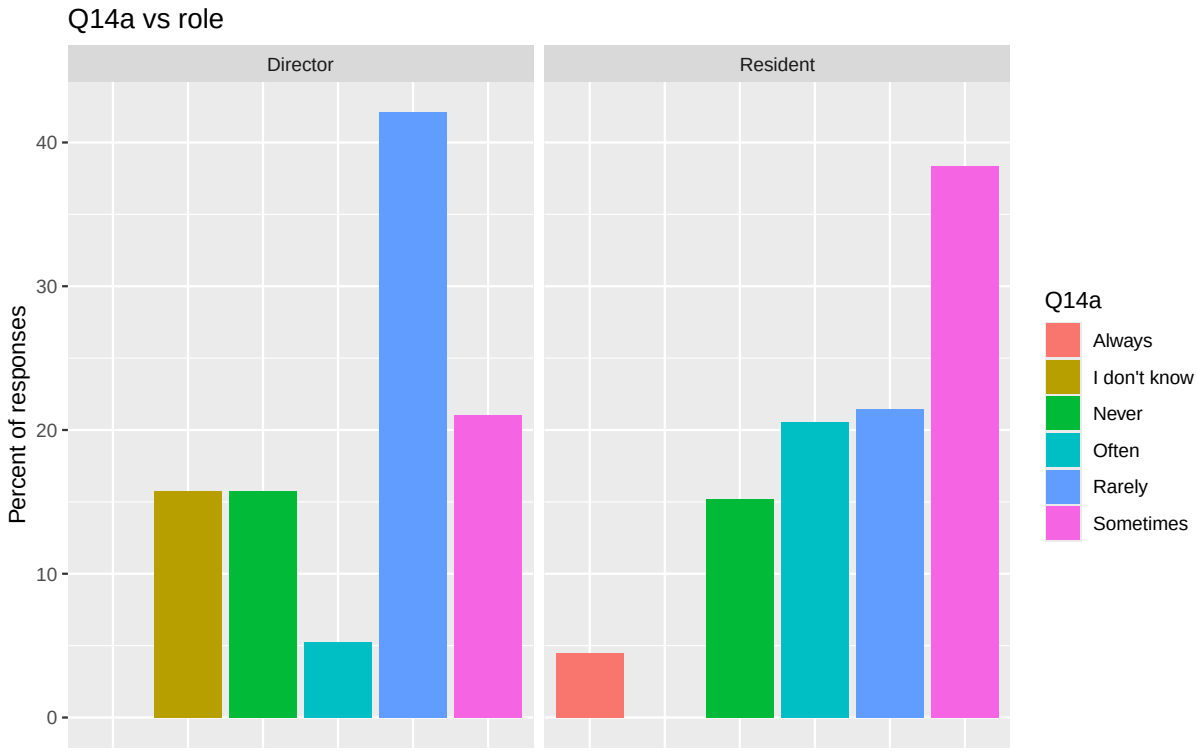

Q13e vs role
Director
Resident
Percent of responses
0
10
20
30
Q13e
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

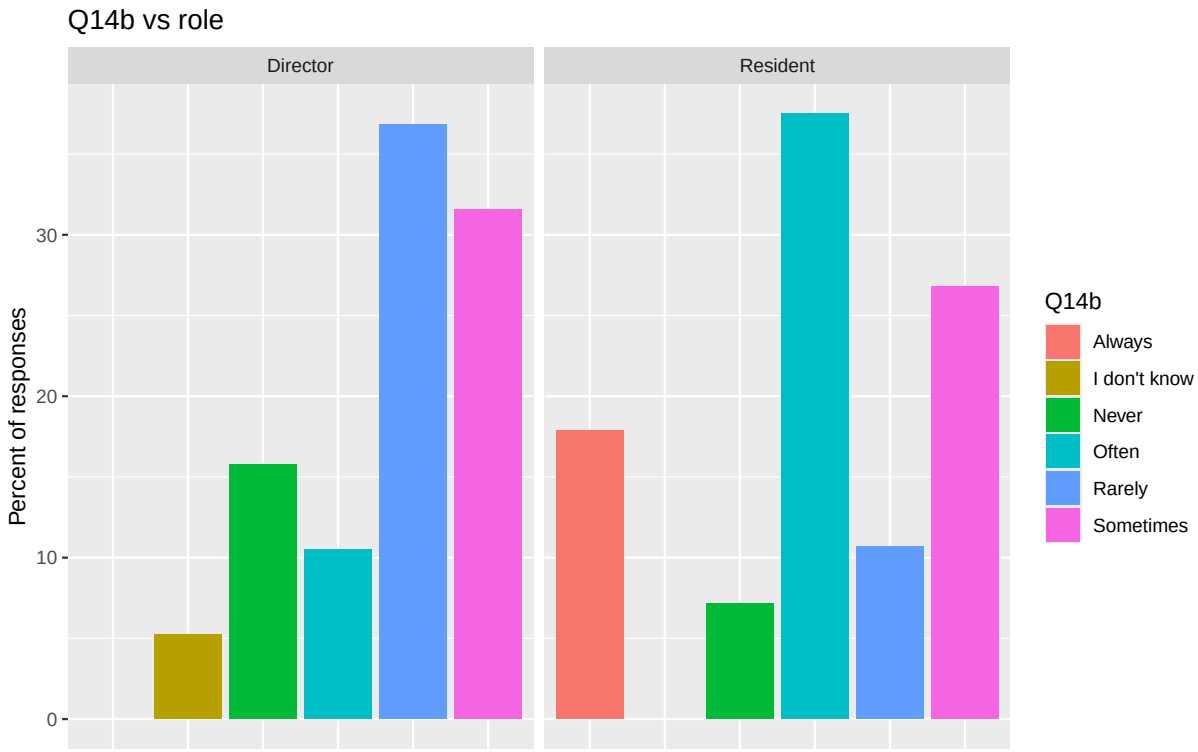

Q13f vs role
Director
Resident
Percent of responses
0
20
40
60
Q13f
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

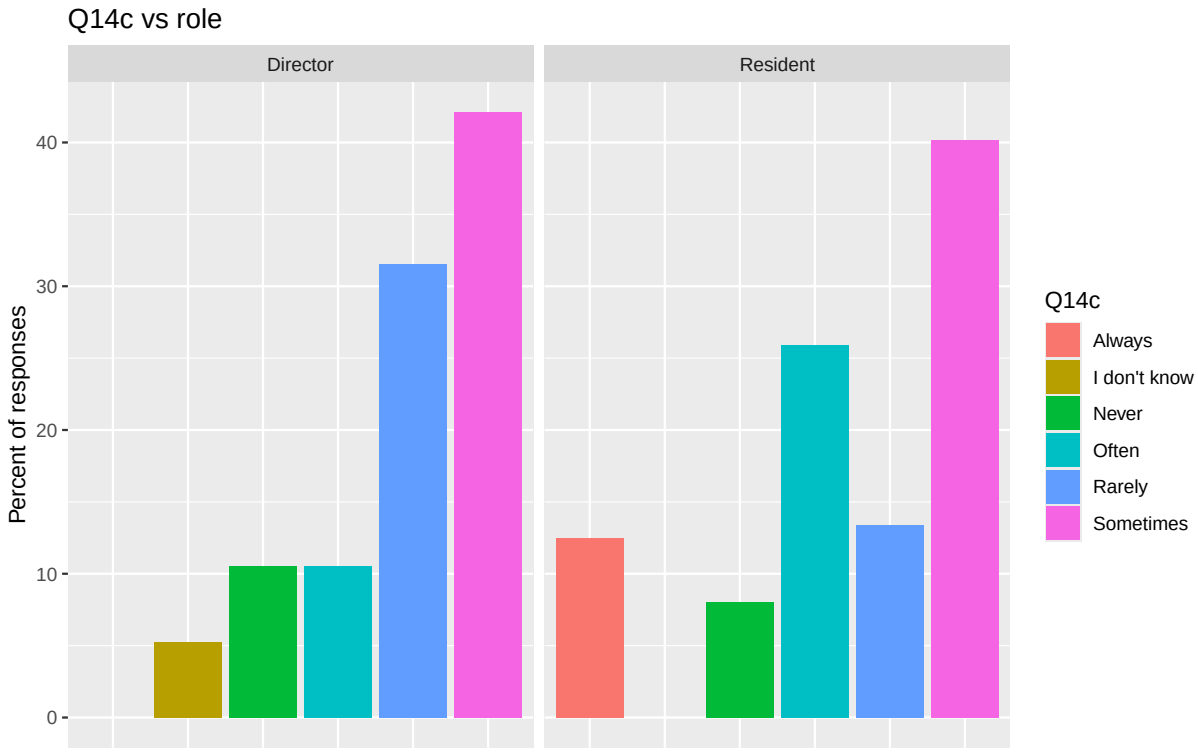

Q13g vs role
Director
Resident
Percent of responses
0
10
20
30
Q13g
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Once per month or less
Once per week

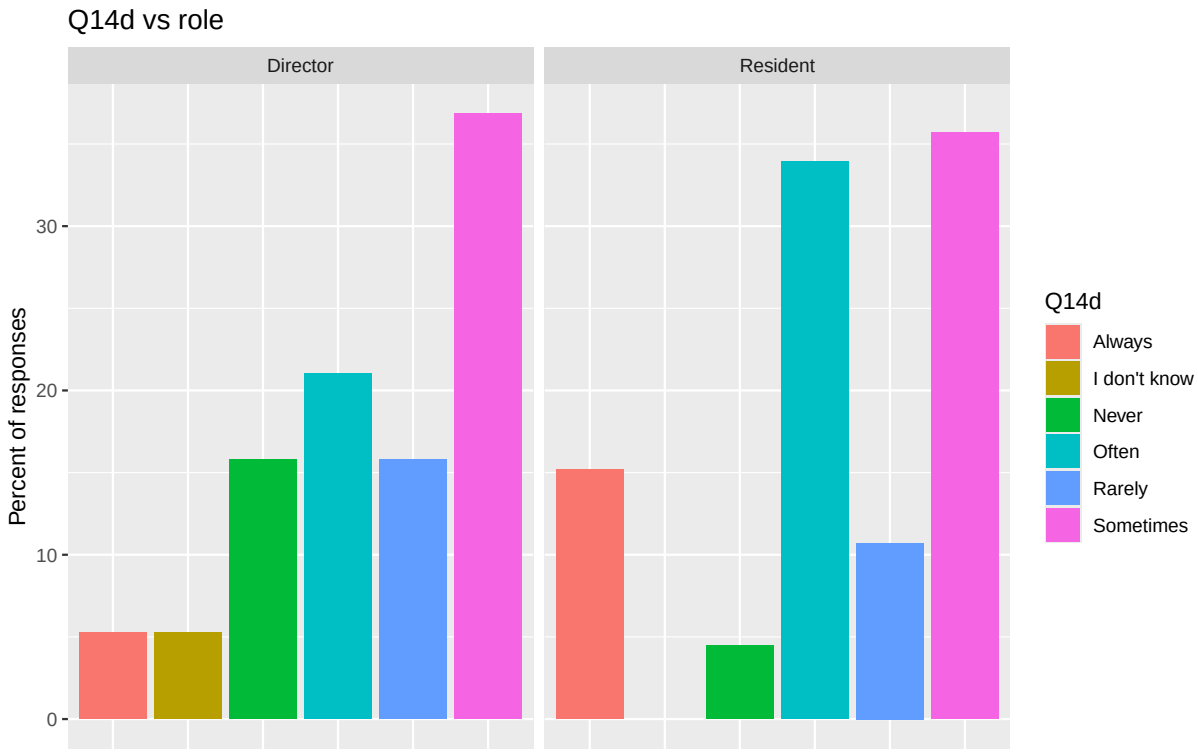

Q13h vs role
Director
Resident
Percent of responses
30
20
10
0
Q13h
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Never
Once per month or less
Once per week

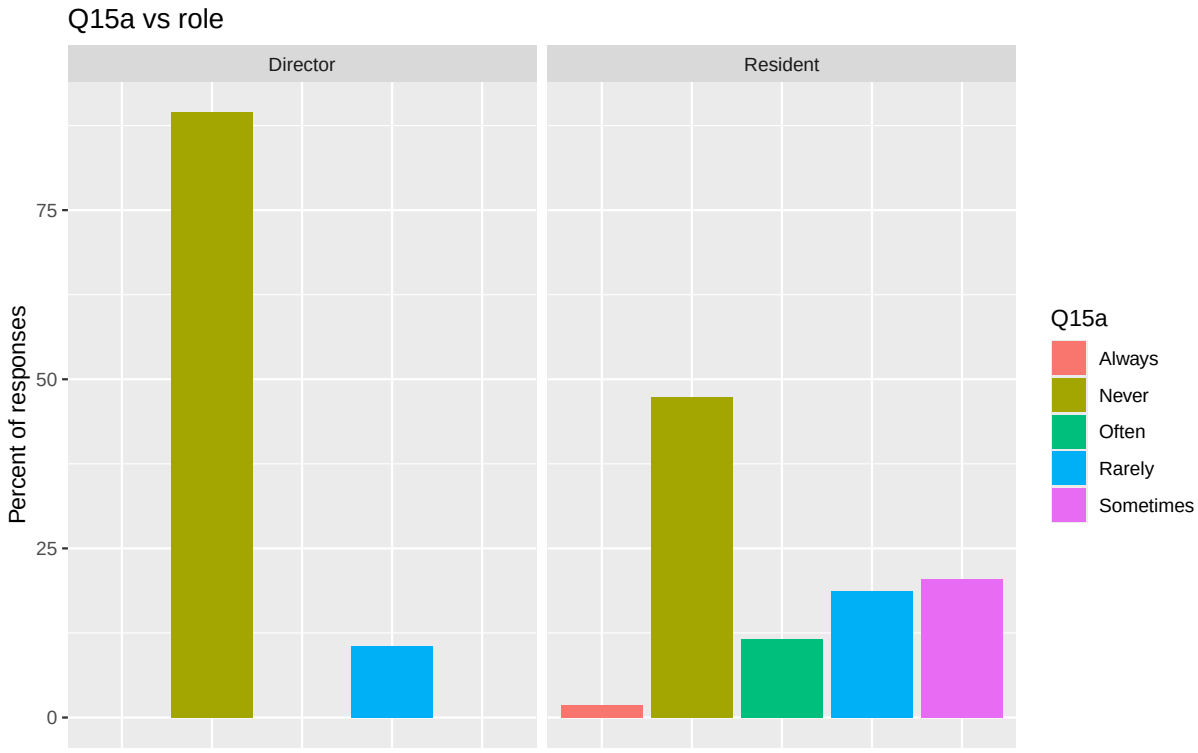

Q13i vs role
Director
Resident
Percent of responses
40
20
0
Q13i
A few times in a month
A few times in a week
A few times in a year
Every day
I don't know
Once per month or less
Once per week

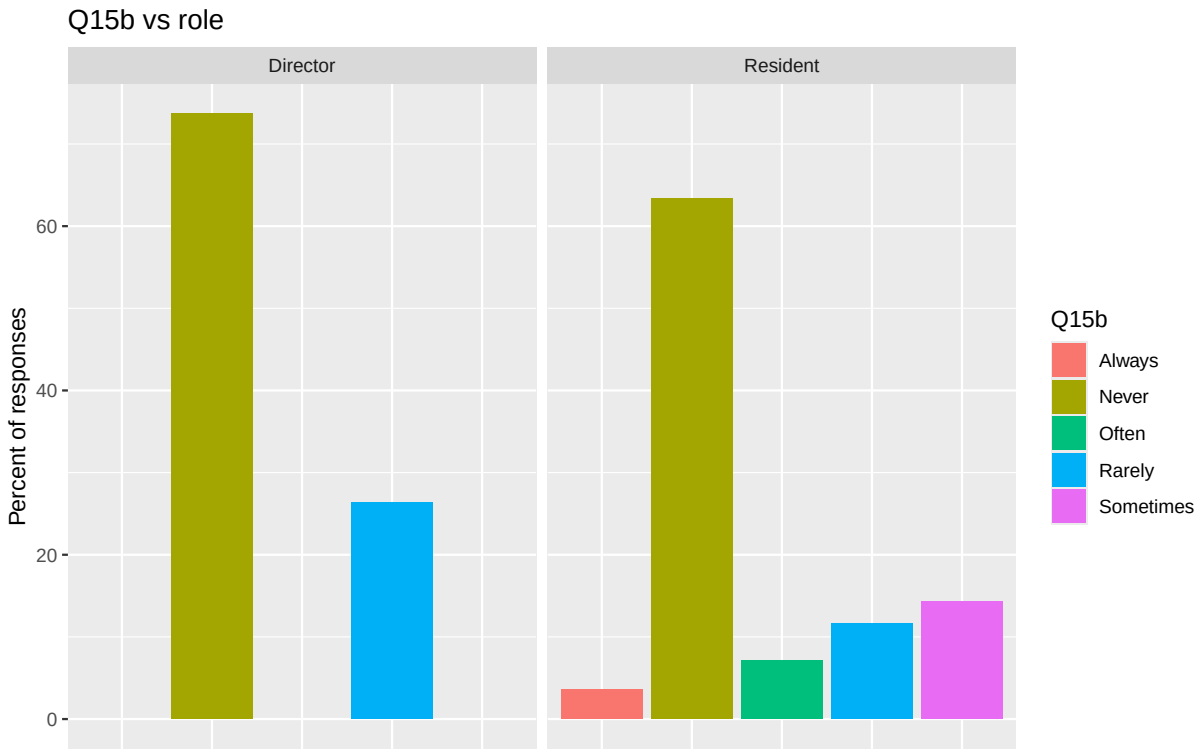

Q14a vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q14a
Always
I don't know
Never
Often
Rarely
Sometimes

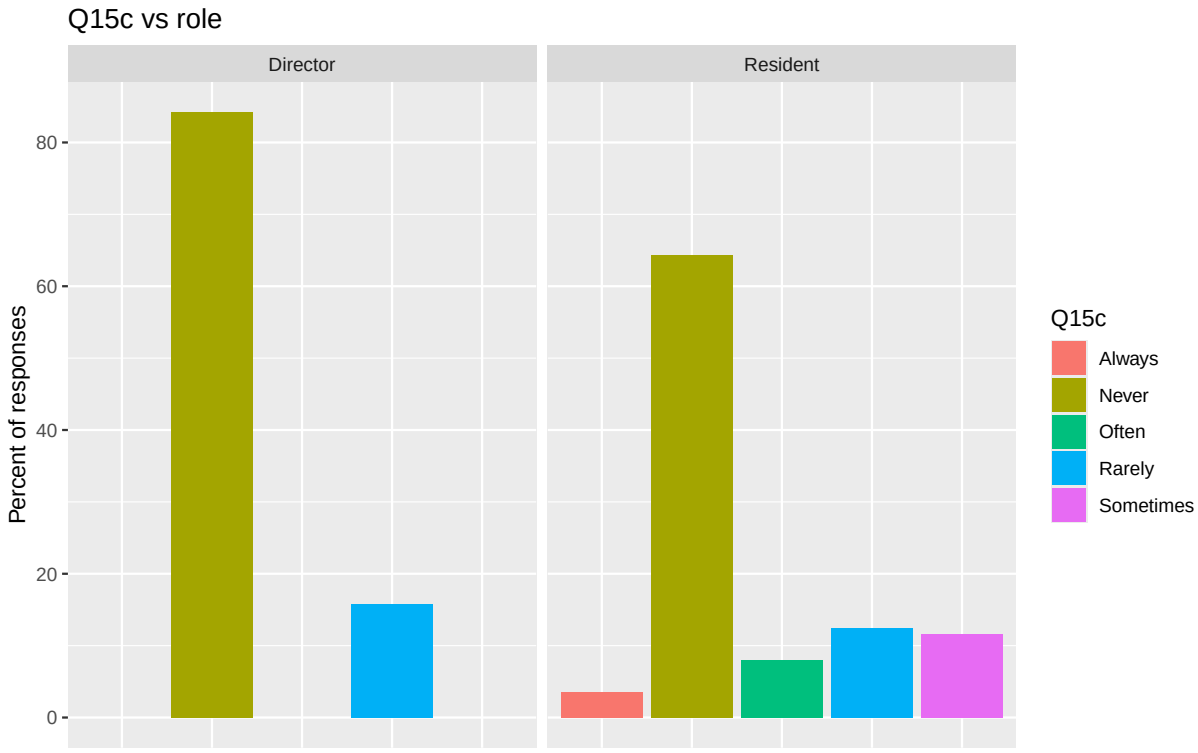

Q14b vs role
Director
Resident
Percent of responses
0
10
20
30
Q14b
Always
I don't know
Never
Often
Rarely
Sometimes

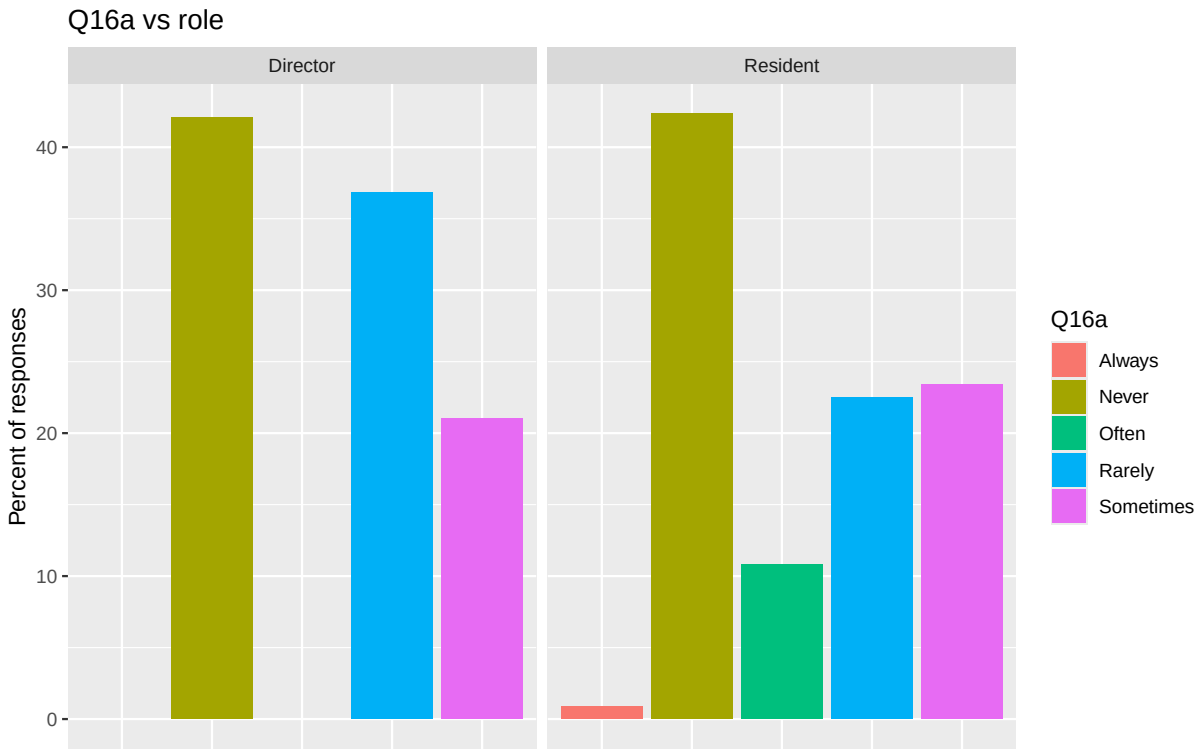

Q14c vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q14c
Always
I don't know
Never
Often
Rarely
Sometimes

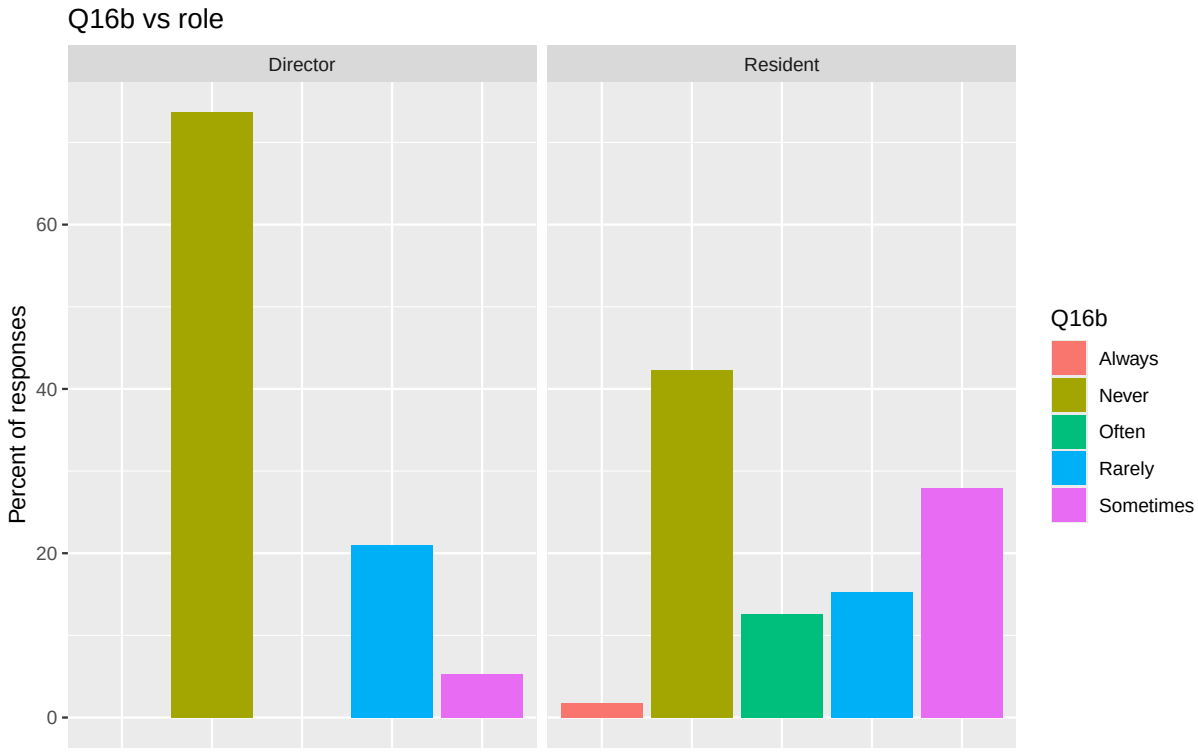

Q14d vs role
Director
Resident
Percent of responses
0
10
20
30
Q14d
Always
I don't know
Never
Often
Rarely
Sometimes

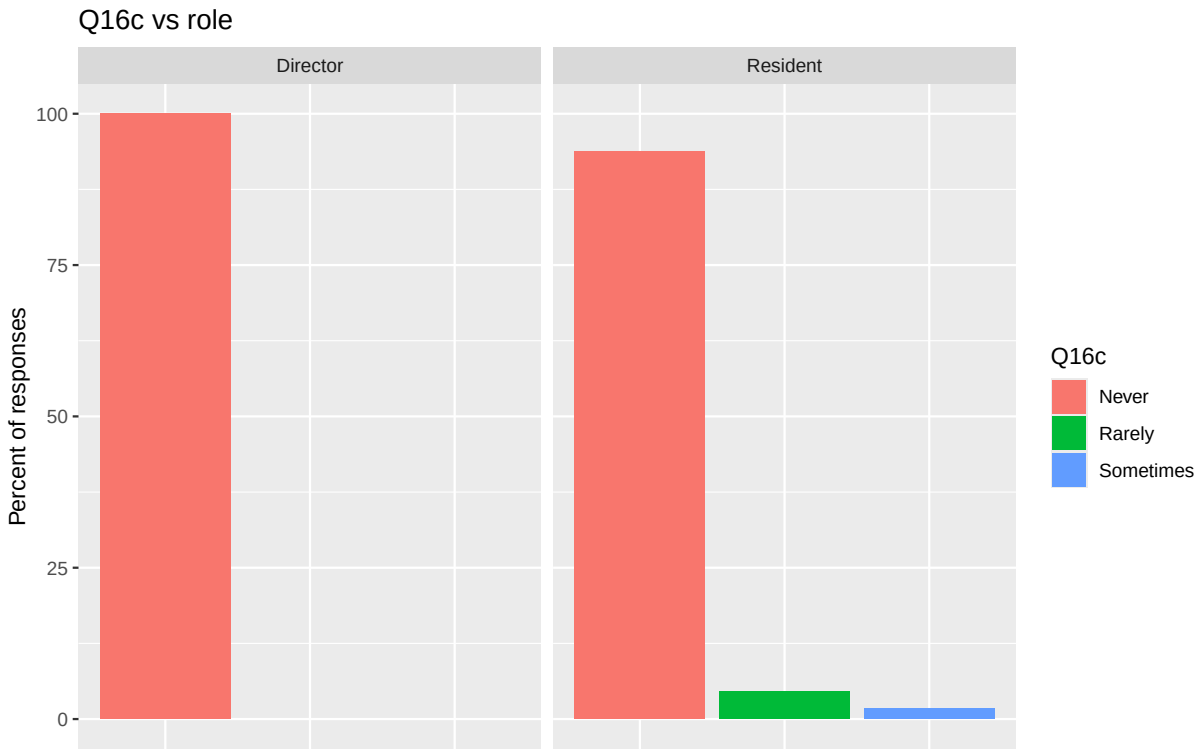

Q15a vs role
Director
Resident
Percent of responses
75
50
25
0
Q15a
Always
Never
Often
Rarely
Sometimes

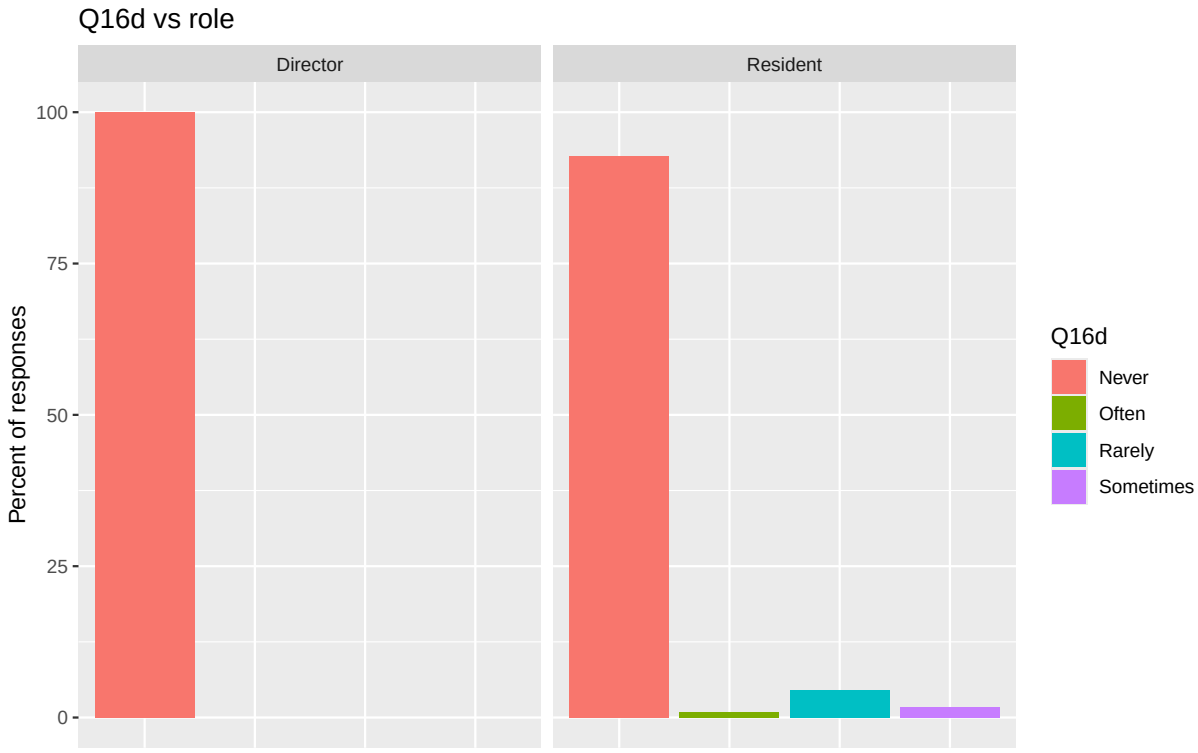

Q15b vs role
Director
Resident
Percent of responses
60
40
20
0
Q15b
Always
Never
Often
Rarely
Sometimes

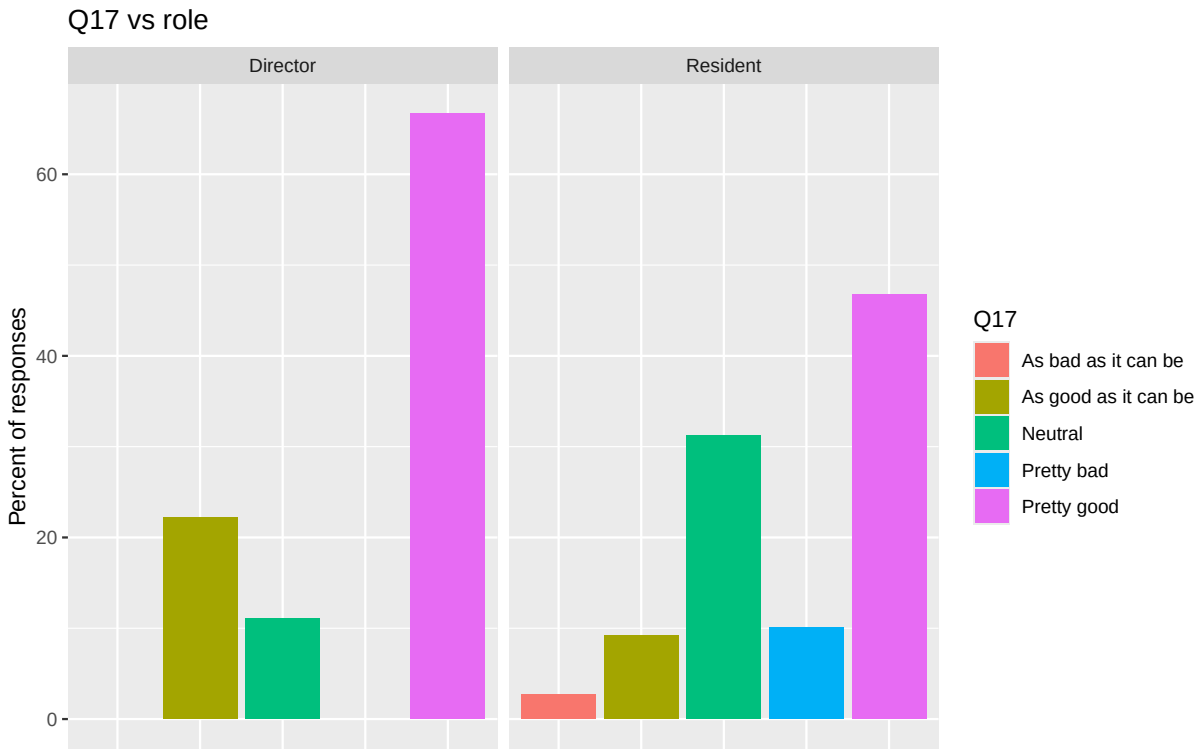

Q15c vs role
Director
Resident
Percent of responses
0
20
40
60
80
Q15c
Always
Never
Often
Rarely
Sometimes

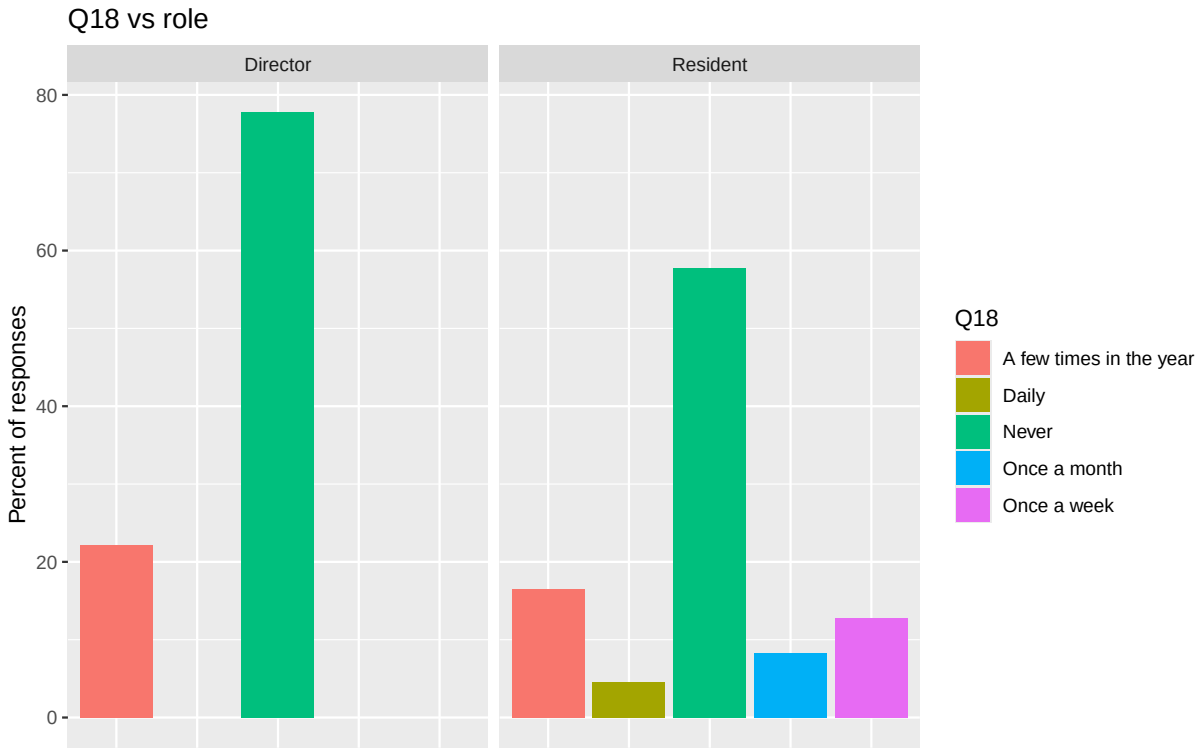

Q16a vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q16a
Always
Never
Often
Rarely
Sometimes

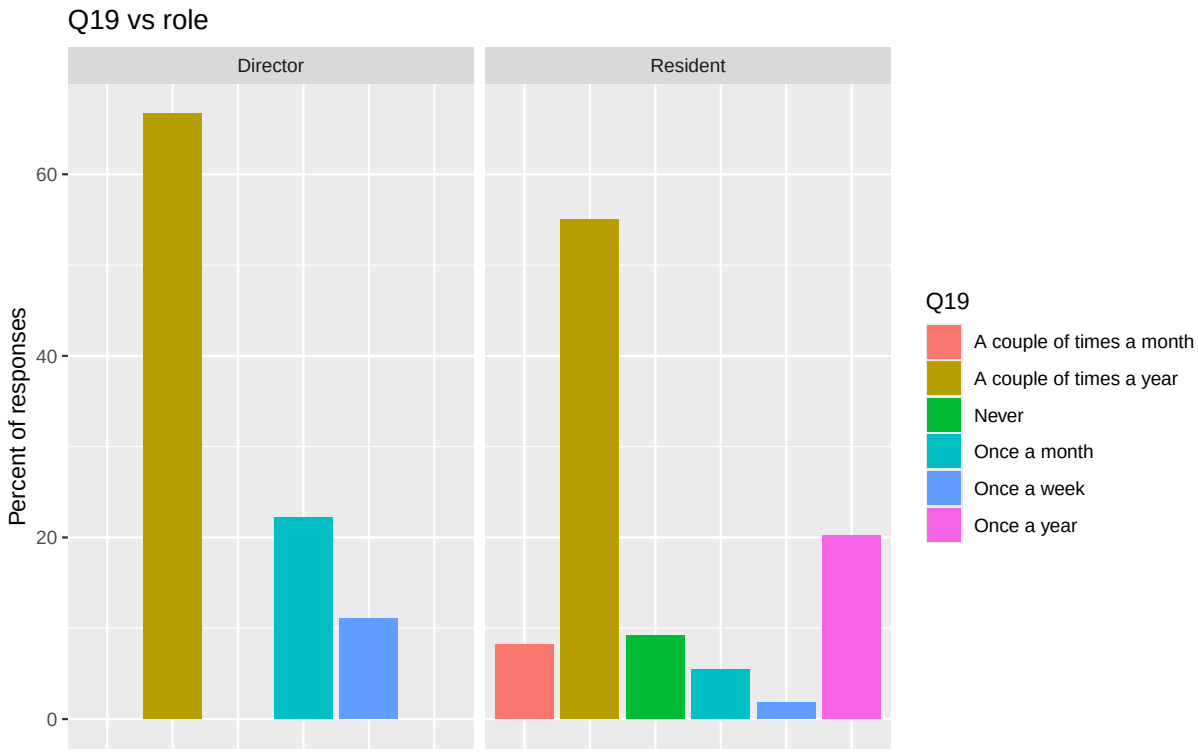

Q16b vs role
Director
Resident
Percent of responses
0
20
40
60
Q16b
Always
Never
Often
Rarely
Sometimes

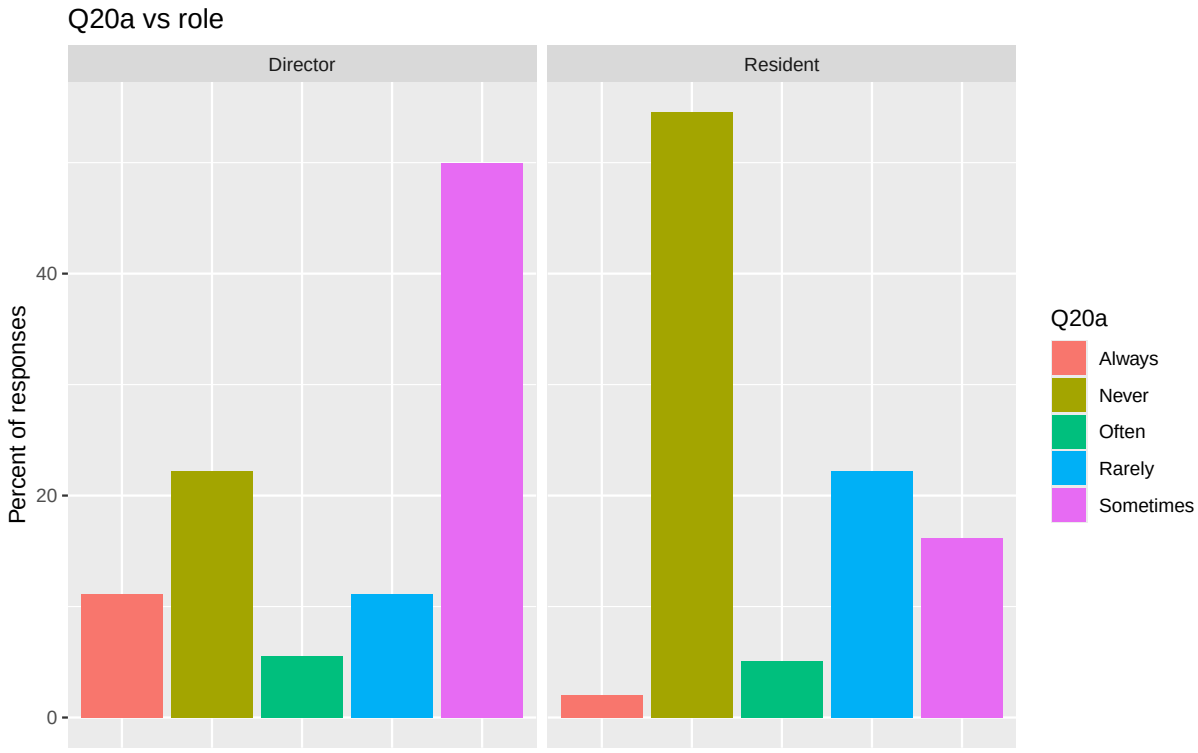

Q16c vs role
Director
Resident
Percent of responses
0
25
50
75
100
Q16c
Never
Rarely
Sometimes

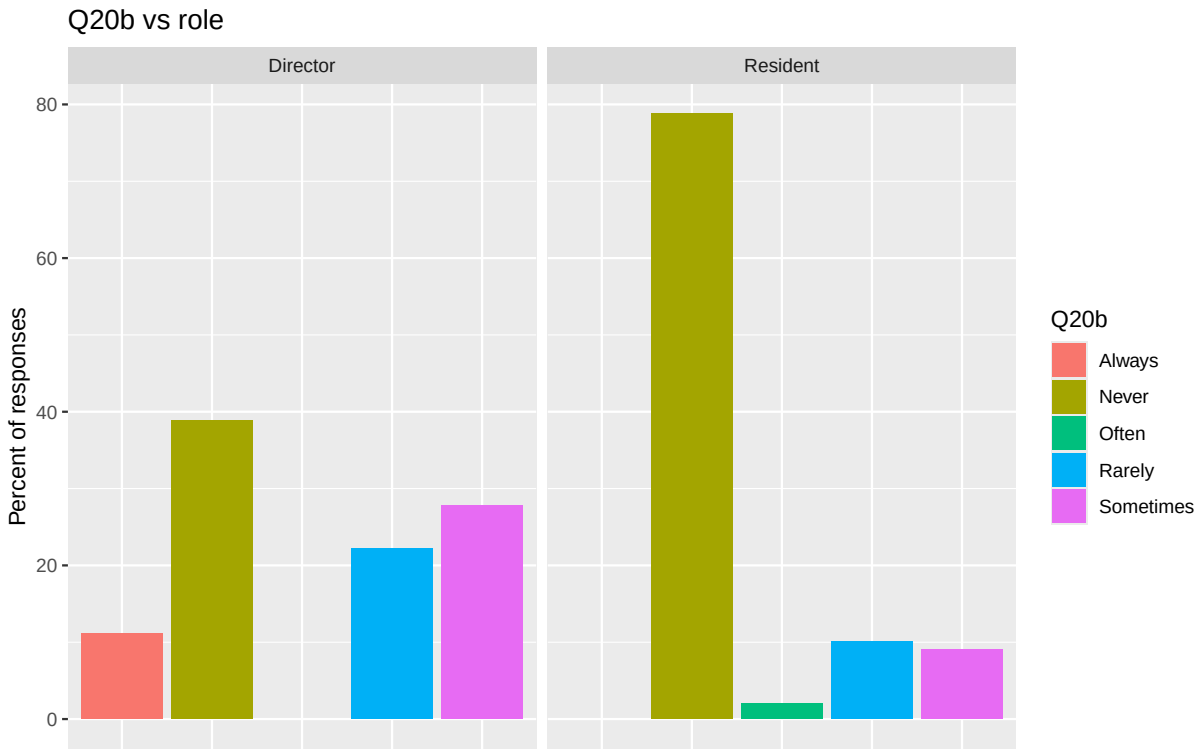

Q16d vs role
Director
Resident
Percent of responses
100
75
50
25
0
Q16d
Never
Often
Rarely
Sometimes

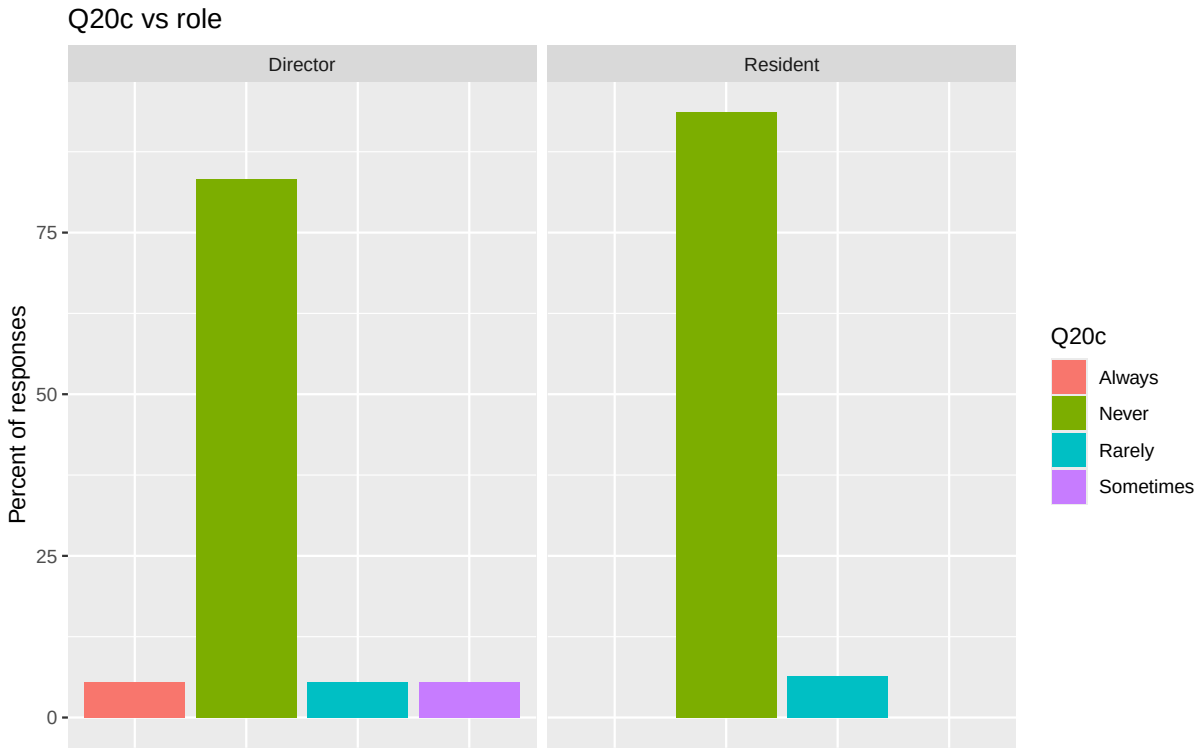

Q17 vs role
Director
Resident
Percent of responses
60
40
20
0
Q17
As bad as it can be
As good as it can be
Neutral
Pretty bad
Pretty good

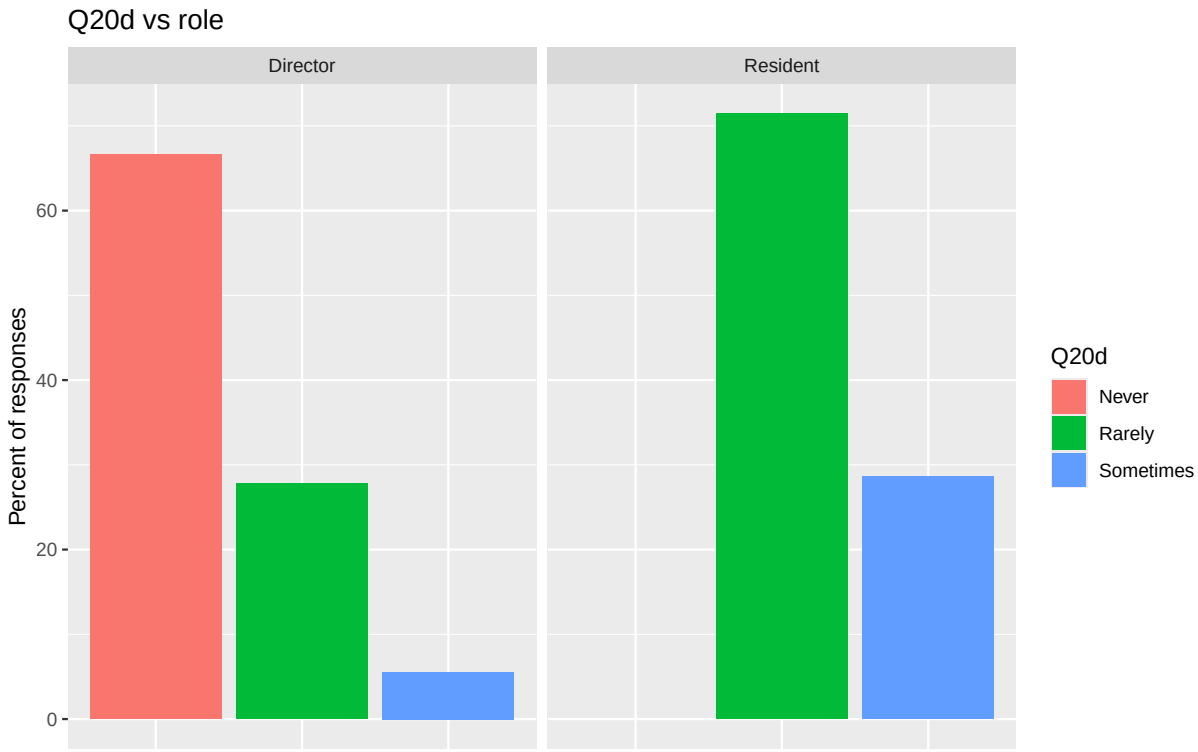

Q18 vs role
Director
Resident
Percent of responses
80
60
40
20
0
Q18
A few times in the year
Daily
Never
Once a month
Once a week


Q19 vs role
Director
Resident
Percent of responses
60
40
20
0
Q19
A couple of times a month
A couple of times a year
Never
Once a month
Once a week
Once a year

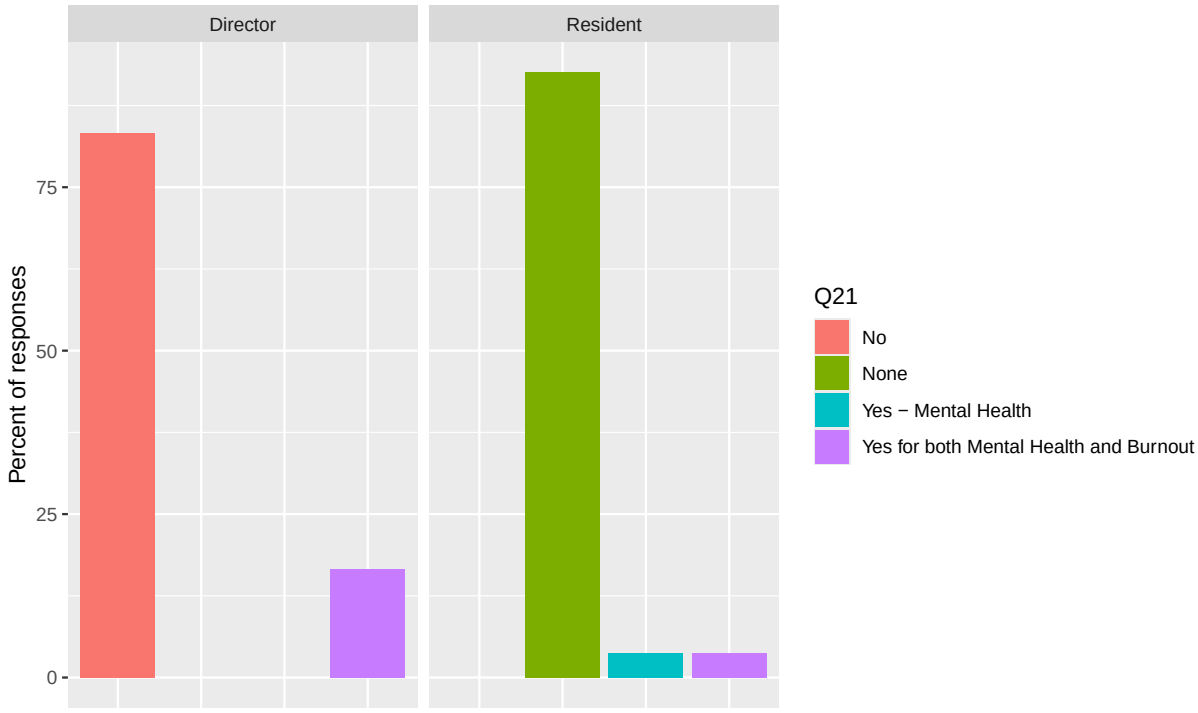

Q20a vs role
Director
Resident
Percent of responses
0
20
40
Q20a
Always
Never
Often
Rarely
Sometimes


Q20b vs role
Director
Resident
Percent of responses
0
20
40
60
80
Q20b
Always
Never
Often
Rarely
Sometimes

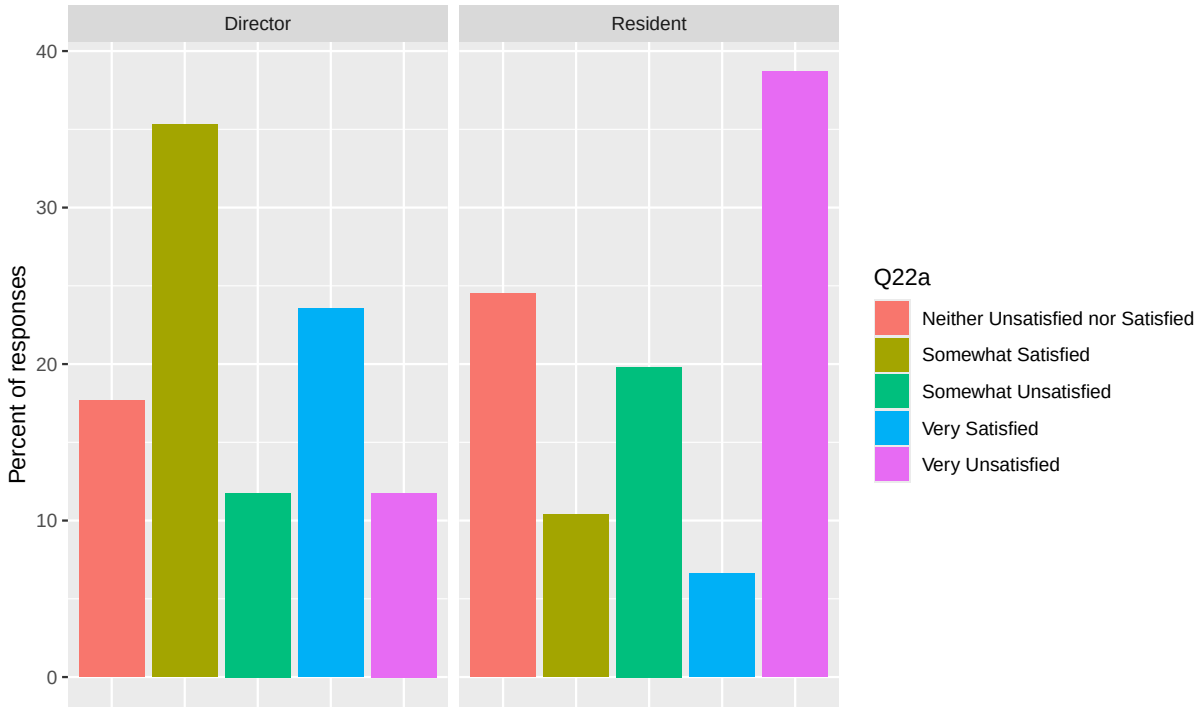

Q20c vs role
Director
Resident
Percent of responses
75
50
25
0
Q20c
Always
Never
Rarely
Sometimes

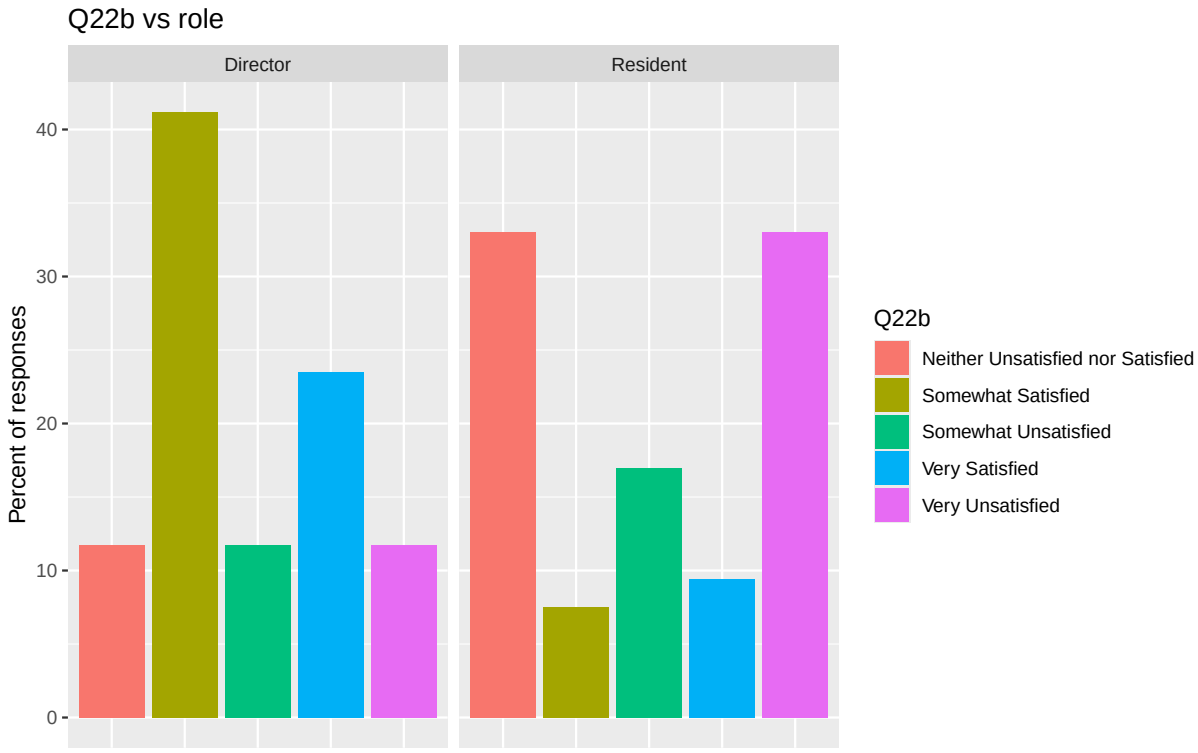

Q20d vs role
Director
Resident
Percent of responses
60
40
20
0
Q20d
Never
Rarely
Sometimes

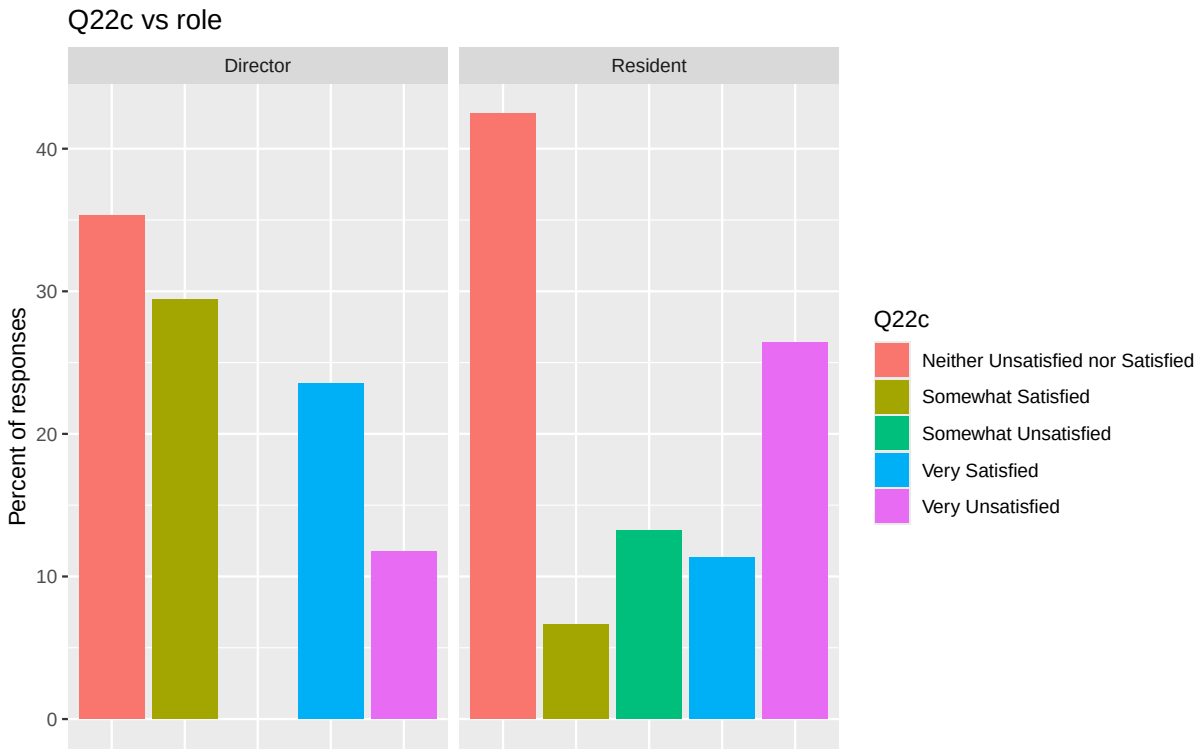
Q21 vs role

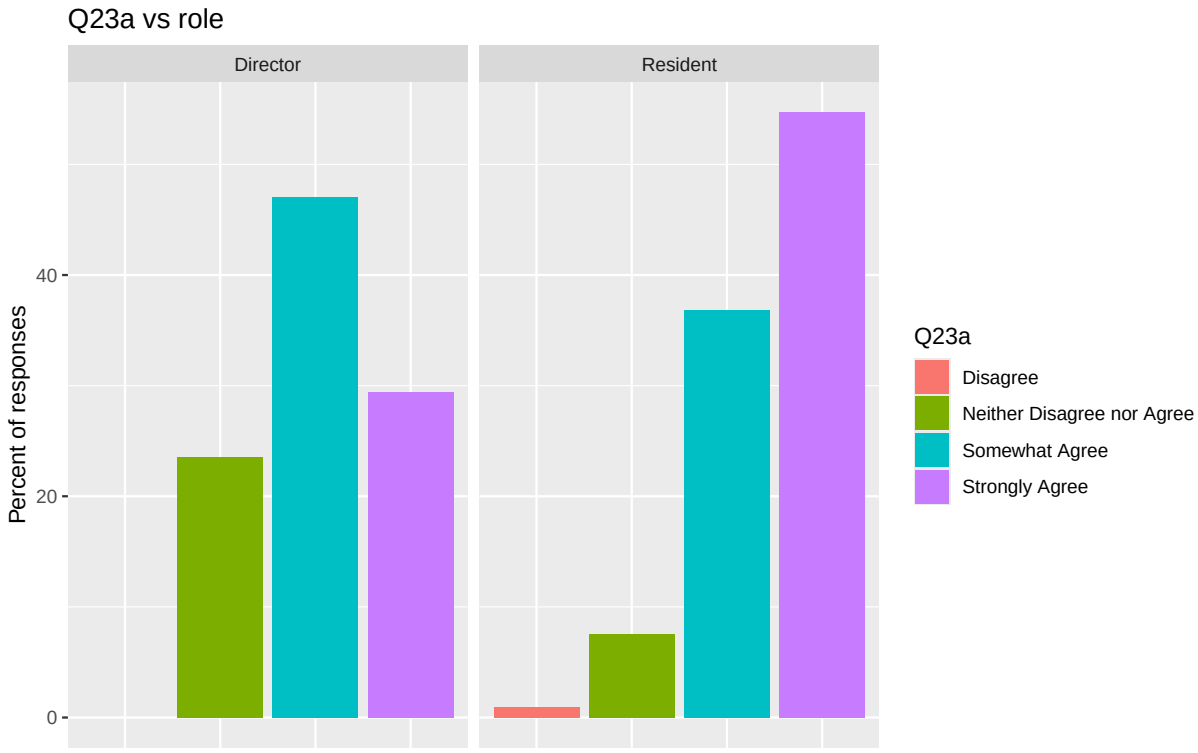


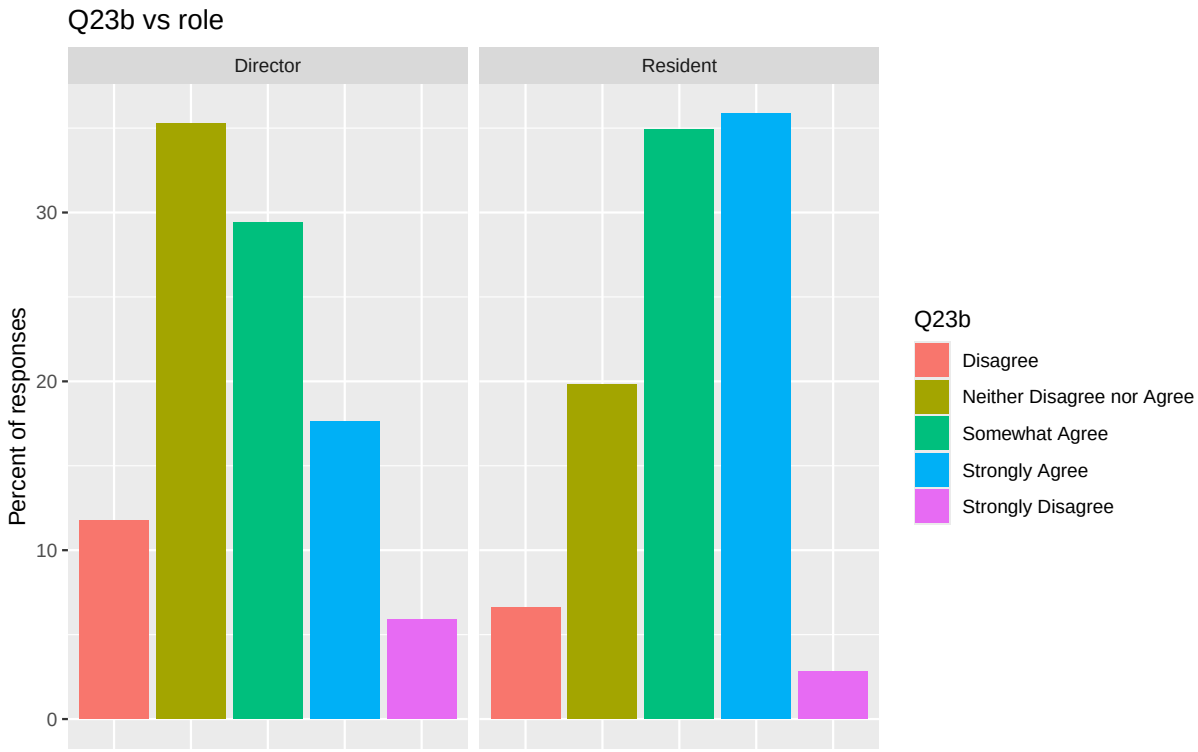
Q22a vs role

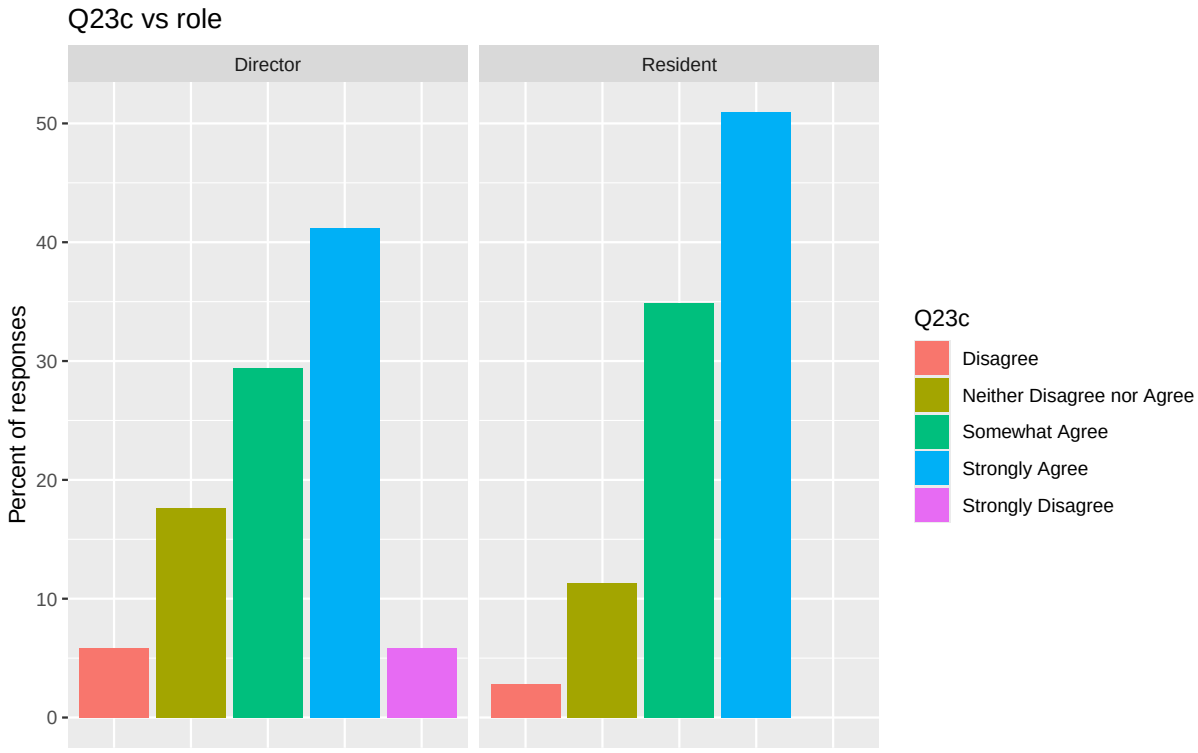


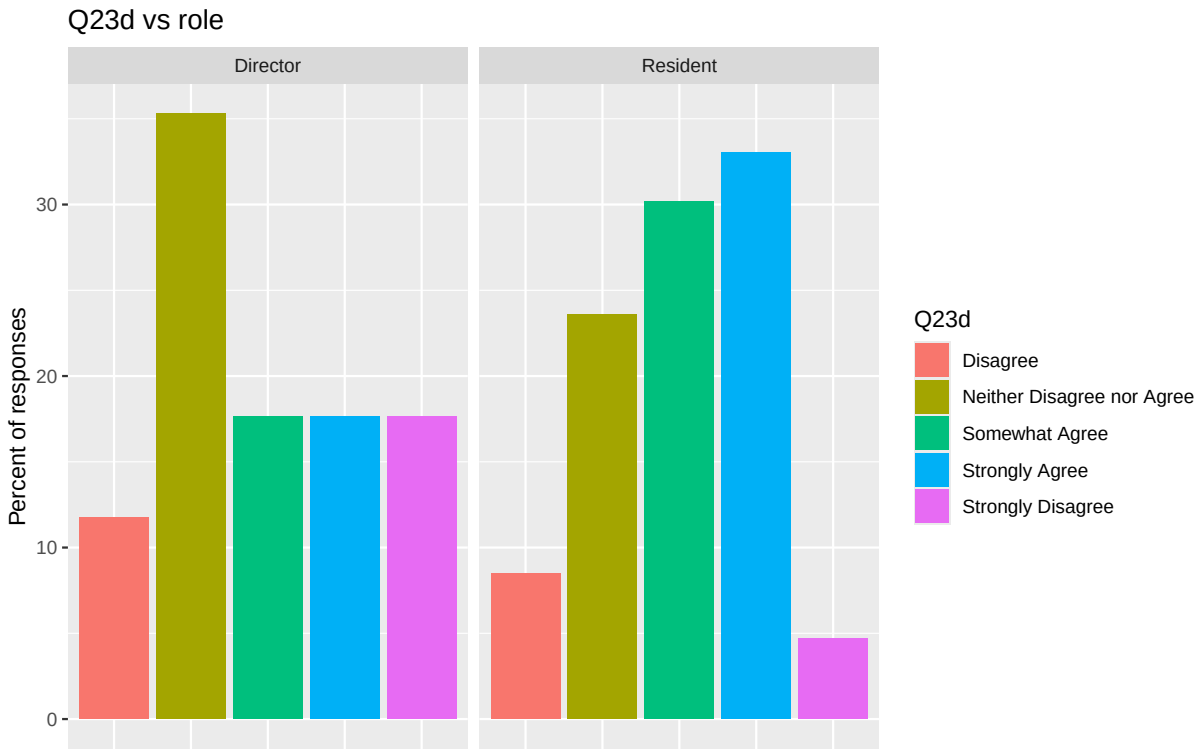

Q22b vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q22b
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Somewhat Unsatisfied
Very Satisfied
Very Unsatisfied

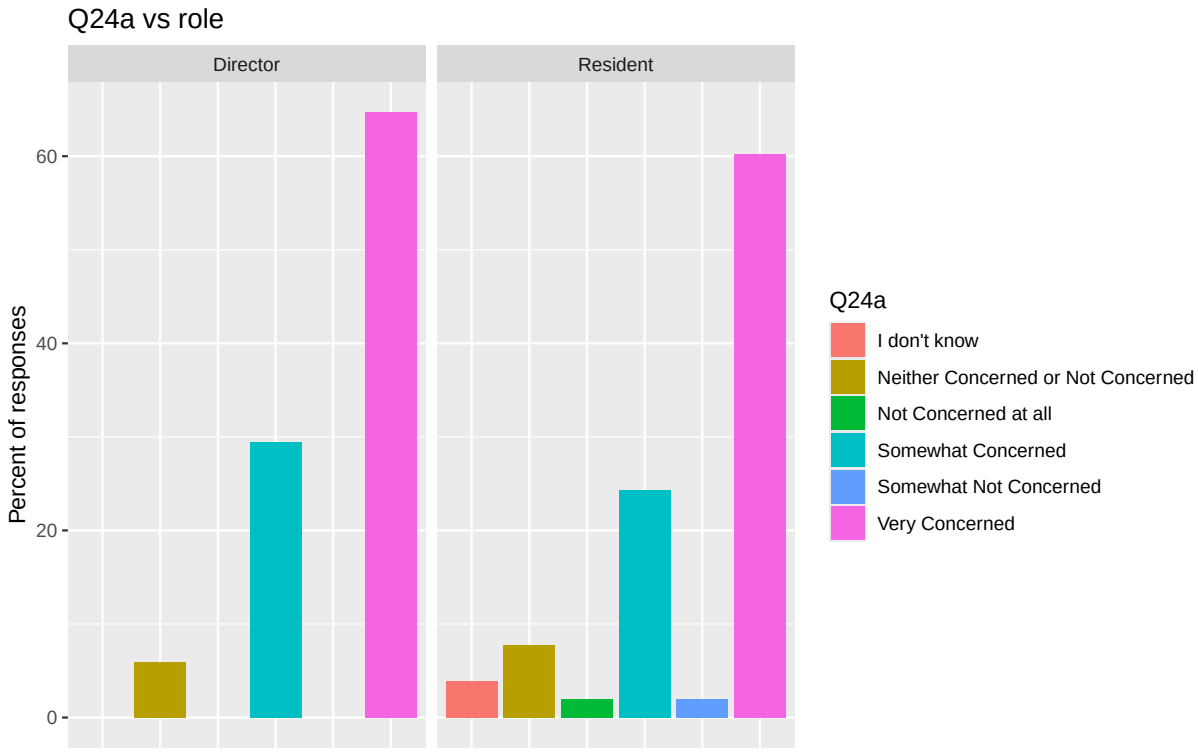

Q22c vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q22c
Neither Unsatisfied nor Satisfied
Somewhat Satisfied
Somewhat Unsatisfied
Very Satisfied
Very Unsatisfied

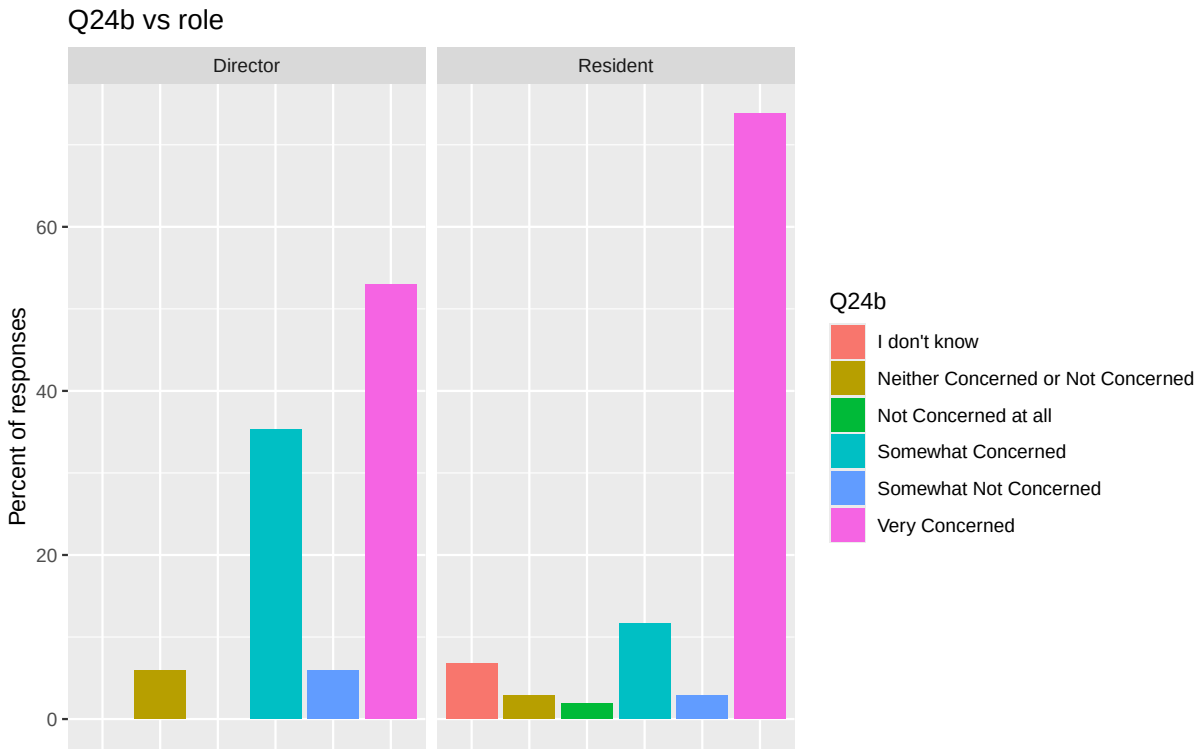

Q23a vs role
Director
Resident
Percent of responses
40
20
0
Q23a
Disagree
Neither Disagree nor Agree
Somewhat Agree
Strongly Agree

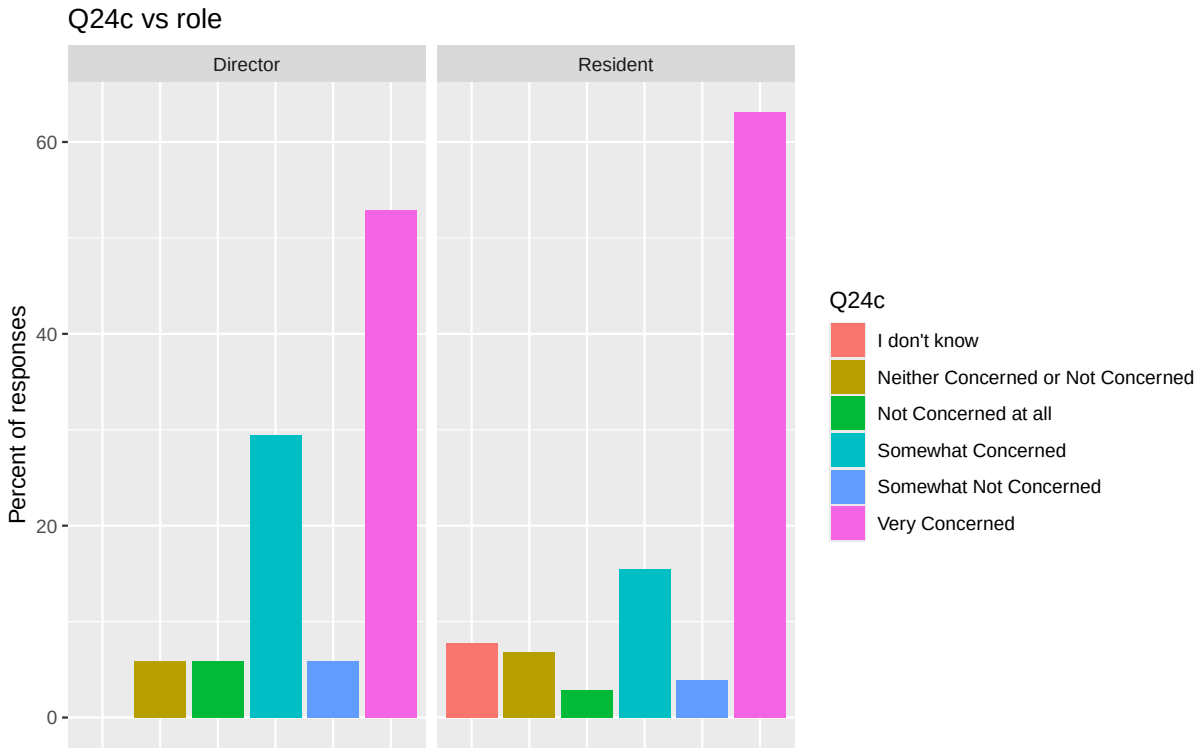

Q23b vs role
Director
Resident
Percent of responses
30
20
10
0
Q23b
Disagree
Neither Disagree nor Agree
Somewhat Agree
Strongly Agree
Strongly Disagree

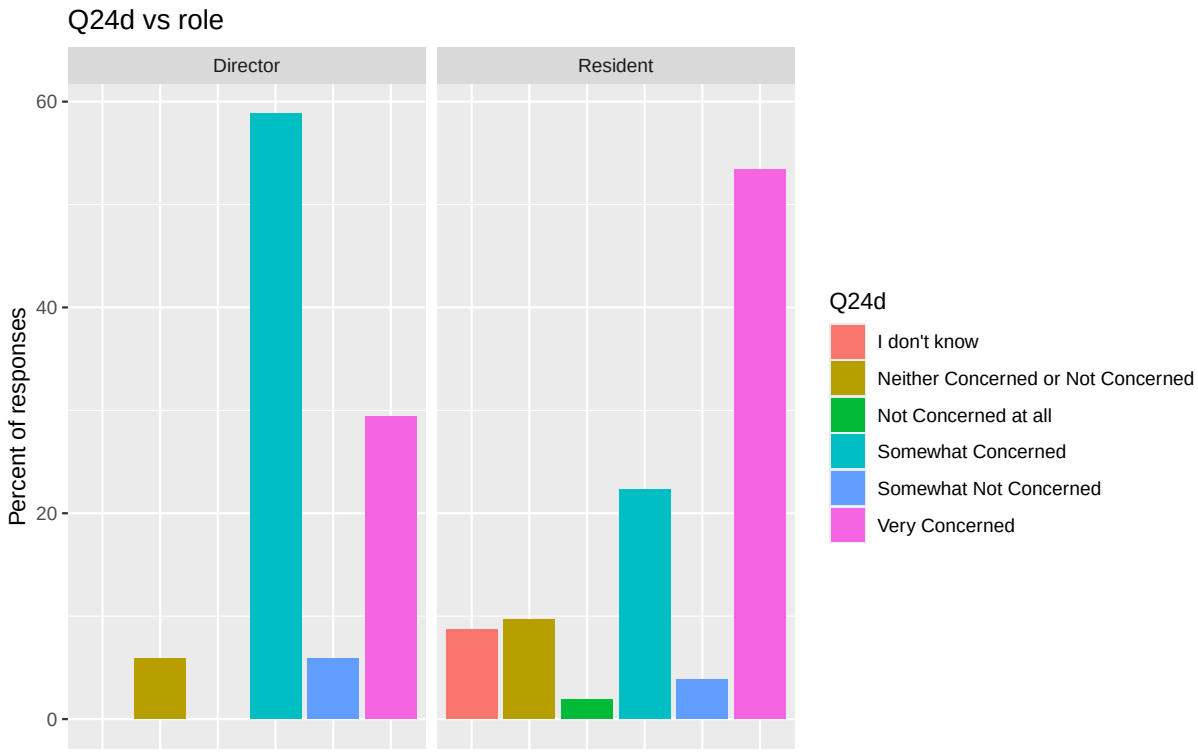

Q23c vs role
Director
Resident
Percent of responses
0
10
20
30
40
50
Q23c
Disagree
Neither Disagree nor Agree
Somewhat Agree
Strongly Agree
Strongly Disagree

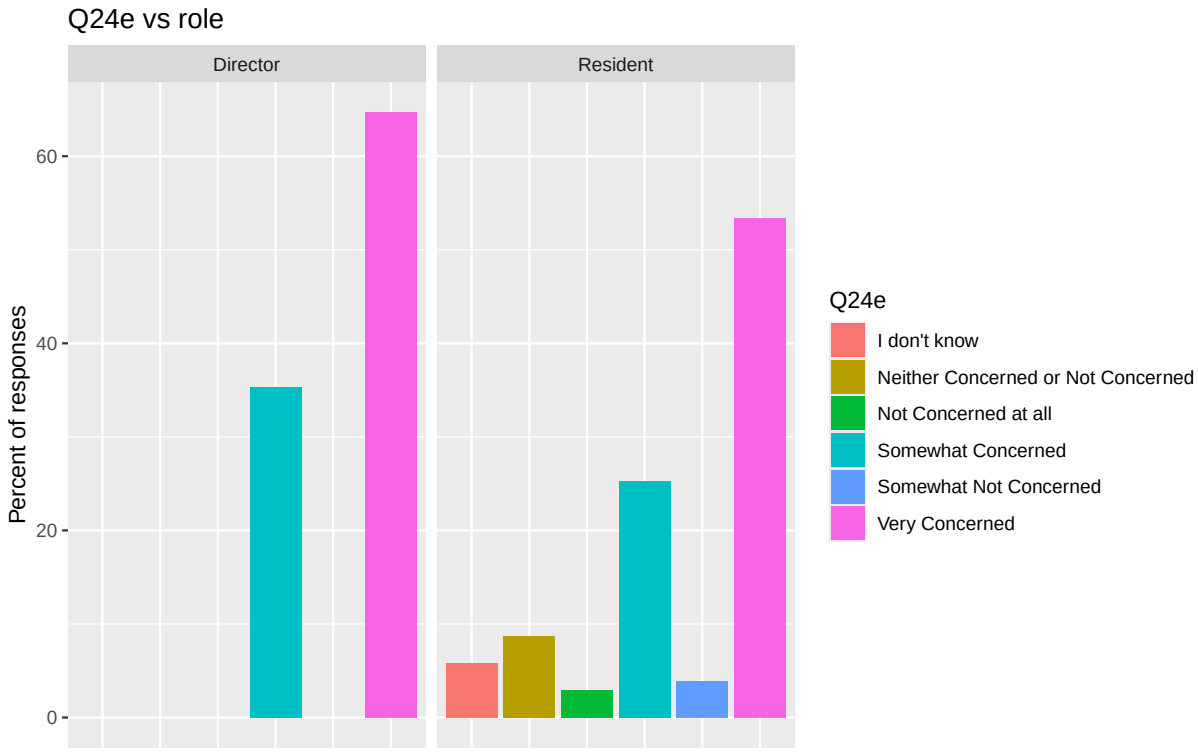

Q23d vs role
Director
Resident
Percent of responses
0
10
20
30
Q23d
Disagree
Neither Disagree nor Agree
Somewhat Agree
Strongly Agree
Strongly Disagree

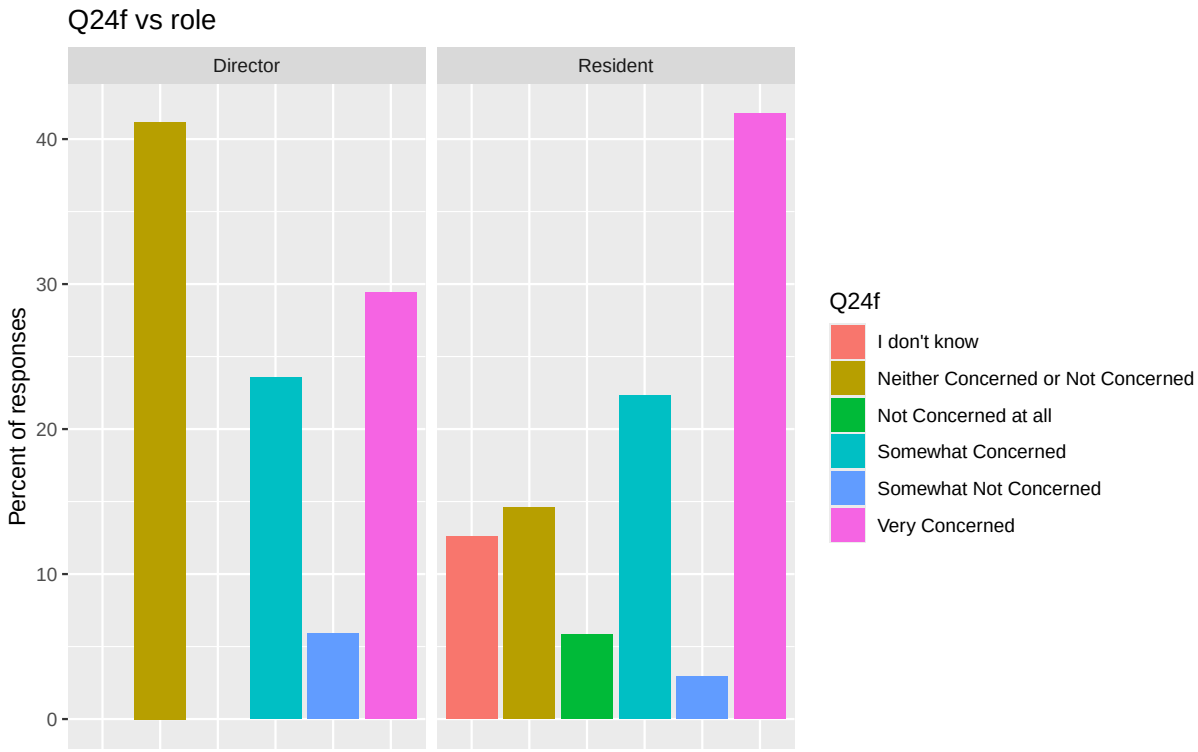

Q24a vs role
Director
Resident
Percent of responses
0
20
40
60
Q24a
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned

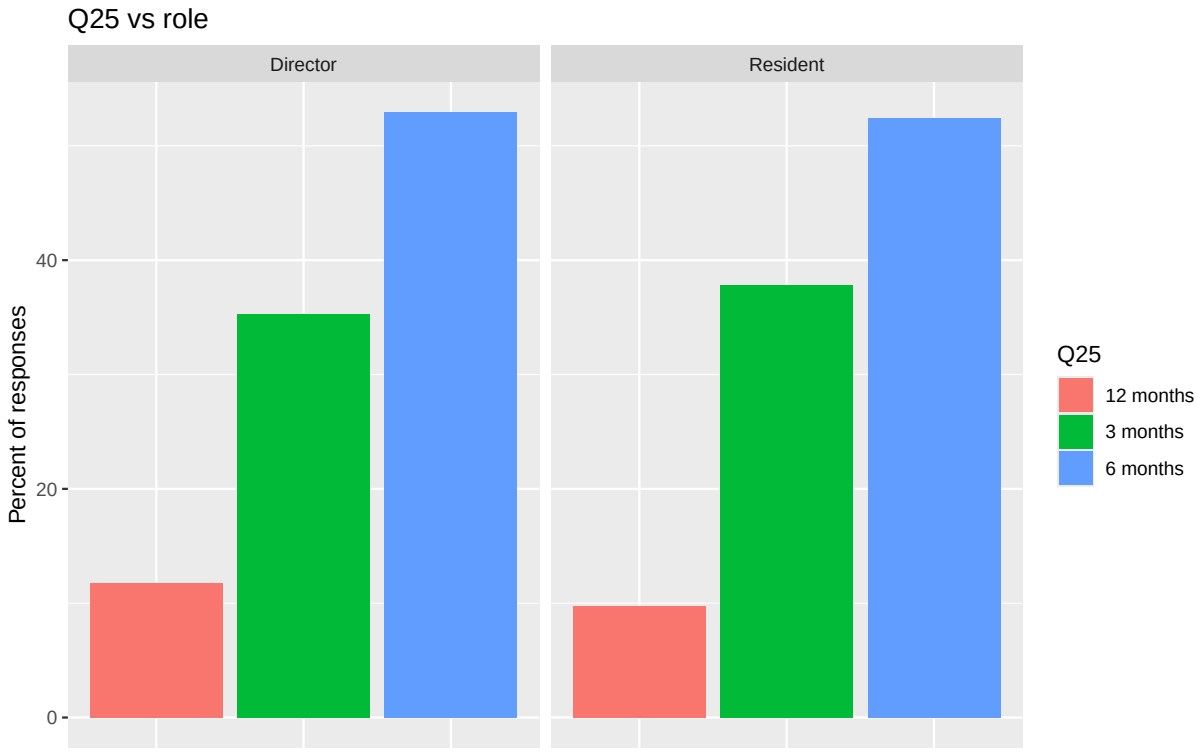

Q24b vs role
Director
Resident
Percent of responses
0
20
40
60
Q24b
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned

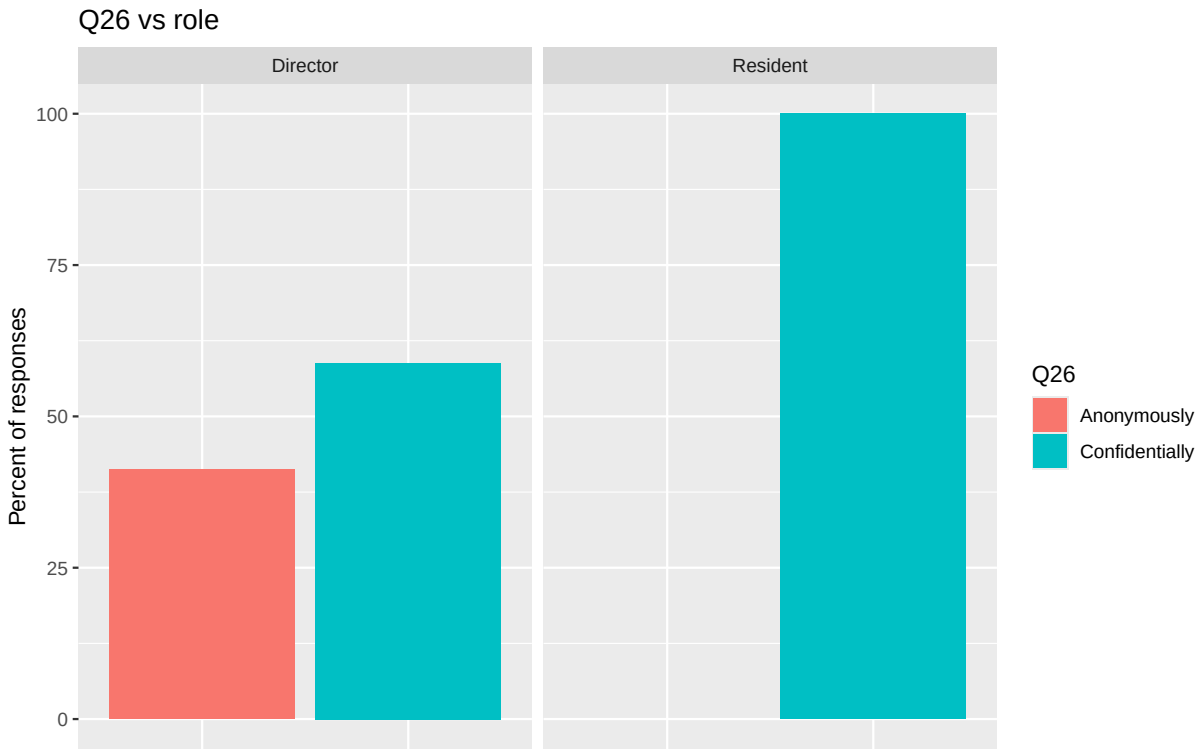

Q24c vs role
Director
Resident
Percent of responses
60
40
20
0
Q24c
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned


Q24d vs role
Director
Resident
Percent of responses
60
40
20
0
Q24d
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned


Q24e vs role
Director
Resident
Percent of responses
0
20
40
60
Q24e
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned


Q24f vs role
Director
Resident
Percent of responses
0
10
20
30
40
Q24f
I don't know
Neither Concerned or Not Concerned
Not Concerned at all
Somewhat Concerned
Somewhat Not Concerned
Very Concerned


Q25 vs role
Director
Resident
Percent of responses
40
20
0
Q25
12 months
3 months
6 months


Q26 vs role
Director
Resident
Percent of responses
100
75
50
25
0
Q26
Anonymously
Confidentially



Appendix B: Code

```
library(tidyverse)
library(zoo)
library(stringr)
library(gt)

if (Sys.info()[["user"]] == "gregorymatthews"){
  headers_raw <- read_csv("/Users/gregorymatthews/Dentist Survey Levine/Falchook/Resident Exp
} else {
  headers_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/Resident Expanded(in).csv"
}
questions <- as.character(headers_raw[1,])

answer_choices <- as.character(headers_raw[2,])

questions_filled <- na.locf(ifelse(questions == "" | is.na(questions), NA, questions))

clean_headers <- paste(questions_filled, answer_choices, sep = "_")
```

```r
if (Sys.info()[["user"]] == "gregorymatthews"){
  f_res_raw <- read_csv("/Users/gregorymatthews/Dentist Survey Levine/Falchook/Resident Expa
} else {
  f_res_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/Resident Expanded(in).csv", 
}

f_res <- f_res_raw %>%
  select(c(10:ncol(f_res_raw)))

f_res <- f_res %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )

f_res_q <- f_res %>%
  mutate(Q1 = do.call(coalesce, pick(1:3))) %>%
  mutate(Q2 = do.call(coalesce, pick(4:6))) %>%
  mutate(Q3 = do.call(coalesce, pick(7:9))) %>%
  mutate(Q4 = do.call(coalesce, pick(10:10))) %>%
  mutate(Q5 = do.call(coalesce, pick(11:15))) %>%
  mutate(Q6 = do.call(coalesce, pick(16:17))) %>%
  mutate(Q7 = do.call(coalesce, pick(18:18))) %>%
  mutate(Q8 = do.call(coalesce, pick(19:23))) %>%
  #mutate(Q9 = do.call(coalesce, pick(24:27))) %>%
  mutate(Q10 = do.call(coalesce, pick(28:32))) %>%
  mutate(Q11 = do.call(coalesce, pick(33:38))) %>%
  mutate(Q12 = do.call(coalesce, pick(39:42))) %>%
  mutate(Q13a = do.call(coalesce, pick(43:49))) %>%
  mutate(Q13b = do.call(coalesce, pick(50:56))) %>%
  mutate(Q13c = do.call(coalesce, pick(57:63))) %>%
  mutate(Q13d = do.call(coalesce, pick(64:70))) %>%
  mutate(Q13e = do.call(coalesce, pick(71:77))) %>%
  mutate(Q13f = do.call(coalesce, pick(78:84))) %>%
  mutate(Q13g = do.call(coalesce, pick(85:91))) %>%
  mutate(Q13h = do.call(coalesce, pick(92:98))) %>%
  mutate(Q13i = do.call(coalesce, pick(99:105))) %>%
  mutate(Q14a = do.call(coalesce, pick(106:110))) %>%
  mutate(Q14b = do.call(coalesce, pick(111:115))) %>%
  mutate(Q14c = do.call(coalesce, pick(116:120))) %>%
```

```
  mutate(Q14d = do.call(coalesce, pick(121:125))) %>%
  mutate(Q15a = do.call(coalesce, pick(126:130))) %>%
  mutate(Q15b = do.call(coalesce, pick(131:135))) %>%
  mutate(Q15c = do.call(coalesce, pick(136:140))) %>%
  mutate(Q16a = do.call(coalesce, pick(141:145))) %>%
  mutate(Q16b = do.call(coalesce, pick(146:150))) %>%
  mutate(Q16c = do.call(coalesce, pick(151:155))) %>%
  mutate(Q16d = do.call(coalesce, pick(156:160))) %>%
  mutate(Q17 = do.call(coalesce, pick(161:165))) %>%
  mutate(Q18 = do.call(coalesce, pick(166:170))) %>%
  mutate(Q19 = do.call(coalesce, pick(171:177))) %>%
  mutate(Q20a = do.call(coalesce, pick(178:182))) %>%
  mutate(Q20b = do.call(coalesce, pick(183:187))) %>%
  mutate(Q20c = do.call(coalesce, pick(188:192))) %>%
  mutate(Q20d = do.call(coalesce, pick(193:196))) %>%
  mutate(Q21 = do.call(coalesce, pick(197:200))) %>%
  mutate(Q22a = do.call(coalesce, pick(201:205))) %>%
  mutate(Q22b = do.call(coalesce, pick(206:210))) %>%
  mutate(Q22c = do.call(coalesce, pick(211:215))) %>%
  mutate(Q23a = do.call(coalesce, pick(216:220))) %>%
  mutate(Q23b = do.call(coalesce, pick(221:225))) %>%
  mutate(Q23c = do.call(coalesce, pick(226:230))) %>%
  mutate(Q23d = do.call(coalesce, pick(231:235))) %>%
  mutate(Q24a = do.call(coalesce, pick(236:241))) %>%
  mutate(Q24b = do.call(coalesce, pick(242:247))) %>%
  mutate(Q24c = do.call(coalesce, pick(248:253))) %>%
  mutate(Q24d = do.call(coalesce, pick(254:259))) %>%
  mutate(Q24e = do.call(coalesce, pick(260:265))) %>%
  mutate(Q24f = do.call(coalesce, pick(266:271))) %>%
  mutate(Q25 = do.call(coalesce, pick(272:275))) %>%
  mutate(Q26 = do.call(coalesce, pick(276:278))) %>%
  #mutate(Q27 = do.call(coalesce, pick(279:289))) %>%
  mutate(Q28 = do.call(coalesce, pick(290:290)))

f_res_coalesced_partial <- f_res_q %>%
  select(291:298, 24:27, 299:344, 279:289, 345)

f_res_coalesced <- f_res_coalesced_partial %>%
  mutate(Q9 = pmap_chr(select(., 9:12), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q27 = pmap_chr(select(., 59:69), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  select(1:8, 71, 13:58, 72, 70)
```

```r
f_res_coalesced$role <- "Resident"
f_res_coalesced <- f_res_coalesced %>%
  mutate(Q26 = case_when(
  Q26 == "Confidentially (one-on-one meetings)" ~ "Confidentially",
  Q26 == "Anonymously (online questionnaire)" ~ "Anonymously"
  ))

if (Sys.info()[["user"]] == "gregorymatthews"){
  dir_headers_raw <- read_csv("/Users/gregorymatthews/Dentist Survey Levine/Falchook/PD expa
} else {
  dir_headers_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/PD expanded(in).csv", 
}

dir_questions <- as.character(dir_headers_raw[1,])

dir_answer_choices <- as.character(dir_headers_raw[2,])

dir_questions_filled <- na.locf(ifelse(dir_questions == "" | is.na(dir_questions), NA, dir_q

dir_clean_headers <- paste(dir_questions_filled, dir_answer_choices, sep = "_")

if (Sys.info()[["user"]] == "gregorymatthews"){
  f_dir_raw <- read_csv("/Users/gregorymatthews/Dentist Survey Levine/Falchook/PD expanded.cs
    f_dir_raw <- read_csv("/Users/jdumanski24/Desktop/CDSC/Levine/PD expanded(in).csv", skip
  }

f_dir <- f_dir_raw %>%
  select(c(10:ncol(f_dir_raw)))
#view(f_dir)

f_dir <- f_dir %>%
  mutate(
    across(
      everything(),
      ~ na_if(as.character(.), "")
    )
  )

f_dir_q <- f_dir %>%
  mutate(Q1 = do.call(coalesce, pick(1:2))) %>%
  mutate(Q2 = do.call(coalesce, pick(3:5))) %>%
  mutate(Q3 = do.call(coalesce, pick(6:6))) %>%
```

```
  mutate(Q4 = do.call(coalesce, pick(7:9))) %>%
  mutate(Q5 = do.call(coalesce, pick(10:10))) %>%
  mutate(Q6 = do.call(coalesce, pick(11:11))) %>%
  mutate(Q7 = do.call(coalesce, pick(12:16))) %>%
  #mutate(Q8 = do.call(coalesce, pick(17:20))) %>%
  mutate(Q9 = do.call(coalesce, pick(21:25))) %>%
  mutate(Q10 = do.call(coalesce, pick(26:32))) %>%
  mutate(Q11 = do.call(coalesce, pick(33:37))) %>%
  mutate(Q12a = do.call(coalesce, pick(38:45))) %>%
  mutate(Q12b = do.call(coalesce, pick(46:53))) %>%
  mutate(Q12c = do.call(coalesce, pick(54:61))) %>%
  mutate(Q12d = do.call(coalesce, pick(62:69))) %>%
  mutate(Q12e = do.call(coalesce, pick(70:77))) %>%
  mutate(Q12f = do.call(coalesce, pick(78:85))) %>%
  mutate(Q12g = do.call(coalesce, pick(86:93))) %>%
  mutate(Q12h = do.call(coalesce, pick(94:101))) %>%
  mutate(Q12i = do.call(coalesce, pick(102:109))) %>%
  mutate(Q13a = do.call(coalesce, pick(110:115))) %>%
  mutate(Q13b = do.call(coalesce, pick(116:121))) %>%
  mutate(Q13c = do.call(coalesce, pick(122:127))) %>%
  mutate(Q13d = do.call(coalesce, pick(128:133))) %>%
  mutate(Q14a = do.call(coalesce, pick(134:138))) %>%
  mutate(Q14b = do.call(coalesce, pick(139:143))) %>%
  mutate(Q14c = do.call(coalesce, pick(144:148))) %>%
  mutate(Q15a = do.call(coalesce, pick(149:153))) %>%
  mutate(Q15b = do.call(coalesce, pick(154:158))) %>%
  mutate(Q15c = do.call(coalesce, pick(159:163))) %>%
  mutate(Q15d = do.call(coalesce, pick(164:168))) %>%
  mutate(Q16 = do.call(coalesce, pick(169:173))) %>%
  mutate(Q17 = do.call(coalesce, pick(174:178))) %>%
  mutate(Q18 = do.call(coalesce, pick(179:184))) %>%
  mutate(Q19a = do.call(coalesce, pick(185:189))) %>%
  mutate(Q19b = do.call(coalesce, pick(190:194))) %>%
  mutate(Q19c = do.call(coalesce, pick(195:199))) %>%
  mutate(Q19d = do.call(coalesce, pick(200:204))) %>%
  mutate(Q20 = do.call(coalesce, pick(205:208))) %>%
  mutate(Q21 = do.call(coalesce, pick(209:210))) %>%
  mutate(Q22a = do.call(coalesce, pick(211:215))) %>%
  mutate(Q22b = do.call(coalesce, pick(216:220))) %>%
  mutate(Q22c = do.call(coalesce, pick(221:225))) %>%
  mutate(Q23a = do.call(coalesce, pick(226:230))) %>%
  mutate(Q23b = do.call(coalesce, pick(231:235))) %>%
```

```
  mutate(Q23c = do.call(coalesce, pick(236:240))) %>%
  mutate(Q23d = do.call(coalesce, pick(241:245))) %>%
  mutate(Q24a = do.call(coalesce, pick(246:250))) %>%
  mutate(Q24b = do.call(coalesce, pick(251:255))) %>%
  mutate(Q24c = do.call(coalesce, pick(256:260))) %>%
  mutate(Q24d = do.call(coalesce, pick(261:265))) %>%
  mutate(Q24e = do.call(coalesce, pick(266:270))) %>%
  mutate(Q24f = do.call(coalesce, pick(271:275))) %>%
  mutate(Q25 = do.call(coalesce, pick(276:278))) %>%
  mutate(Q26 = do.call(coalesce, pick(279:281))) %>%
  #mutate(Q27 = do.call(coalesce, pick(282:292))) %>%
  mutate(Q28 = do.call(coalesce, pick(293:293)))

f_dir_coalesced_partial <- f_dir_q %>%
  select(294:300, 17:20, 301:347, 282:292, 348)

f_dir_coalesced <- f_dir_coalesced_partial %>%
  mutate(Q8 = pmap_chr(select(., 8:11), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  mutate(Q27 = pmap_chr(select(., 59:69), ~ paste(na.omit(c(...)), collapse = "; "))) %>%
  select(1:7, 71, 12:58, 72, 70)
f_dir_coalesced$role <- "Director"
f_dir_coalesced <- f_dir_coalesced %>%
  mutate(Q26 = case_when(
  Q26 == "Confidentially (one-on-one meetings)" ~ "Confidentially",
  Q26 == "Anonymously (online questionnaire)" ~ "Anonymously"
  ))

question1s <- c("Q11", rep("Q13a", 6), rep("Q13b", 6), rep("Q13c", 6), rep("Q13d", 6), rep("
question2s <- c("Q1", rep(c("Q10", "Q11", "Q12", "Q1", "Q5", "Q2"), 9), rep(c("Q5", "Q3"), 4

question_dic_res <- data.frame(question1s, question2s)
# when trying to cross reference one question (question1) by another grouping variable (quest
validation <- function(question1, question2, data){
  df <- data %>%
    transmute(
      response1 = .data[[question1]],
      response2 = .data[[question2]]
    ) %>%
    drop_na()

  comparison_res <- df %>%
    group_by(response1, response2) %>%
```

```
    summarize(counts = n(), .groups = "drop") %>%
    group_by(response2) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  table_comparison <- table(df$response2, df$response1)
#table_hours
  test <- suppressWarnings(chisq.test(table_comparison))

  if (mean(test$expected <= 5) < 0.8 | any(test$expected < 1)){
    fish_test <- fisher.test(table_comparison, simulate.p.value = TRUE)
    if (fish_test$p.value <= 0.05){
     significant <- TRUE
   }
   else {
      significant <- FALSE
   }
    return(c("question1", "question2", round(fish_test$p.value, digits = 4), significant, "fi
  }
  else{
    if (test$p.value <= 0.05){
     significant <- TRUE
   }
   else {
      significant <- FALSE
   }
    }
  return(c("question1", "question2", round(test$p.value, digits = 4), significant, "chisq"))
}


res_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  test = character()
)

for (i in seq_len(nrow(question_dic_res))) {

  res_results[i, ] <- validation(
```

```
    question1 = question_dic_res$question1s[i],
    question2 = question_dic_res$question2s[i],
    data = f_res_coalesced
  )
}

res_results <- res_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = question_dic_res$question1s,
    Question2 = question_dic_res$question2s
  ) %>% arrange(Adj_P)
gt(res_results)
question1s_dir <- c("Q10", "Q12a", "Q12b", "Q12c", "Q12d", "Q12e", "Q12f", "Q12g", "Q12h", "
question2s_dir <- c(rep("Q2",14))

question_dic_dir <- data.frame(question1s_dir, question2s_dir)
dir_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  test = character()
)

for (i in seq_len(nrow(question_dic_dir))) {
  dir_results[i, ] <- validation(
    question1 = question_dic_dir$question1s_dir[i],
    question2 = question_dic_dir$question2s_dir[i],
    data = f_dir_coalesced
  )
}

dir_results <- dir_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = question_dic_dir$question1s,
    Question2 = question_dic_dir$question2s
  ) %>% arrange(Adj_P)
gt(dir_results)
```

```r
common_qs_res <- c("Q1", "Q3", "Q8", "Q9", "Q10", "Q11", "Q12", "Q13a", "Q13b","Q13c","Q13d"
"Q16a", "Q16b", "Q16c", "Q16d", "Q17",  "Q18",  "Q19",  "Q20a", "Q20b", "Q20c", "Q20d", "Q21
common_qs_dir <- c("Q2", "Q4", "Q7", "Q8", "Q9", "Q10", "Q11", "Q12a", "Q12b","Q12c","Q12d",
"Q15a", "Q15b", "Q15c", "Q15d", "Q16",  "Q17",  "Q18",  "Q19a", "Q19b", "Q19c", "Q19d", "Q20

full_dic <- data.frame(common_qs_res, common_qs_dir)

res_common <- f_res_coalesced %>%
  select(all_of(common_qs_res))

dir_common <- f_dir_coalesced %>%
  select(all_of(common_qs_dir)) %>%
  setNames(common_qs_res)
res_dir <- bind_rows(
  res_common,
  dir_common
)
res_dir_results <- data.frame(
  Question1 = character(),
  Question2 = character(),
  P_value = numeric(),
  Significant = logical(),
  test = character()
)

for (i in seq_len(ncol(res_dir)-1)) {

  res_dir_results[i, ] <- validation(
    question1 = colnames(res_dir)[i],
    question2 = colnames(res_dir)[52],
    data = res_dir
  )
}

res_dir_results <- res_dir_results %>%
  mutate(
    Adj_P = round(p.adjust(P_value, method = "BH"), digits = 4),
    Sig_Adj = Adj_P <= 0.05,
    Question1 = colnames(res_dir)[1:51],
    Question2 = "role"
  ) %>% arrange(Adj_P)
gt(res_dir_results)
```

```r
stack_bar <- function(q_reference, q_group, data){
  bar_df <- data %>%
    group_by(.data[[q_reference]], .data[[q_group]]) %>%
    summarize(counts = n(), .groups = "drop") %>%
    drop_na() %>%
    group_by(.data[[q_group]]) %>%
    mutate(percent = counts / sum(counts) * 100) %>%
    ungroup()

  p <- ggplot(bar_df, aes(x = .data[[q_reference]], y = percent, fill = .data[[q_reference]]))
    geom_col() +
    facet_wrap(vars(.data[[q_group]])) +
    labs(
      title = paste(q_reference, "vs", q_group, sep = " "),
      x = NULL,
      y = "Percent of responses"
    ) +
    theme(
      axis.text.x = element_blank(),
      axis.ticks.x = element_blank()
    )
  return(p)
}

plots_res <- pmap(question_dic_res,
              function(question1s, question2s){
                stack_bar(
                  q_reference = question1s,
                  q_group = question2s,
                  data = f_res_coalesced
                )
              })
walk(plots_res, print)

plots_dir <- pmap(question_dic_dir,
                  function(question1s_dir, question2s_dir){
                    stack_bar(
                      q_reference = question1s_dir,
                      q_group = question2s_dir,
                      data = f_dir_coalesced
                    )
                  })
```

```r
walk(plots_dir, print)

question_dic_res_dir <- data.frame(question1s_res_dir = colnames(res_dir)
                                   , question2s_res_dir = rep("role", ncol(res_dir)))
question_dic_res_dir <- question_dic_res_dir %>%
  filter(question1s_res_dir != "role")
plots_comp <- pmap(question_dic_res_dir,
                   function(question1s_res_dir, question2s_res_dir){
                     stack_bar(
                       q_reference = question1s_res_dir,
                       q_group = question2s_res_dir,
                       data = res_dir
                     )
                   })
walk(plots_comp, print)
```